
EXPRESSION OF INTEREST

Kankawee Maksomboon | s4097770



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HYBRID TECTONICS

Roland Snooks | Marc Gibson





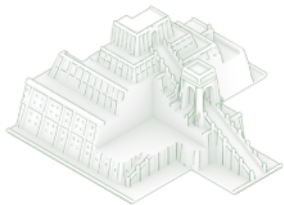
BEYOND MONOLITH

This project demonstrates the transformation of hypertension between two architectural styles: Monolithic Architecture (ancient ruins and postmodern architecture) and the Estranged part. The project explores how these two styles interact through different transformation processes — stepping, merging, blending, and inflation

MONOLITHIC TECTONICS

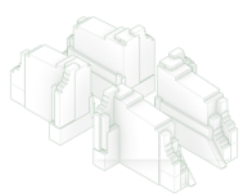
By deconstructing elements from ancient ruins and Postmodern Brutalist architecture, reassembly techniques were applied to introduce a hybridized design language. Combined with monolithic architecture, the project employs stepped tectonic assembly and joinery logic to create a new architectural expression.

A1



ZIGGURAT

A2



KARNAK-TEMPLE

B1



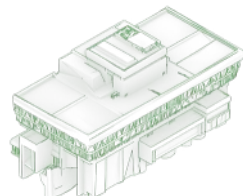
HUMPHREY BUILDING

B2

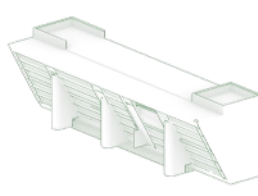


BOSTON CITY HALL

B3

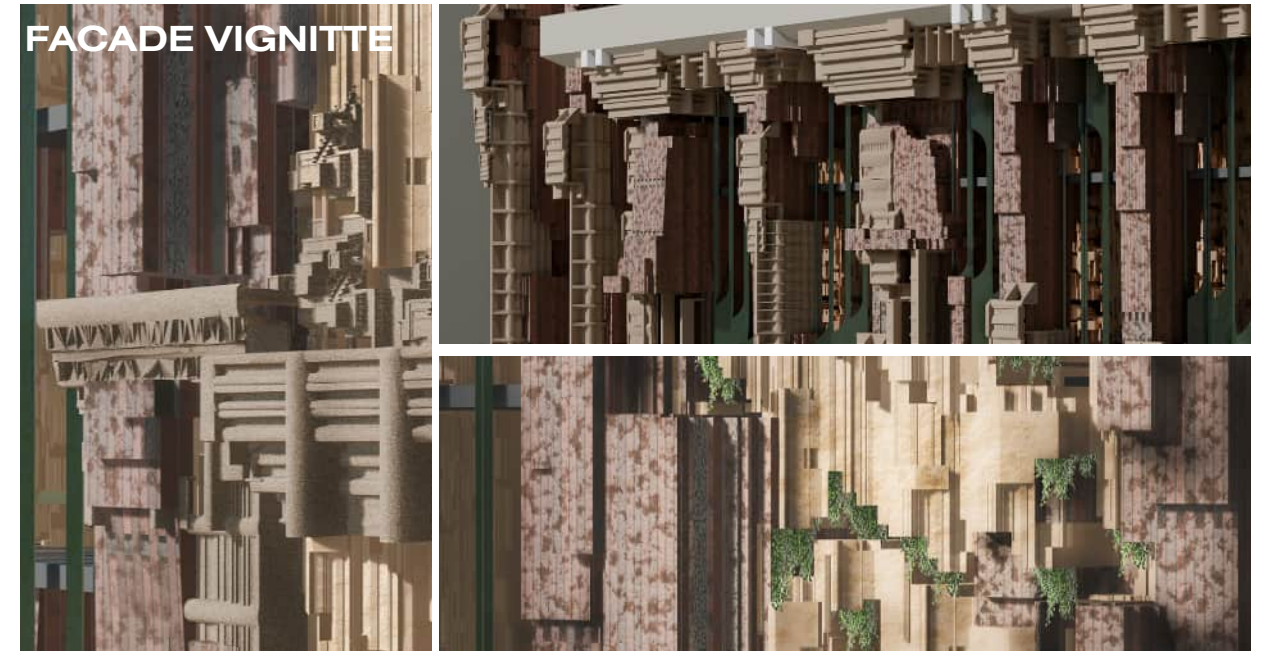


B4



DALLAS CITY HALL

FACADE VIGNETTE



**Color palette inspired
from SEU campus architecture**

Materiality also plays an huge role, using a combination of mono-lithic materials and es-tranged metal materials to express contrast and transition.

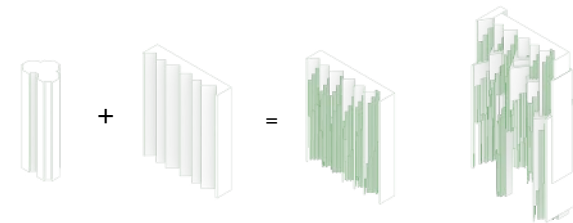


PETRA TOMB, JORDAN | FACADE CONDITION

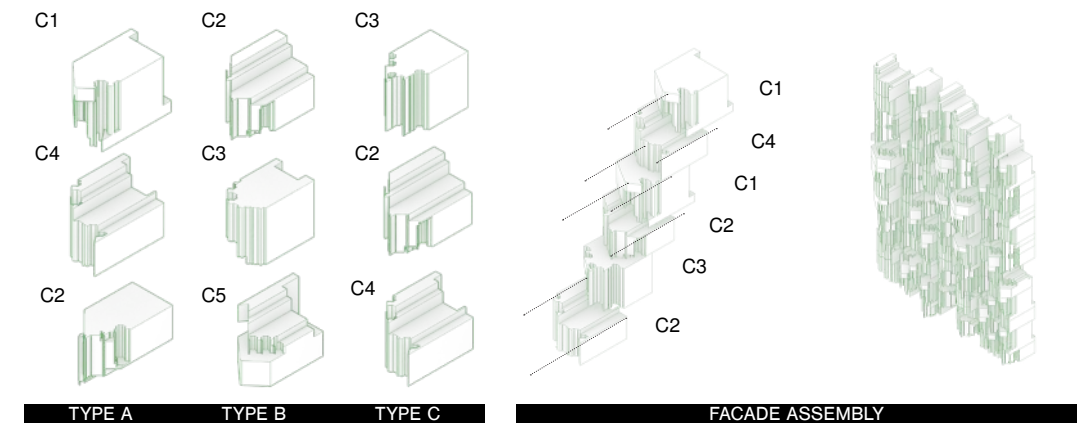
FACADE

Architectural ruin facade Inspired form Petra With Layering effect creates light,- shadow, and depth The tectonic process start form parts together with subtraction & aggregation, resulting in an unique monolithic chunks Then I separate them into difference fragment, then assembly again using stepping tectonic logic (as show in the picture how things join together)

Facade assembly | Stepping tectonic logic



Facade parts process | subtraction & aggregation

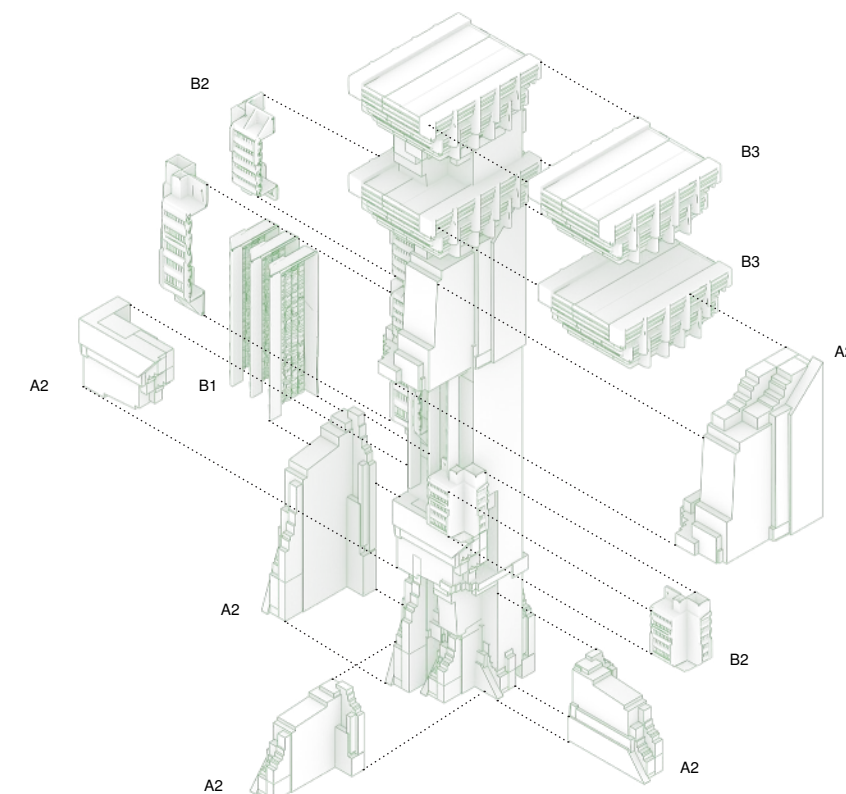
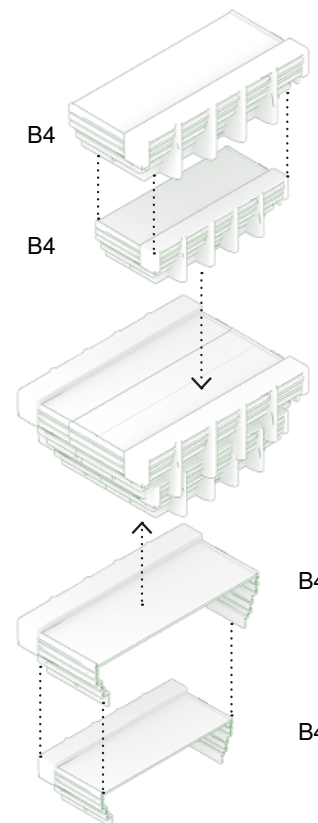


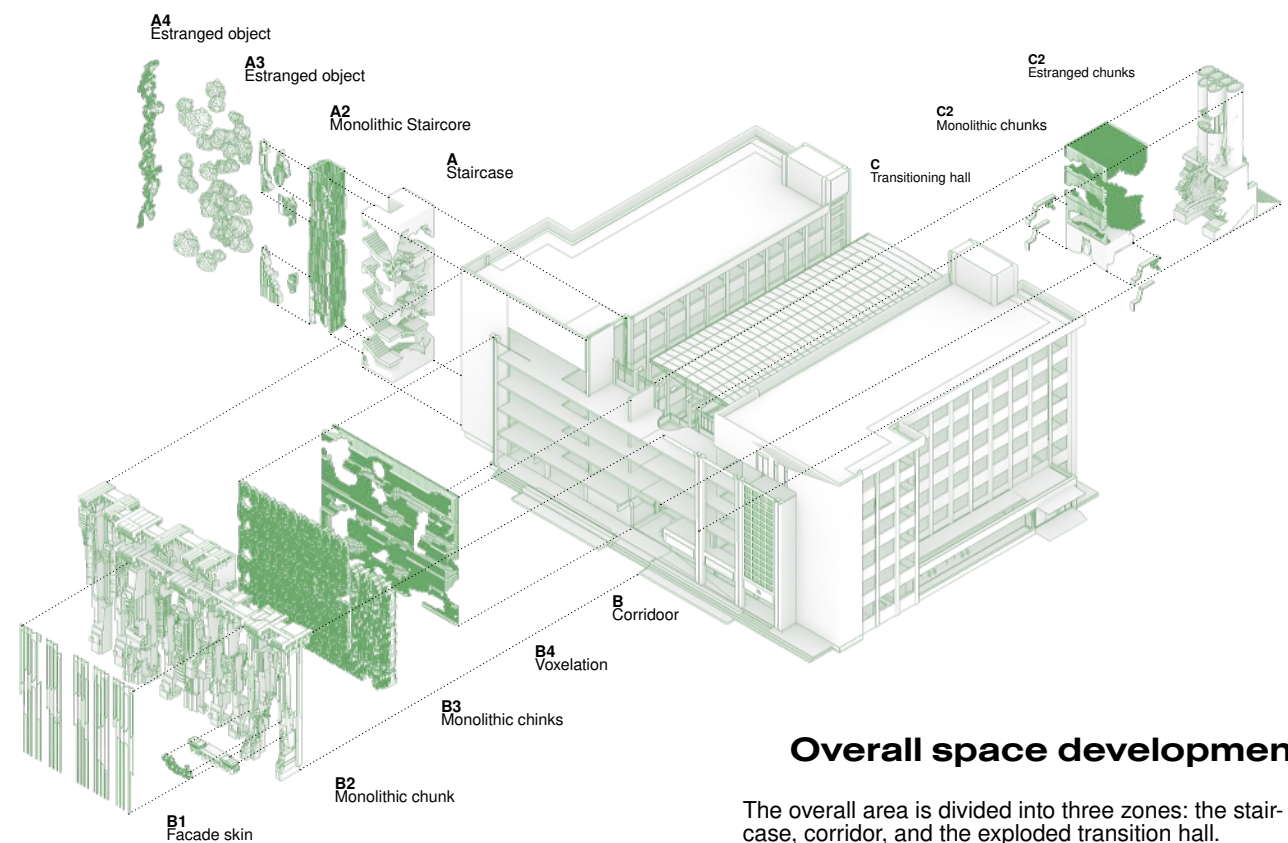
COLUMN ASSEMBLY



Chinese capital

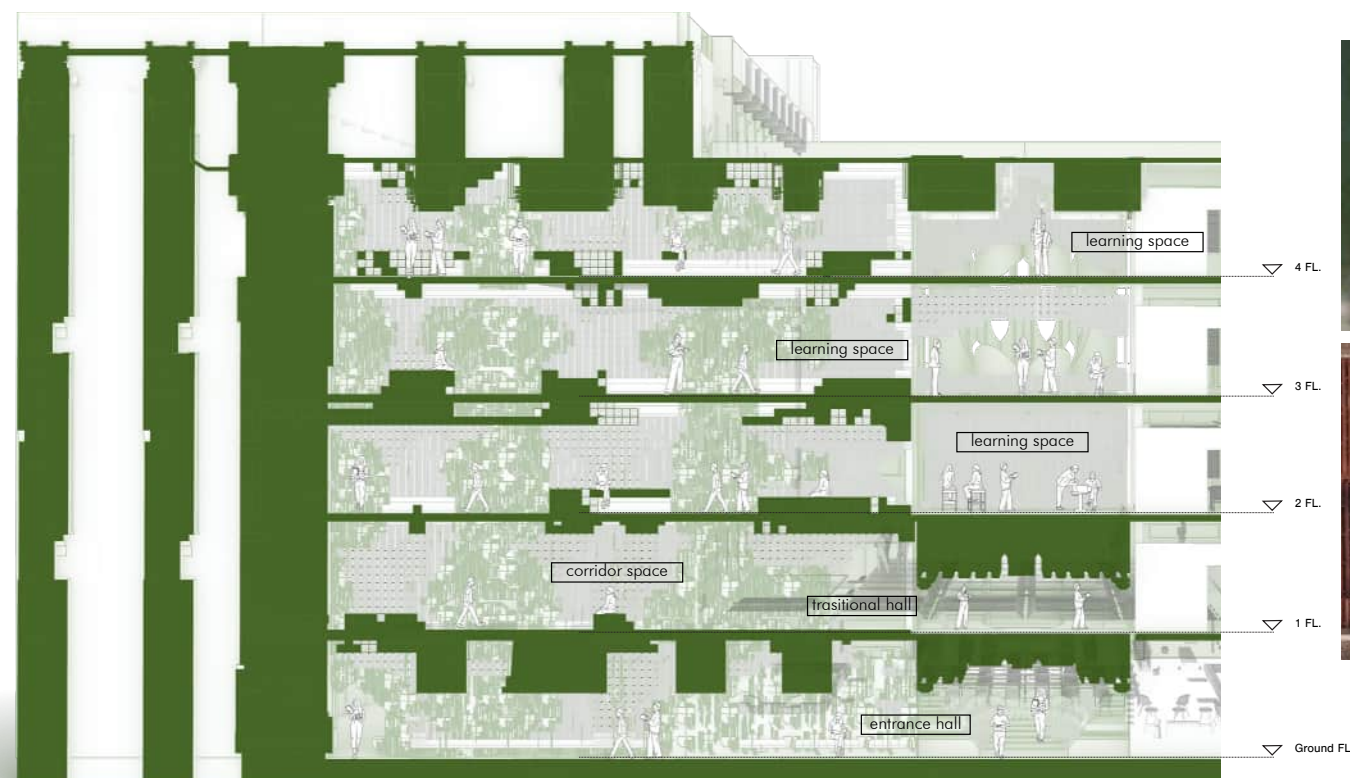
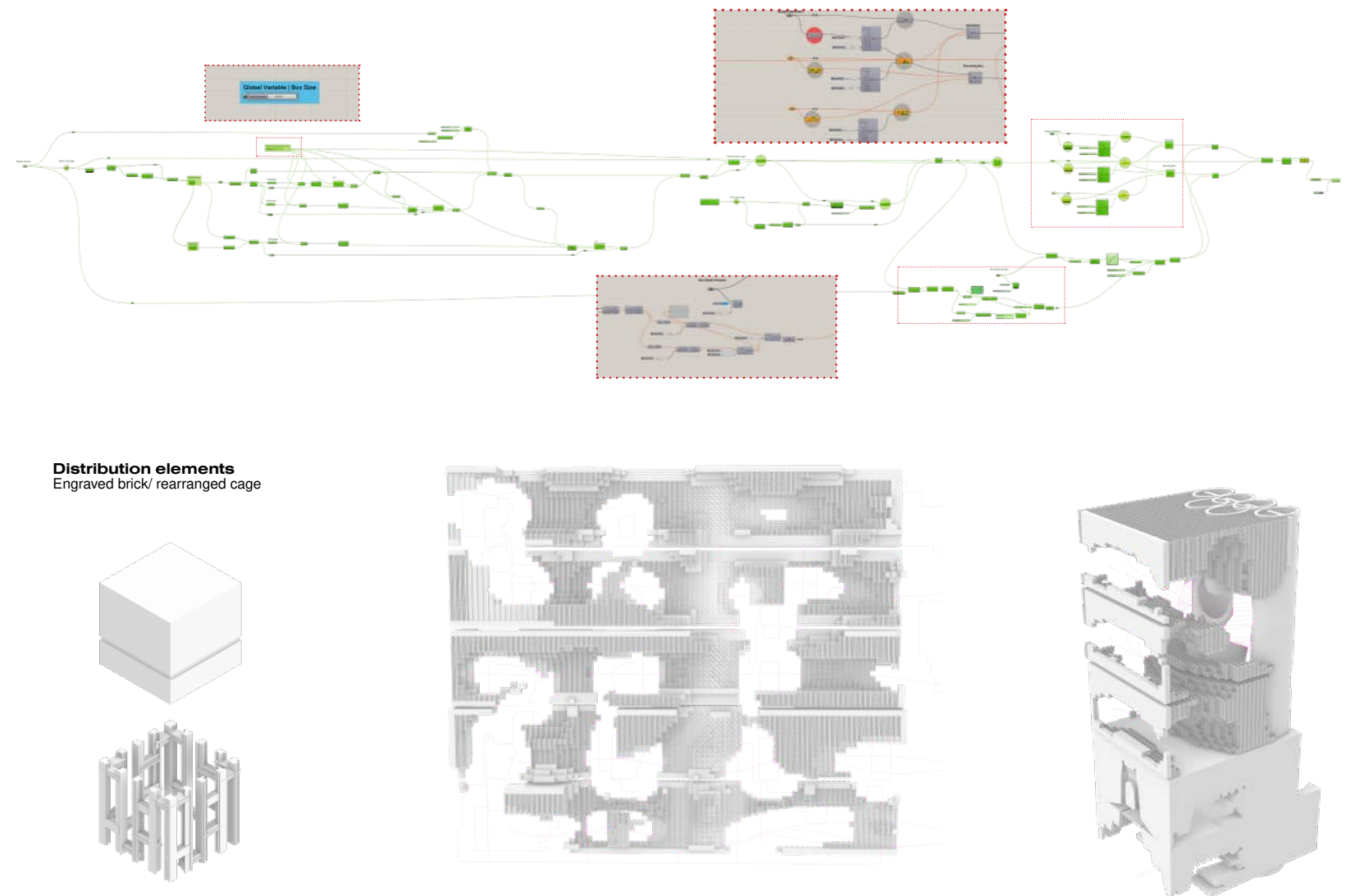
For capital, I use a facade part form post modern architecture, then re-introducing them into column using stacking ancient chinese capital tectonice



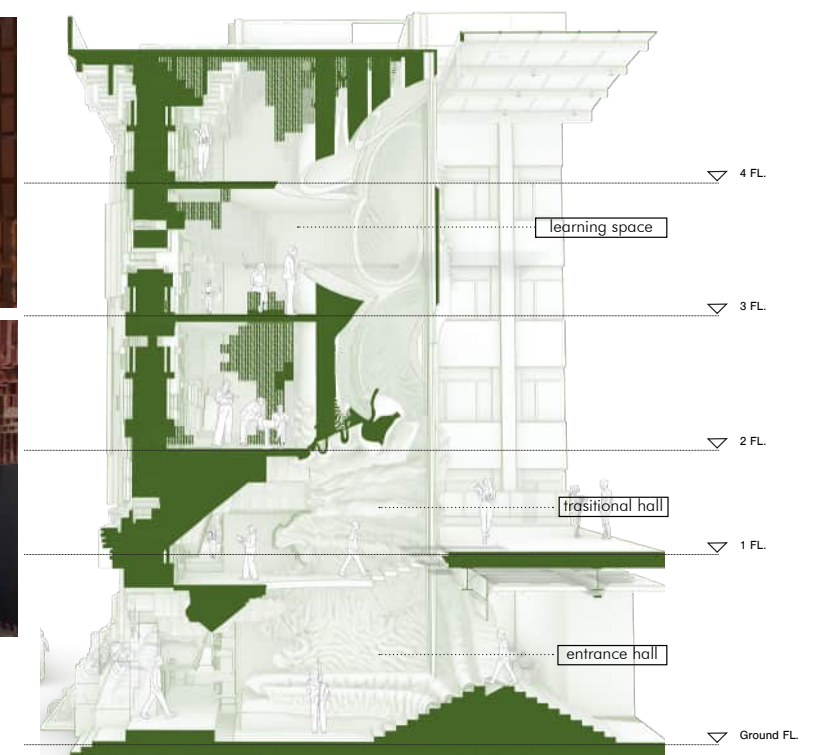


Overall space development

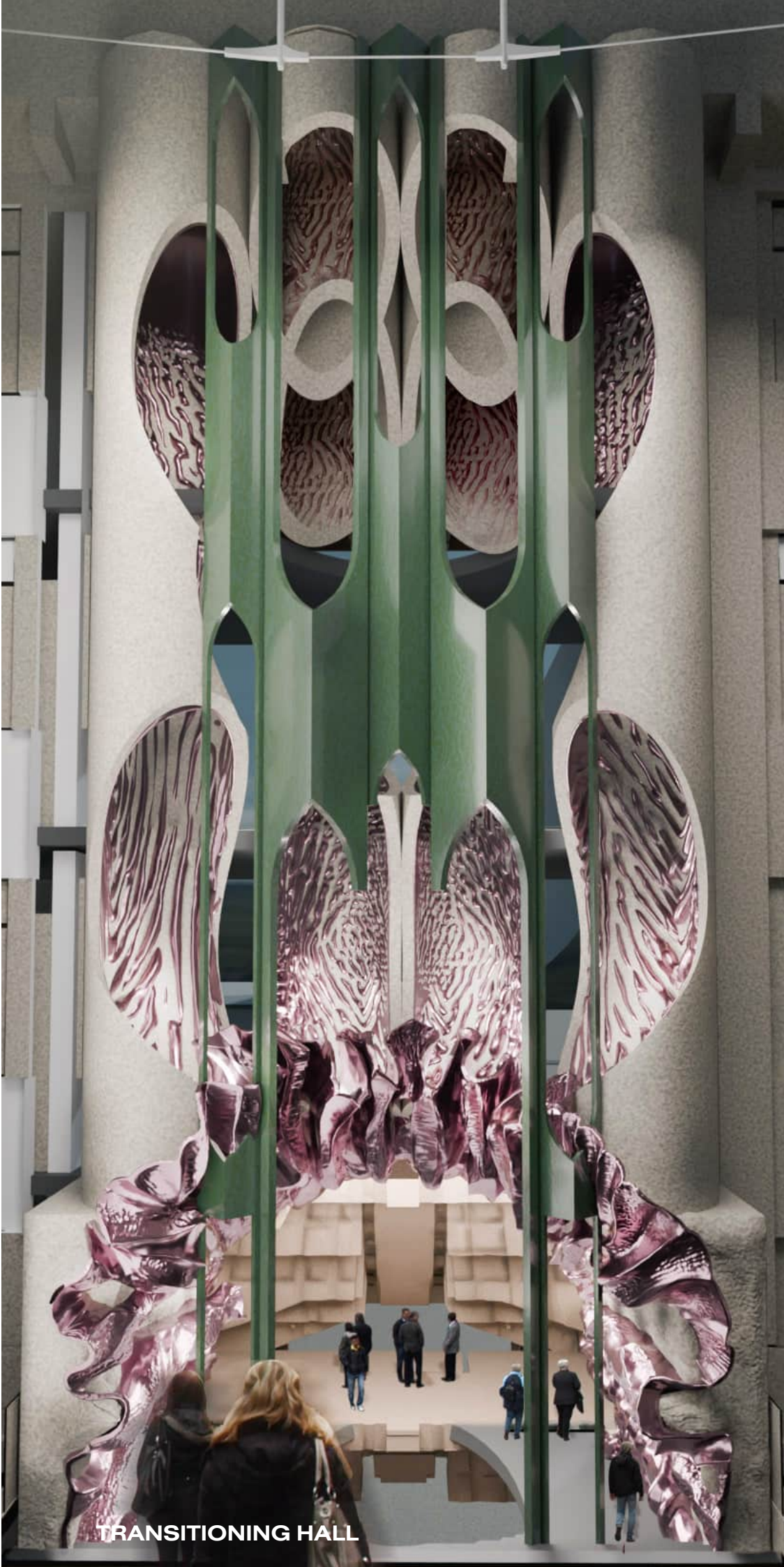
The overall area is divided into three zones: the staircase, corridor, and the exploded transition hall.



Section A



Section B



Material behavior

Transitioning view from small, medium to large gesture.



Edge condition
Folding & interlocking metal

Style transfer
Interwoven materials



Highlight edge

Folding edge, Craggy texture

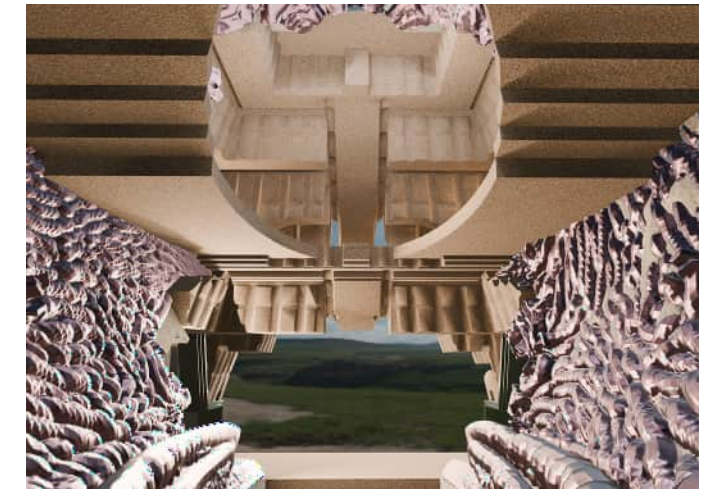


Estranged object 2

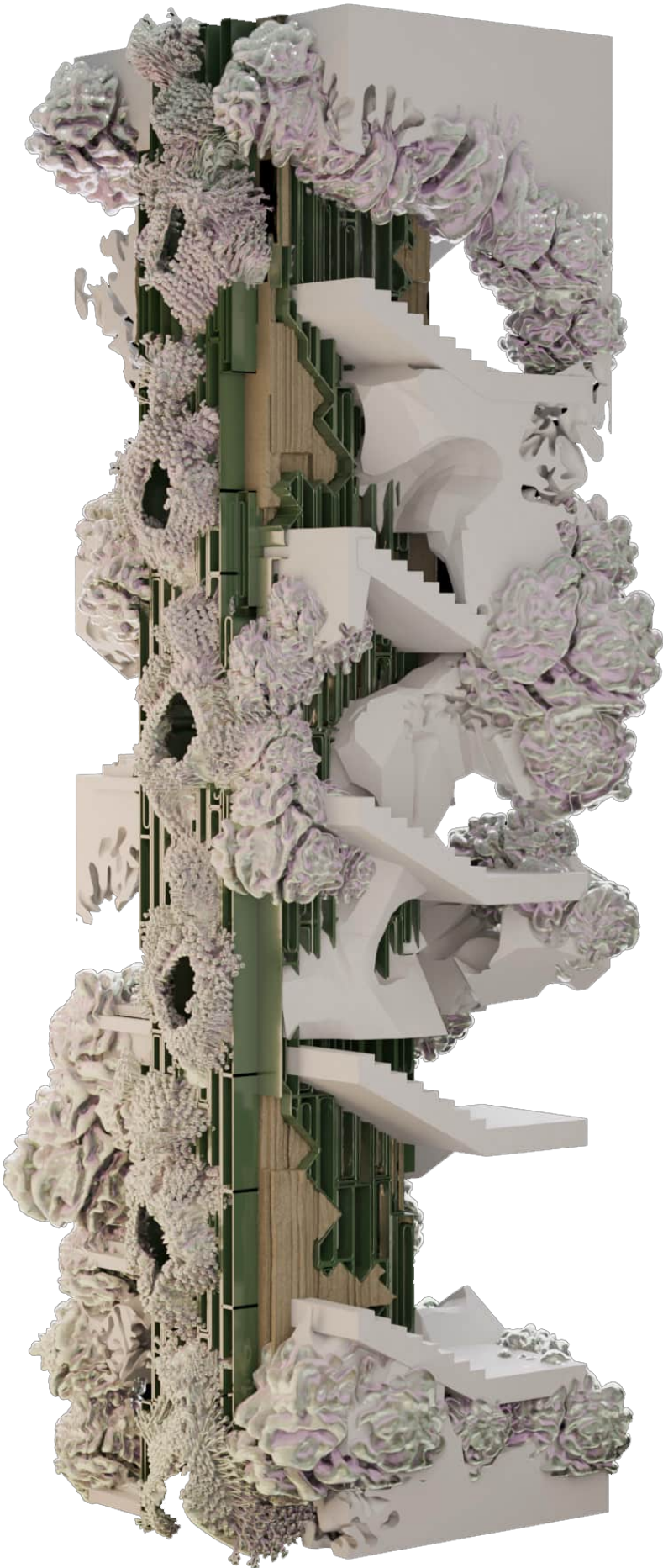
This estranged space represents the core concept of the project — transforming monolithic forms into estranged spaces. It demonstrates how a variety of material transitions, color contrasts, and tectonic shifts can create and define space, connecting both vertical and horizontal dimensions.

Entrance

Style transfer : Monolithic to estrange space



Estranged object 3



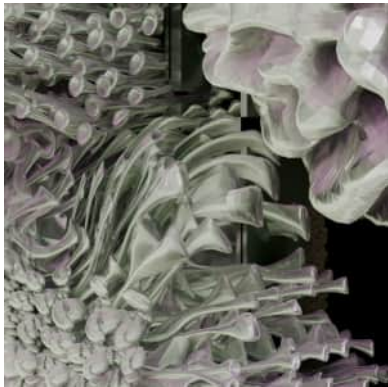
Edge condition
Highlight edge
Grow (Medium-Large)



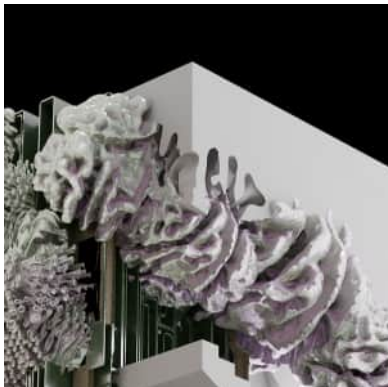
Joint
Shrink (Medium-Small)



Style transfer
Spiral gesture

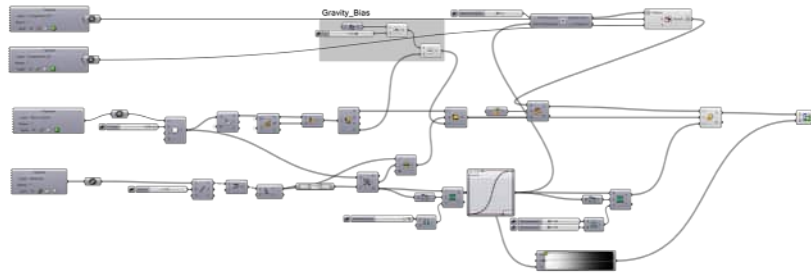


Carving space
Aggrigation & Subtraction
Estrange blobby



Interlocking
Metal- Sandstone
Stair core

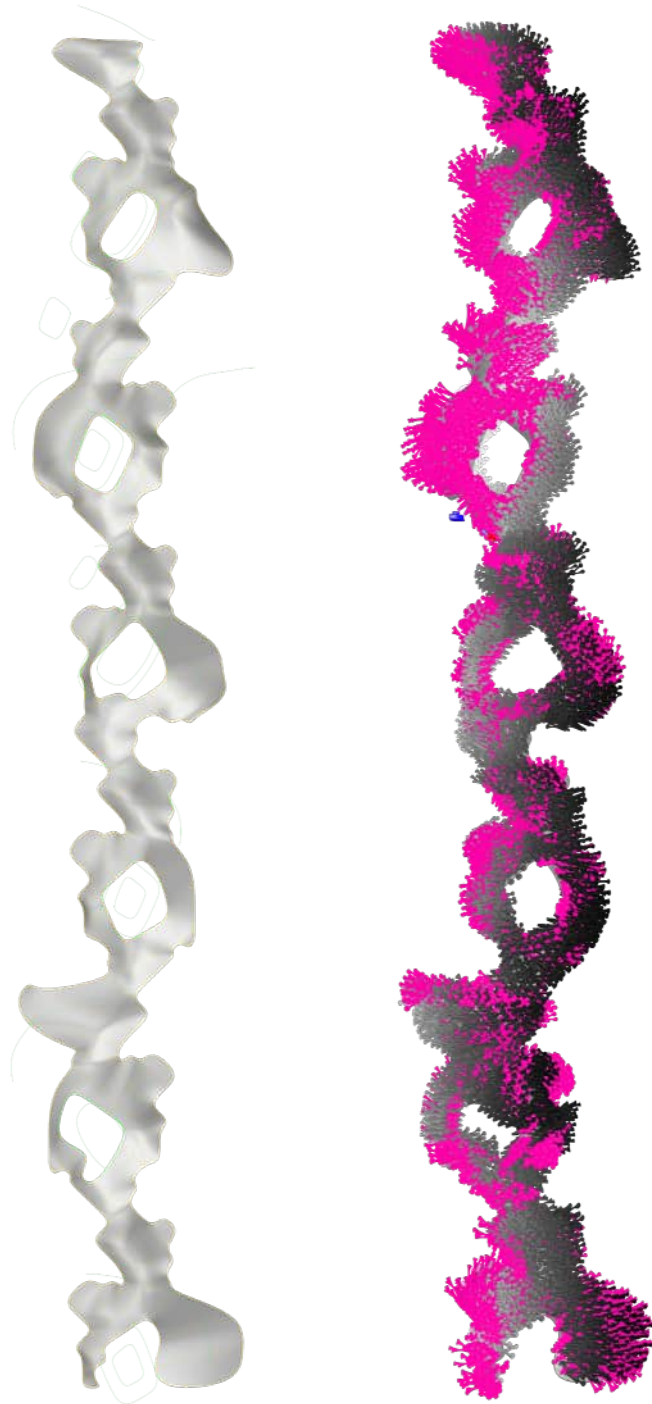
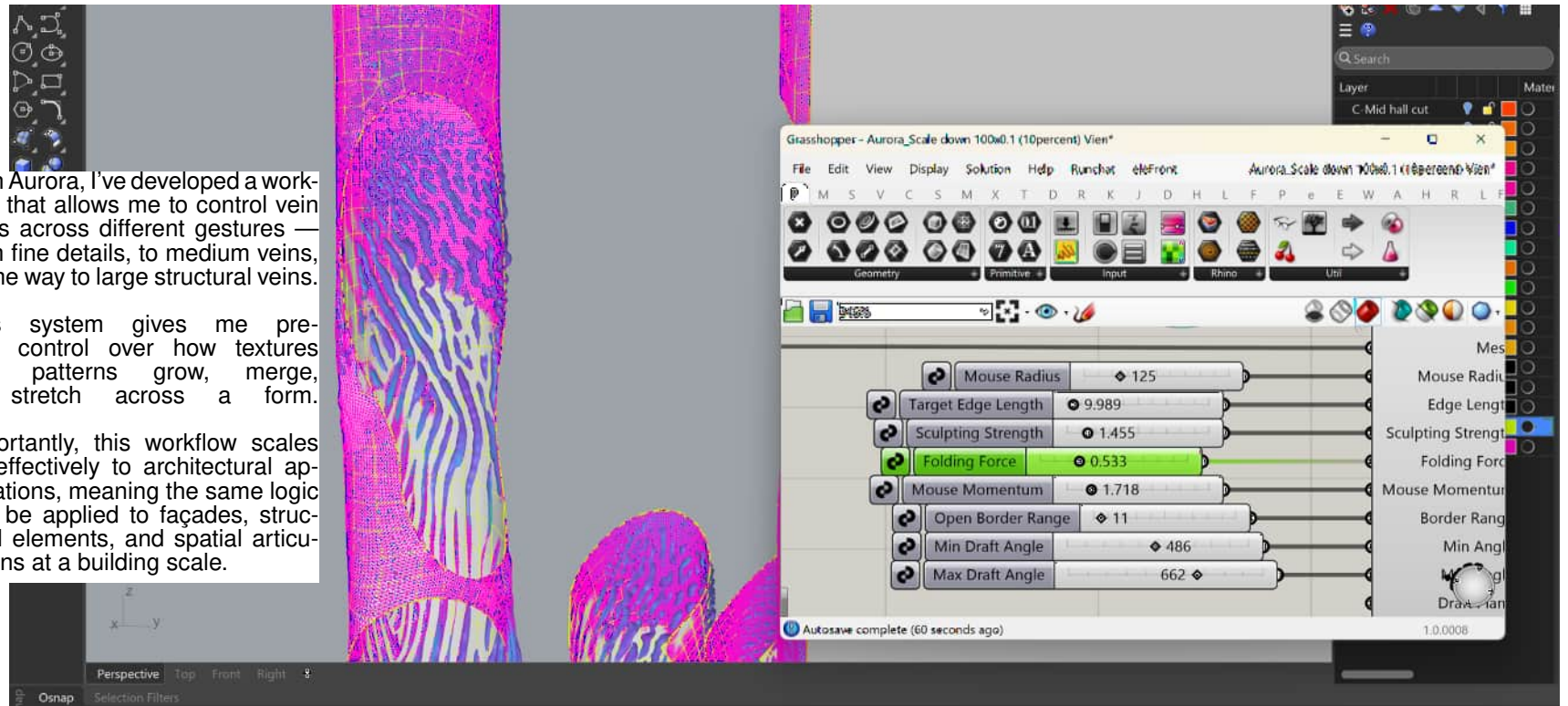




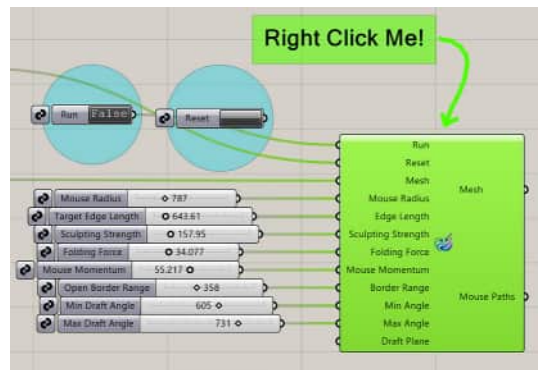
With Aurora, I've developed a workflow that allows me to control vein sizes across different gestures — from fine details, to medium veins, all the way to large structural veins.

This system gives me precise control over how textures and patterns grow, merge, or stretch across a form.

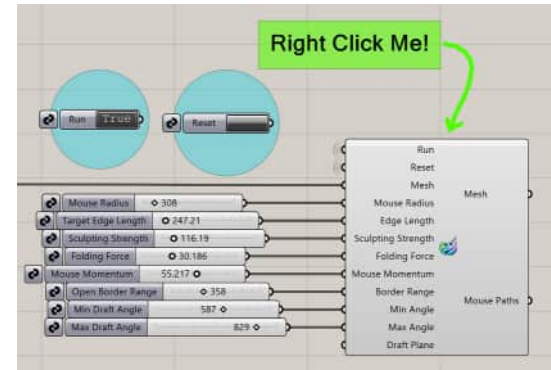
Importantly, this workflow scales up effectively to architectural applications, meaning the same logic can be applied to façades, structural elements, and spatial articulations at a building scale.



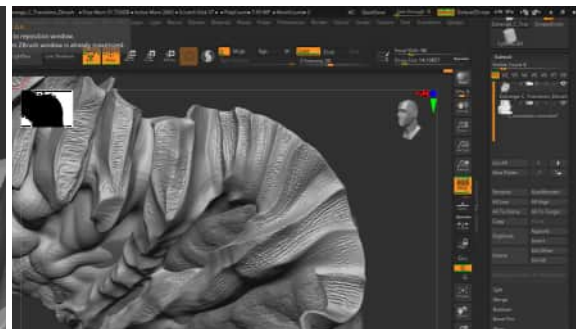
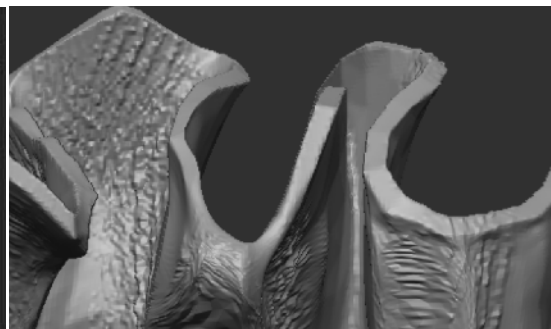
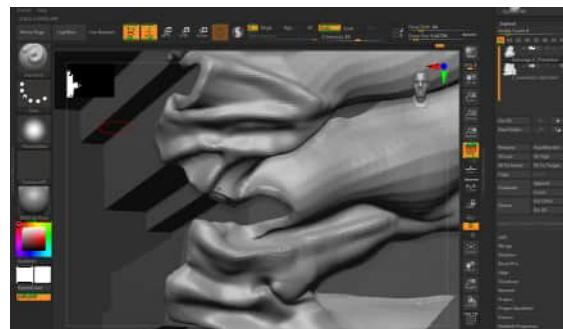
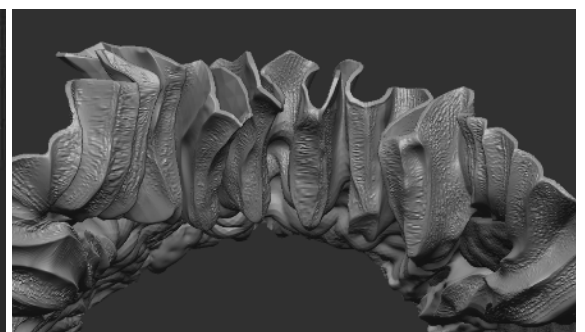
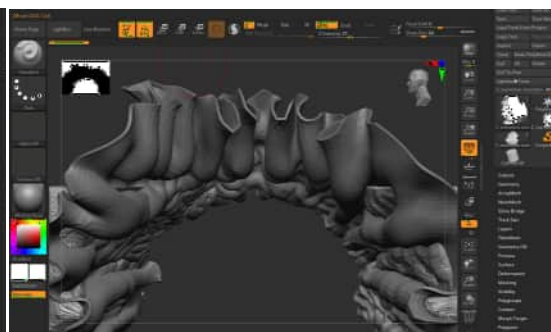
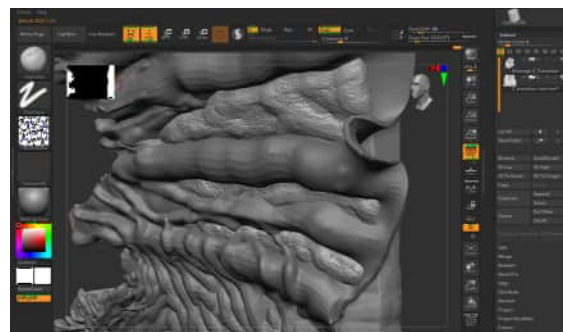
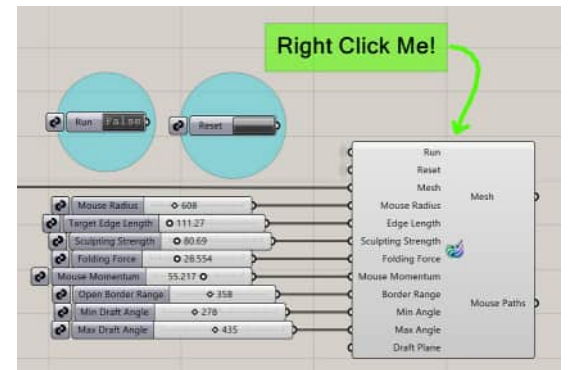
Large vein



Medium vein



Detail



CHINA TRAVELLING STUDIO

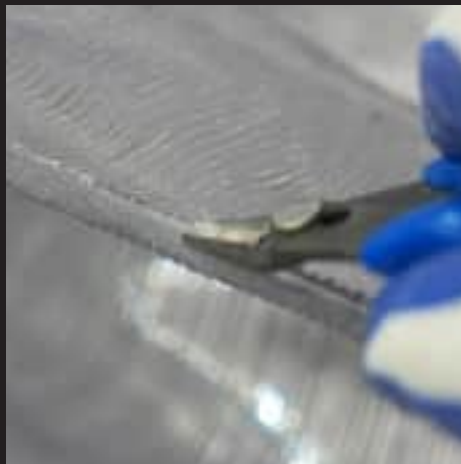


POLYMER FABRICATION

Following the cutting process, the polymer components exhibit irregular geometries, rough surface textures, and numerous discarded residual fragments.

The primary consideration during this refinement is the inherent fragility of the polymer components after segmentation. Due to their reduced size, these pieces are highly susceptible to breakage or damage during the rigorous sanding process. Consequently, the integration of adequate physical support is a crucial prerequisite for maintaining structural integrity and preventing catastrophic failure during post-processing.

As Sanding Assistant, reporting to Rose Yoon Kang, Head of the Polymer Printing Team, I supervised and guided other team members through the sanding process. My primary responsibility was to ensure the surface finish was smooth, requiring delicate work to achieve this without damaging or breaking the core polymer structure.



Observations

Unsmooth surface



Material propertie

Hardness
Heaviness
Brittleness



FOLIO SUBMISSION

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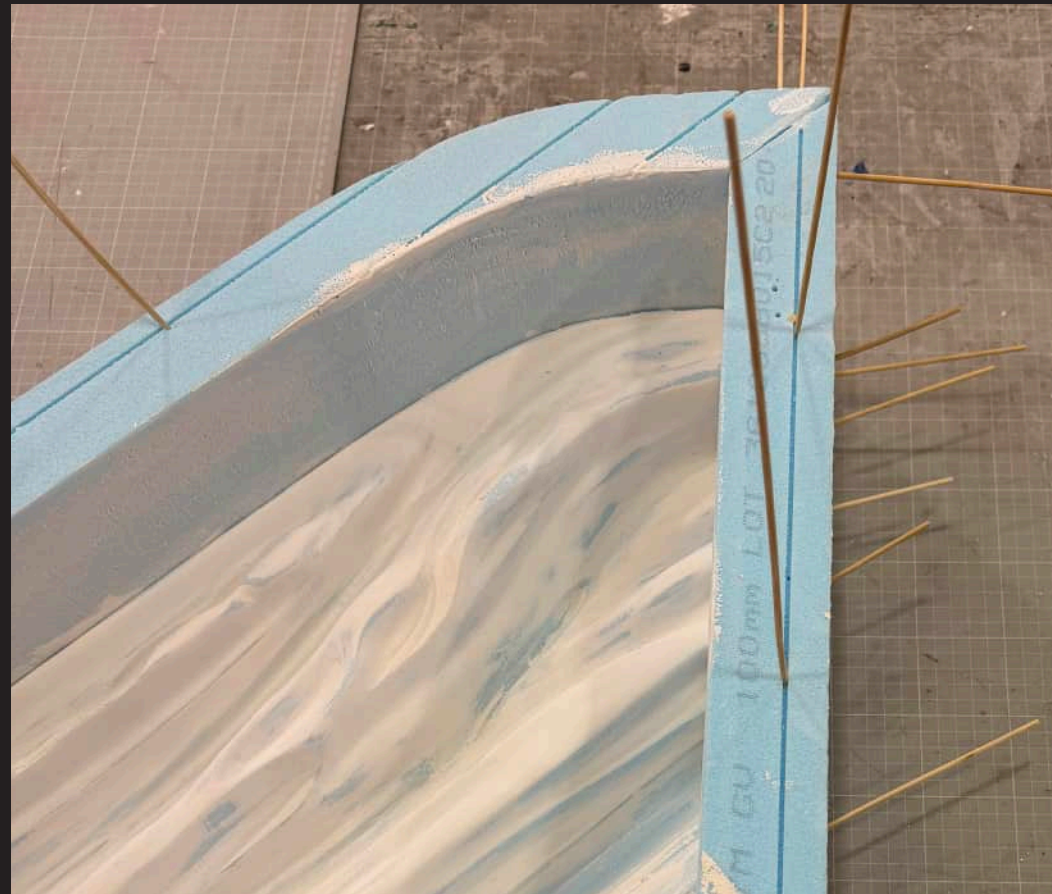
ELECTIVE BASE RESEARCH ASSISTANT

Tectonic formation lab

Marc Gibson

Research team : Charles Boman | Lachlan Buckley
Fabrication team : Tiffany Vuong | Ansel Lin | Kory Ko | Takdanai Phakdee





ASSEMBLY PARTS

PVA GLUE & BAMBOO STICKS

With individual CNC-milled foam components produced, the mould structure was assembled using PVA adhesive at part interfaces, with bamboo sticks inserted as mechanical reinforcement across joints — preventing relative movement during casting and maintaining dimensional accuracy across the assembled mould profile.

Temporary plastic bubble padding was positioned beneath fragile mould sections to prevent deflection or damage during handling and subsequent processing steps.

The assembled mould was then inspected for surface gaps and joint discontinuities, which were addressed in the following plaster patching phase.



PLASTER PATCHING

Following mould assembly, all joint lines, surface voids, and CNC artefacts were filled using Uni-Pro Multi-Purpose Filler. This stage was critical in producing a continuous, seamless mould surface — any discontinuity left unresolved would transfer directly to the cast GFRC panel face.

Plaster was applied by hand and worked into the mould geometry using fingertips and spatulas, carefully following the surface contour. Once set, patched areas were sanded flush to the surrounding foam surface, eliminating any raised edges or surface steps.

This careful, iterative patching process ensured the mould surface was free of joints and defects before resin sealing.



CURING PROCESS

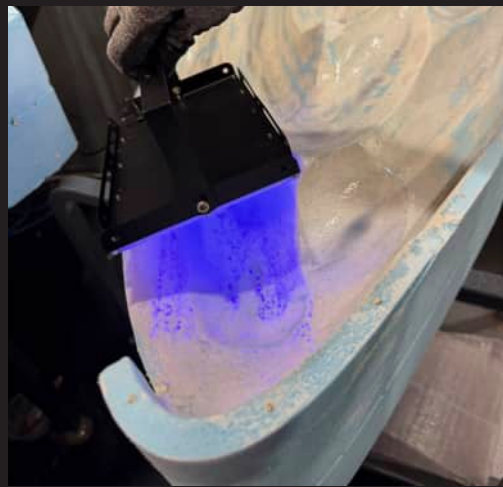
SUNLIGHT

Resin-coated mould sections were transported outside and positioned in direct sunlight for initial cure. Moulds were oriented to maximise UV exposure, with periodic rotation to ensure even cure across all surfaces. Sunlight curing was deployed for full mould sections where direct lamp access was limited by scale.

UV LIGHT

Targeted UV lamp arrays were deployed for controlled, high-intensity curing of individual mould sections and areas requiring additional cure depth. Handheld UV lamps allowed precision application to recessed and complex surface areas unreachable by sunlight alone.

The combination of sunlight and UV lamp curing ensured complete through-cure of the resin layer, producing a hard, stable mould surface ready for the GFRC casting phase.

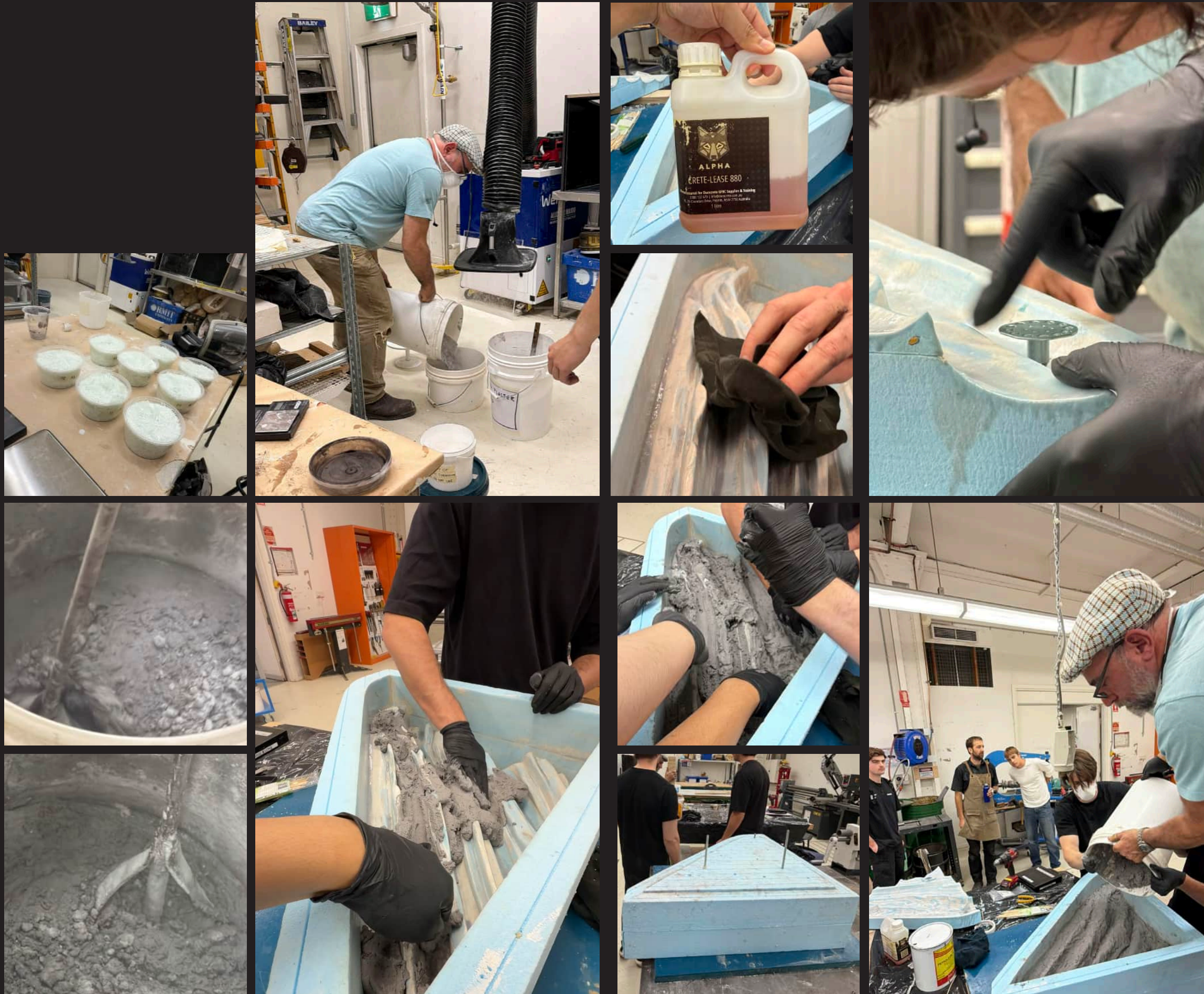


CASTING PHASE

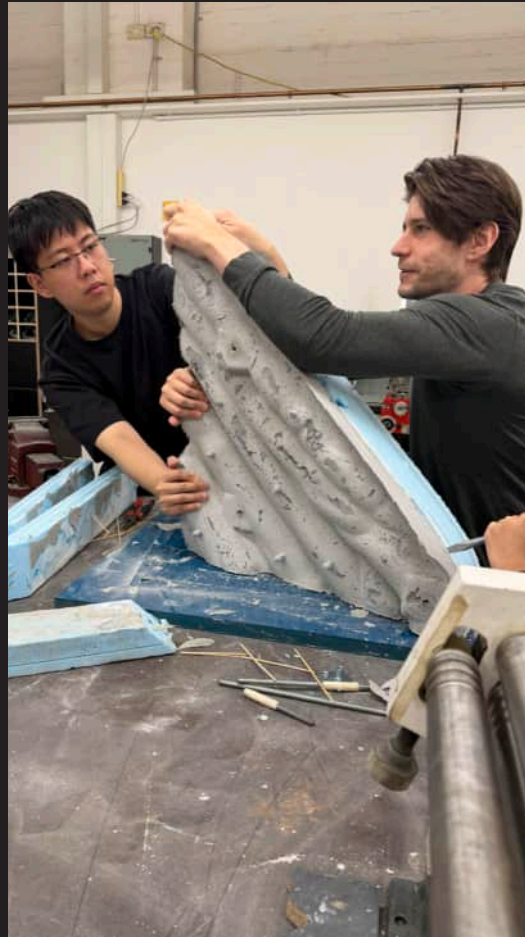
With all mould sections fully prepared, sealed, and inspected, the GFRC casting phase commenced. Alpha Ete-Lease 800 mould release agent was applied to all internal mould surfaces to ensure clean panel separation at demoulding.

The GFRC mix — comprising Portland cement premix, water, iron oxide pigment, and chopped glass fibre strands — was prepared and hand-packed into the prepared moulds. The glass fibres, visible as a loose bundle prior to mixing, provided critical tensile reinforcement for the thin-shell panel geometry.

Casting required coordinated team work to ensure consistent mix application, adequate compaction into the complex surface recesses, and uniform panel thickness throughout. Test discs were cast simultaneously to evaluate mix consistency and surface quality across each batch. Marc Gibson supervised the mixing and casting stages, ensuring correct ratios and technique throughout.



DEMOULDING

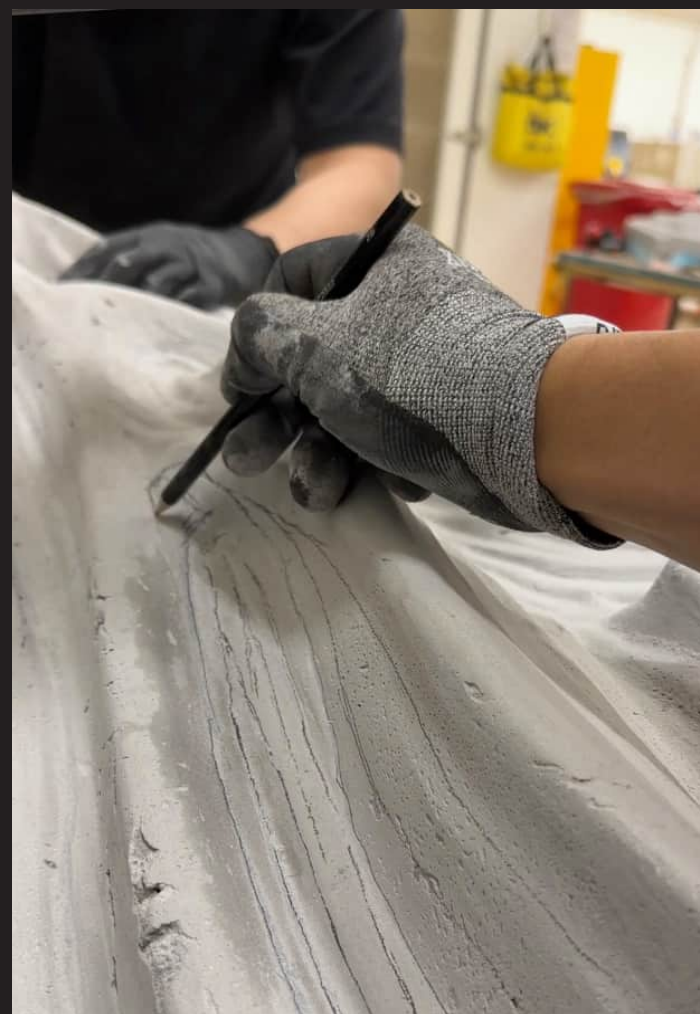


Following cure, cast GFRP panels were carefully separated from their foam mould negatives. Demoulding required progressive manual prying, controlled peeling, and lateral extraction — working from the mould edges inward to release each panel without fracturing the cast face.

The moment of demoulding revealed the direct material translation of the Aurora-generated digital surface into physical GFRP: the continuous folding, draping, and layered surface character of the computational model expressed in cast concrete. The aggregate texture and grey colouration of the GFRP surface produced an immediate material presence consistent with the monolithic language of the facade concept.

Close inspection of the demoulded panels confirmed the quality of the mould preparation — the flowing surface detail was faithfully reproduced, with only minor post-production refinement required.

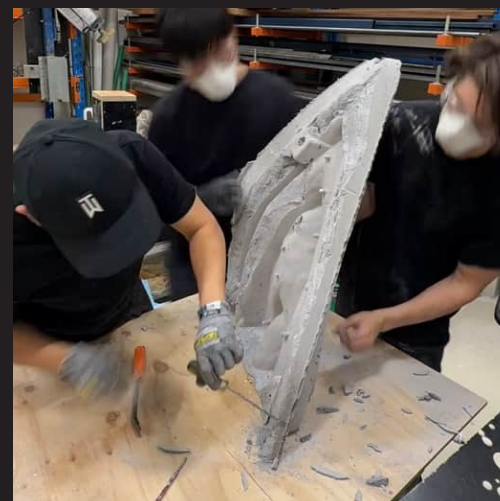




FINAL PROTOTYPE

This stage marked the convergence of digital design development and physical fabrication — the first full-scale realisation of the TFL hybrid tectonic facade prototype

At this stage, the large-format 3D-printed polymer structure — produced by the parallel polymer printing team — was brought into direct juxtaposition with the GFRC panels. The visual contrast between the smooth, draped GFRC surface and the ribbed, directional polymer structure gave physical expression to the core tectonic dialogue of the project.





FINAL INSTALLATION

The final installation brought all fabricated components together at full assembly scale. Two key technical challenges emerged:

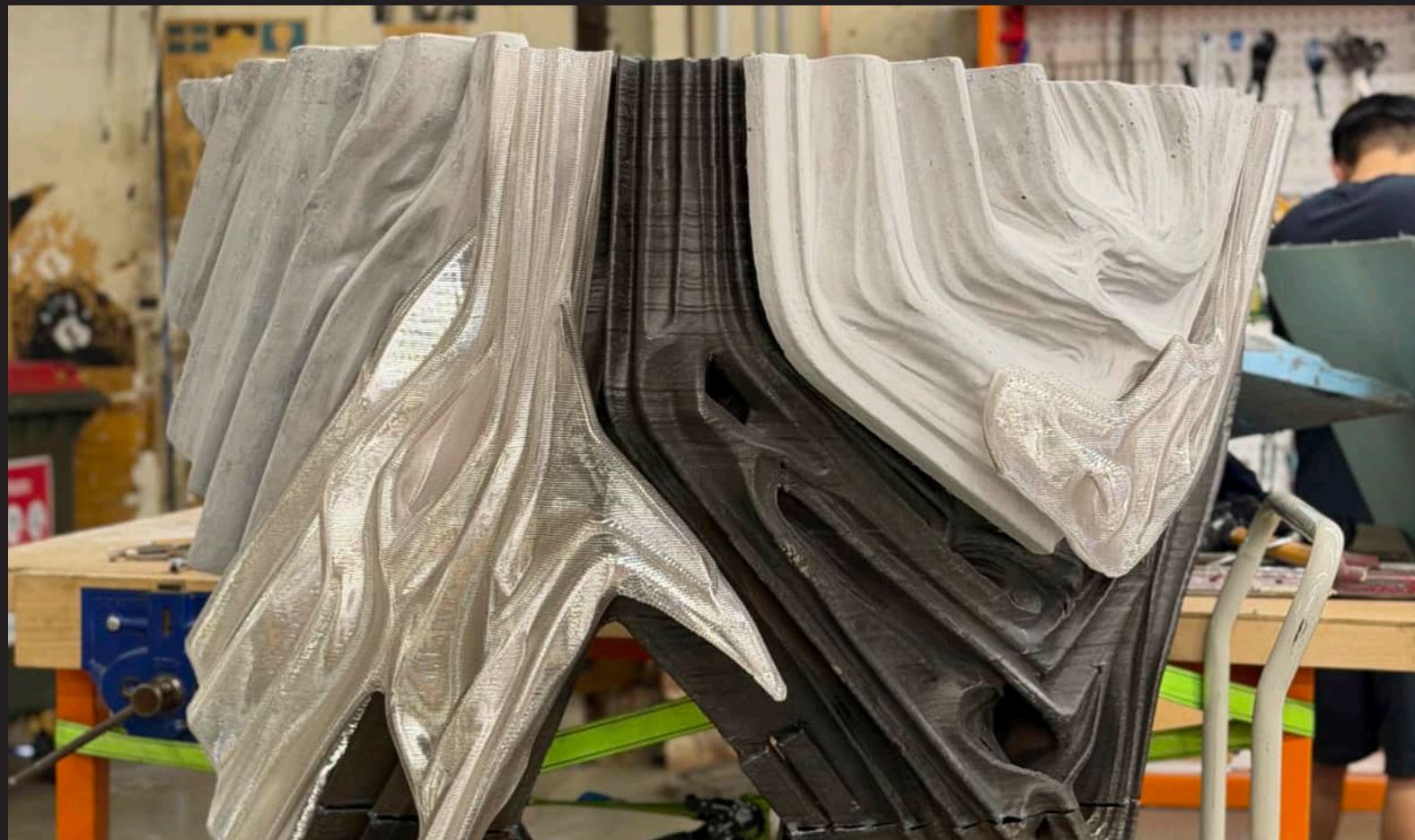
Unsmooth surface: Areas of joint mismatch at panel interfaces were addressed through in-situ patching and grinding to maintain surface continuity across the assembled facade.

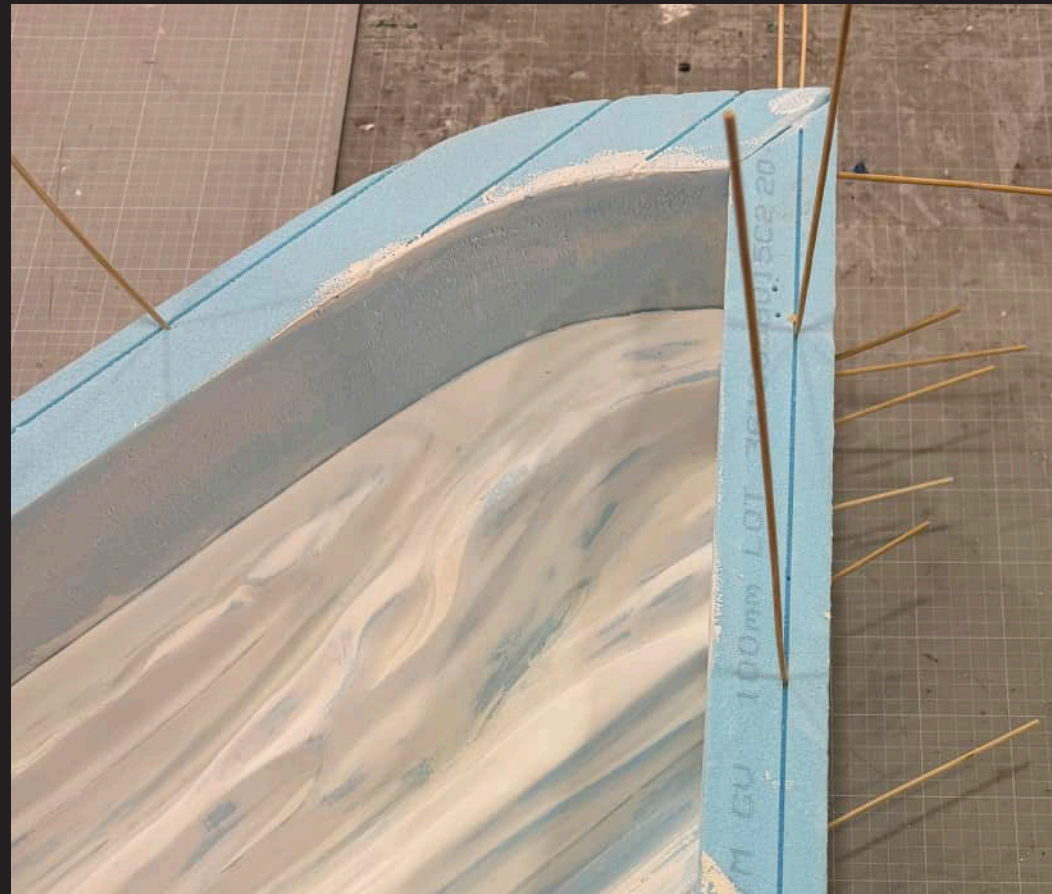
Joint

The structural connection between GFRC panels and the polymer frame was resolved through stainless steel bolt-and-bracket hardware. The joint zone — where the smooth monolithic GFRC surface meets the ribbed polymer structure — became a legible tectonic moment in the final assembly, expressing the hybrid character of the facade system.

Joint

The structural connection between GFRC panels and the polymer frame was resolved through stainless steel bolt-and-bracket hardware. The joint zone — where the smooth monolithic





ASSEMBLY PARTS

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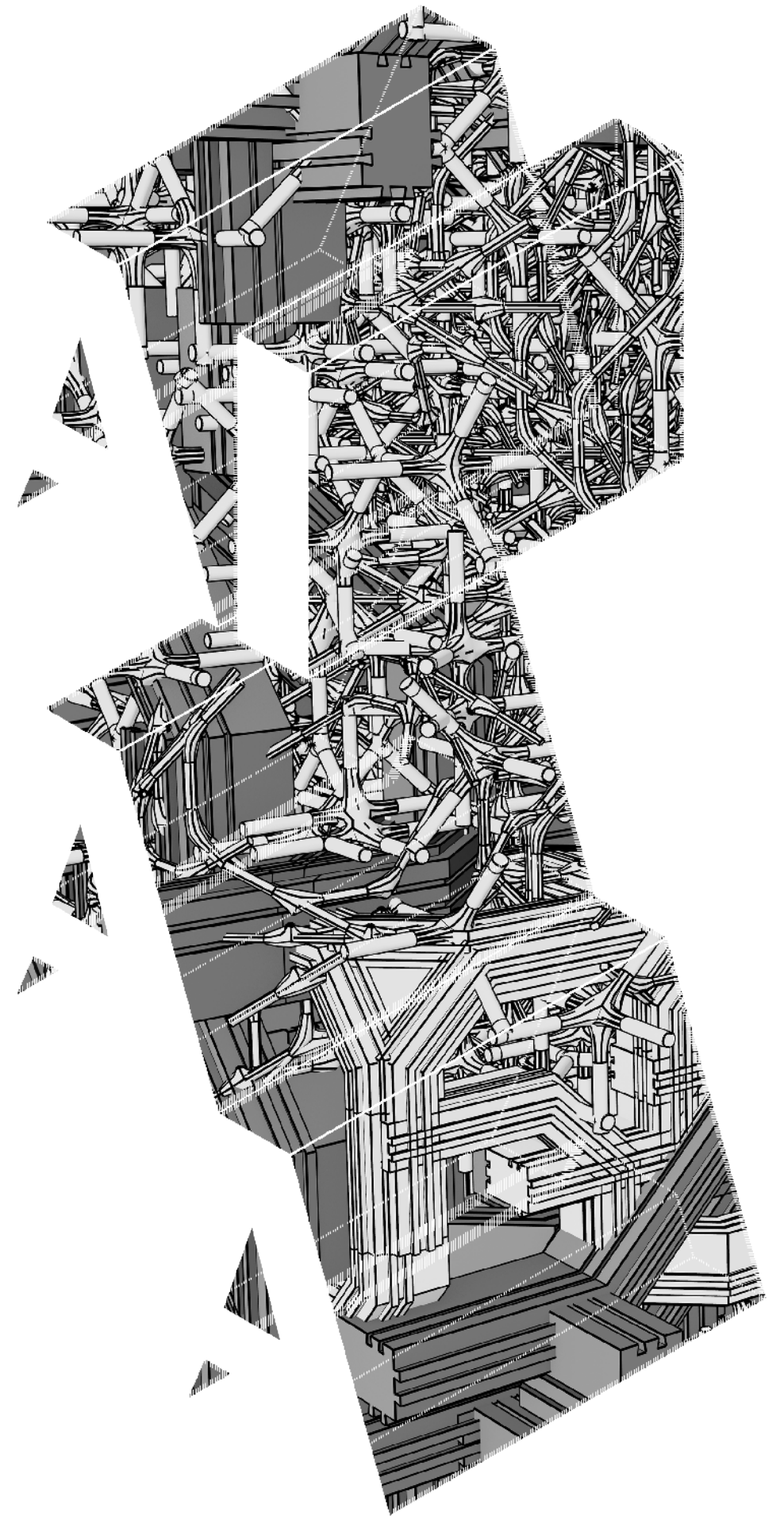
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FOLIO SUBMISSION

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AI ACCELERATED ARCHITECTS

Alisa Andrasek



This studio engages architecture as a form of hybrid cognition, where human designers collaborate directly with AI agents to construct generative tectonic systems. Through agent-assisted modelling and conversational (“vibe-coded”) workflows, students operate at the level of behavioural logic – designing the rules and interactions from which architectural intelligence emerges.

Working on a strategic development site within the RMIT City Campus – one of the university’s designated grounds for its next generation Learning–Teaching–Research Hub – students deploy their generative tectonic systems at full architectural scale, transforming an active institutional frontier into a live testbed for cognitive architecture.

Alisa Andrasek

ARCHITECT IN THE CRACK

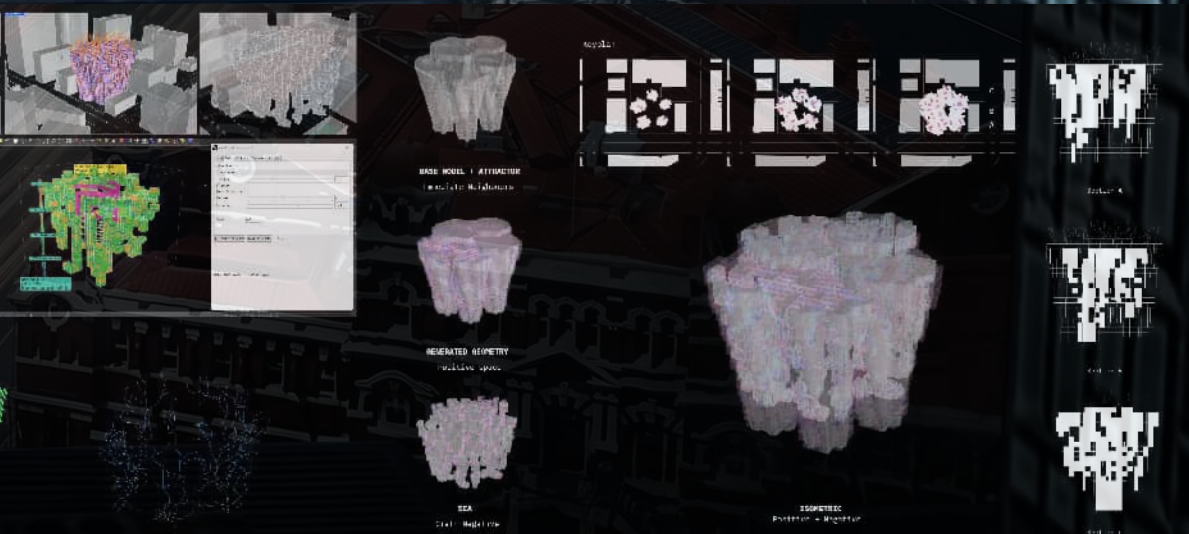
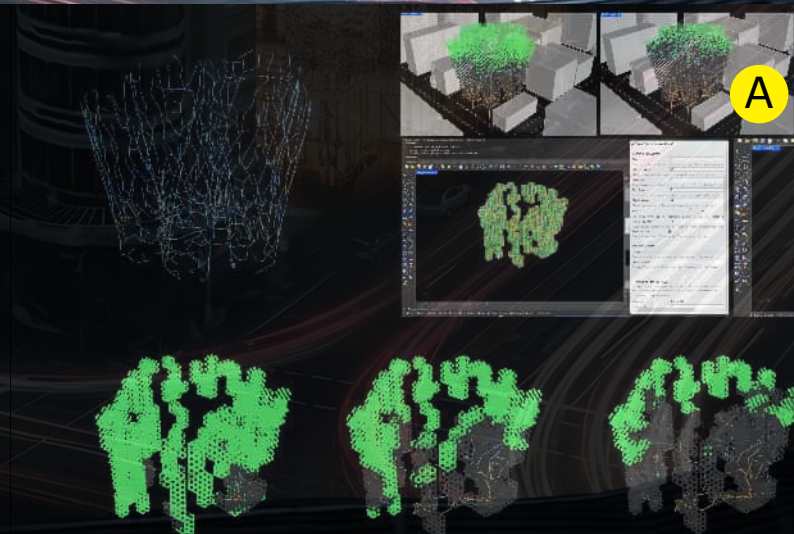
This discrete timber project investigates how a research building for machine learning can simultaneously provide a comfortable learning environment and an isolated refuge from Melbourne's extreme urban climate — through architecture grown by multi-agents across a pipeline of five computational scripts, not designed conventionally.

A building is not shaped by its walls — it is shaped by where light breaks through them.”
“What if the crack in the building is not a failure — it is the architecture?”

● Script stages
● Room locations

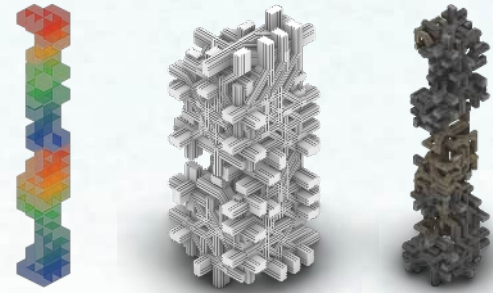
CLIMATE CONTEXT

Site Melbourne presents an extreme climate: hot summers and cold winters. The high-rise shadow zone blocks daylight during the day. The central design challenge: *how can a building provide shade while still achieving suitable indirect light and passive ventilation?*



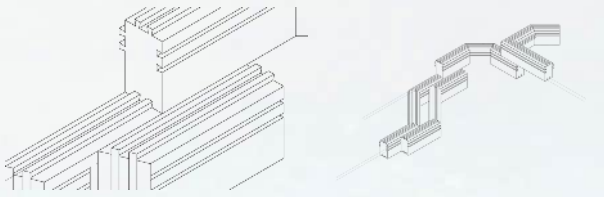
Discrete Element Study

The discrete element investigation focuses on timber and metal, studying three joinery conditions: interlocking, tongue and groove, and metal plate and spike connectors. Parts were manually assembled into chunks to understand how components fit together under different tectonic conditions — floor, wall, column, and facade. This manual process established the assembly logic before scripting. Hierarchy: Parts → Chunks → Architecture. Three output modes: Manual assembly / Chunks placed on field / Procedurally generated on the field.”

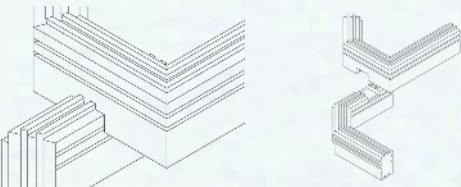


//COLUMN CHUNK

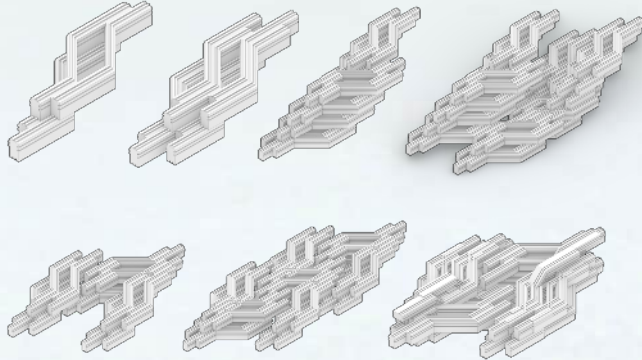
Timber tectonic | Groove and tongue



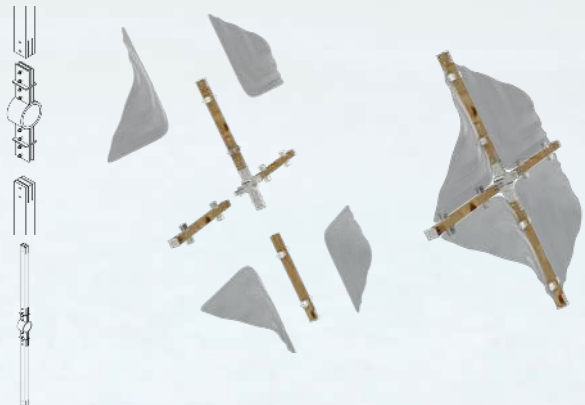
Timber tectonic | Interlocking



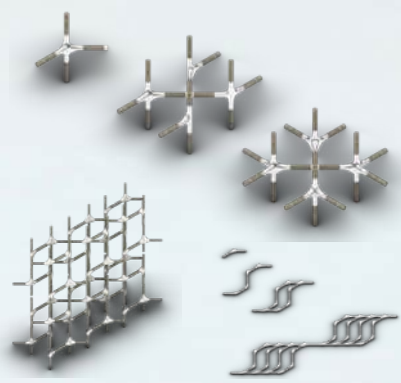
Chunks iterations (outputs)



Metal plate and spike connector



Chunks iterations (outputs)



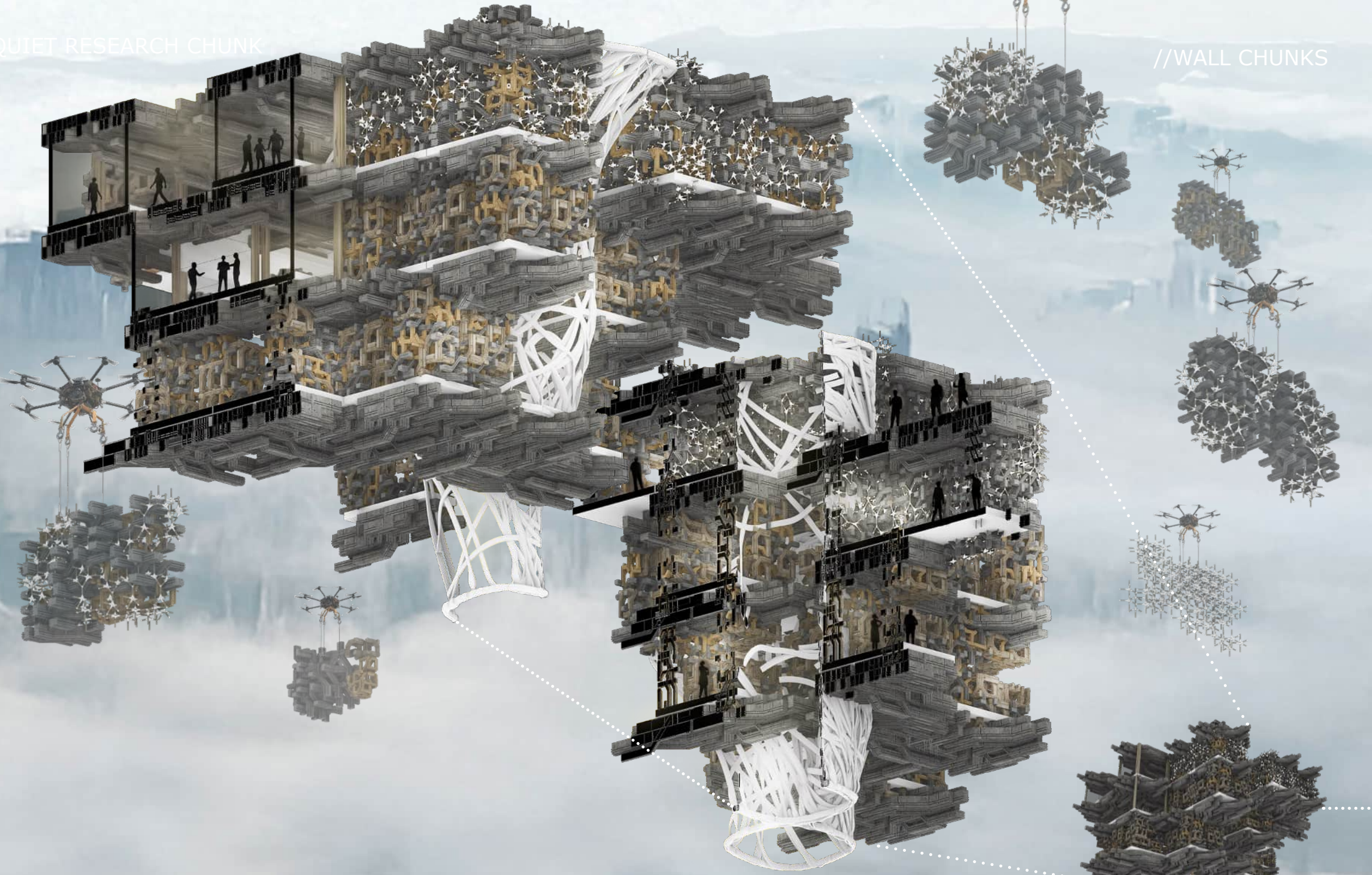
//QUIET RESEARCH CHUNK

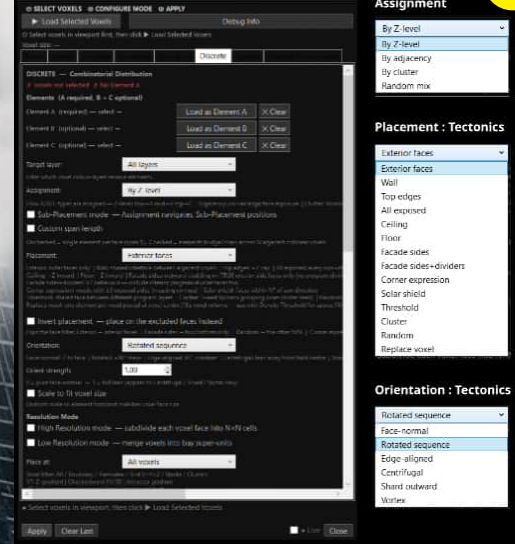
//WALL CHUNKS

DISTRIBUTION DISCRET ELEMENT

Distribution / Aggregation

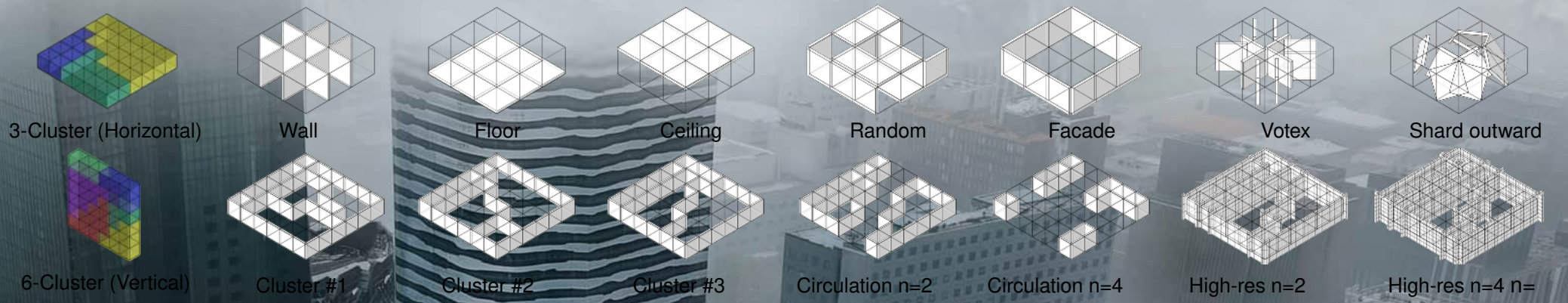
The distribution agent reads the voxel field as its base and populates discrete element chunks across it, responding to attractor values and programmatic zones. Through the combination of Voronoi clustering and K-means, alongside multi-agent room logic, the script drives variation in cluster type, room resolution, and tectonic placement across the building.



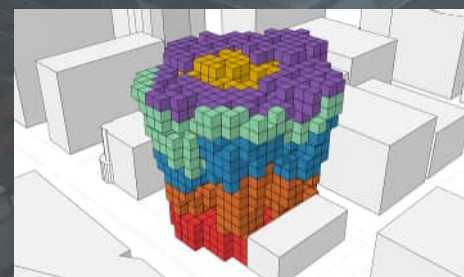


Distribution | Tectonic logics

Distribution base : Voxel field



E



PROGRAM CLASIFIER SCRIPT

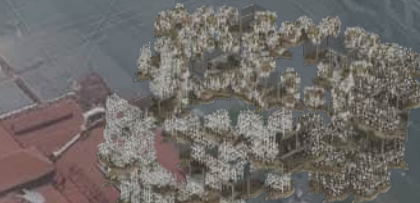
How It Works: Tags voxel clusters into one of seven program types (Robotics Lab, Digital Fab, Makerspace, Seminar, Social Breakout, Quiet Research, Learning Refuge) based on cluster size/shape and multi-agent Voronoi + K-means clustering. Agents: Voxel clusters (connected components) Behaviour: Evaluate clustering rules → assign program label

Together with the Program Classifier (E) and Distribution Discrete Elements (D) scripts, each function was assigned different logics — including assignment, tectonic placement and orientation, and Voronoi cluster logic. We can then configure a variety of architectural outcomes to achieve the desired effect. For instance: wall thickness responding to the climate attractor, variable wall resolution (Low-High), volume height, circulation separation, or open floor plan.

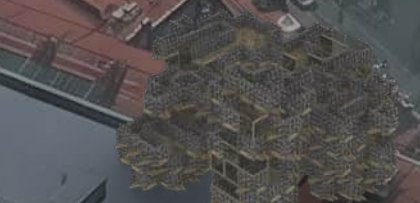
ROBOTIC LAB



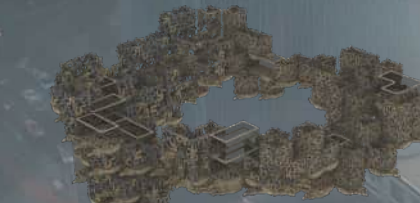
SEMINAR



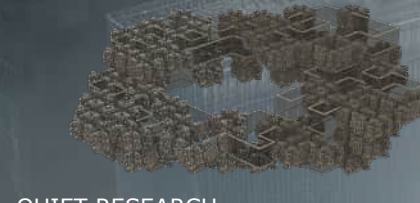
DIGITAL LAB



QUIET RESEARCH



CLASSROOM

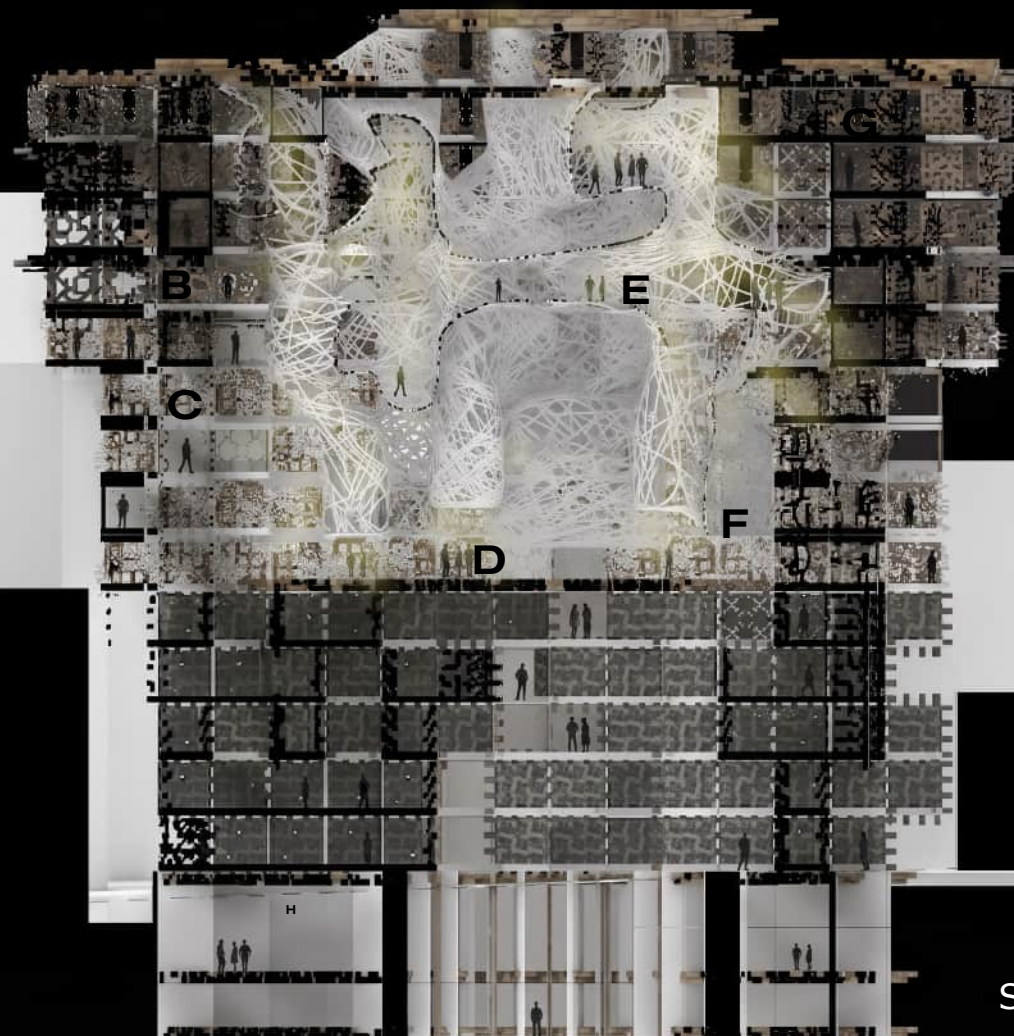


- QUIET RESEARCH
- COMPLEX STUDY ROOM
- 1 ARCHIVE STUDY ROOM
- 2 PRIVATE STUDY CARREL
- 3 RARE BOOK COLLECTION
- 4 DIGITAL ARCHIVING LAB
- 5 REFERENCE ALCOVE
- 6 FOCUS CARREL
- 7 SMALL SEMINAR ROOM
- 8 LIBRARIAN OFFICE
- 9 EQUIPMENT STORAGE
- 10 DATA SERVER ROOM



//SOCIAL ATRIUM_F

GENERATIVE
TIMBER & METAL
PARTS



SECTION



//SOCIAL ATRIUM_D



//SEMINAR_C



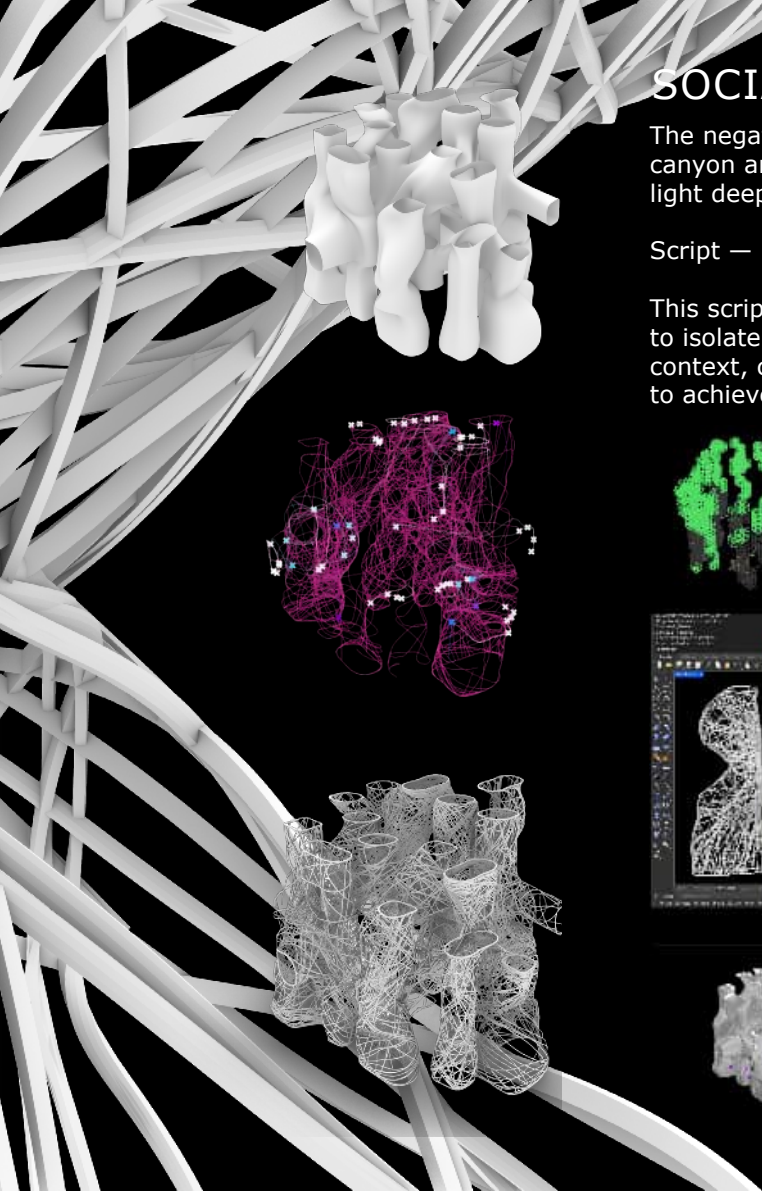
//SOCIAL HIDEOUT_E

//PRIVATE STUDY_B

//ROBOTIC LAB_H

//CLASSROOM_G



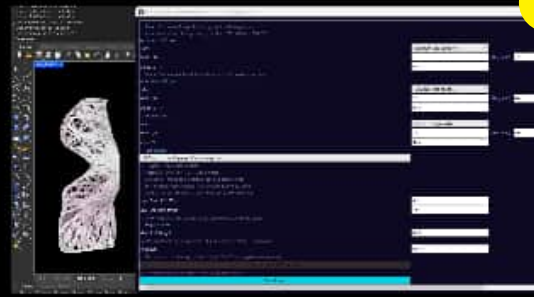
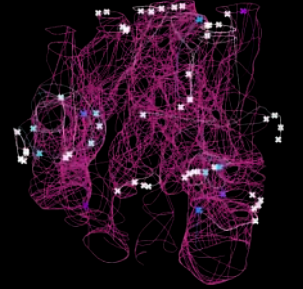


SOCIAL HIDEOUT

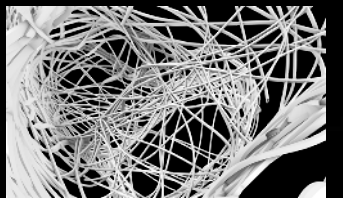
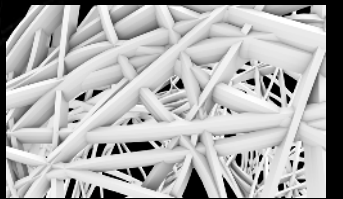
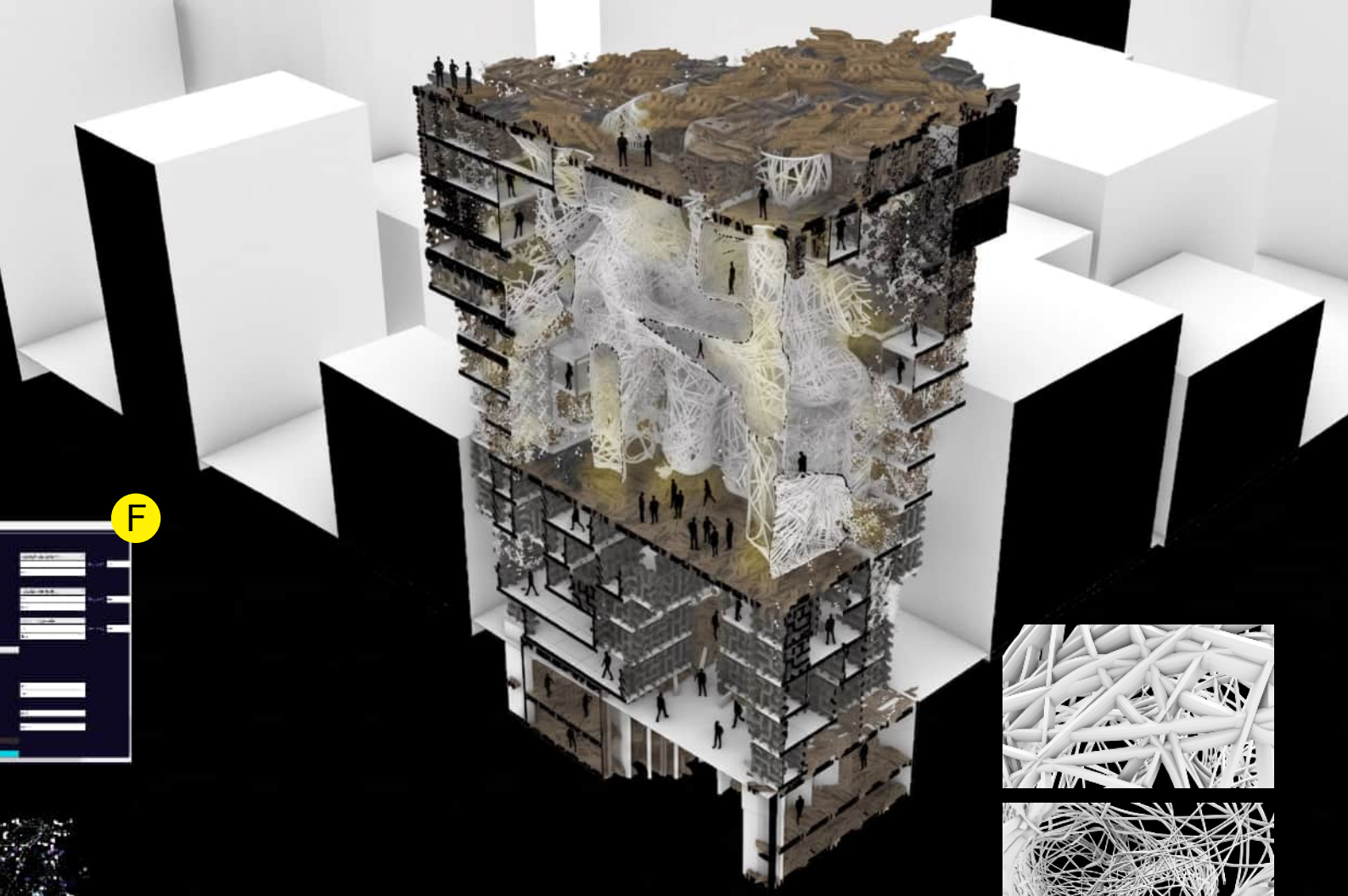
The negative space were carve a rift through the mass, mimicking canyon and cliff formations found in nature, channelling indirect light deep into the whole building.

Script — Flocking Boids

This script generates a porous structure across the mass in order to isolate the noisy environment of the Robotic Lab from the urban context, creating filtered light and noise absorption. The purpose is to achieve the most comfortable learning environment.

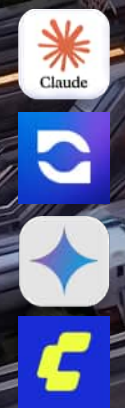


F



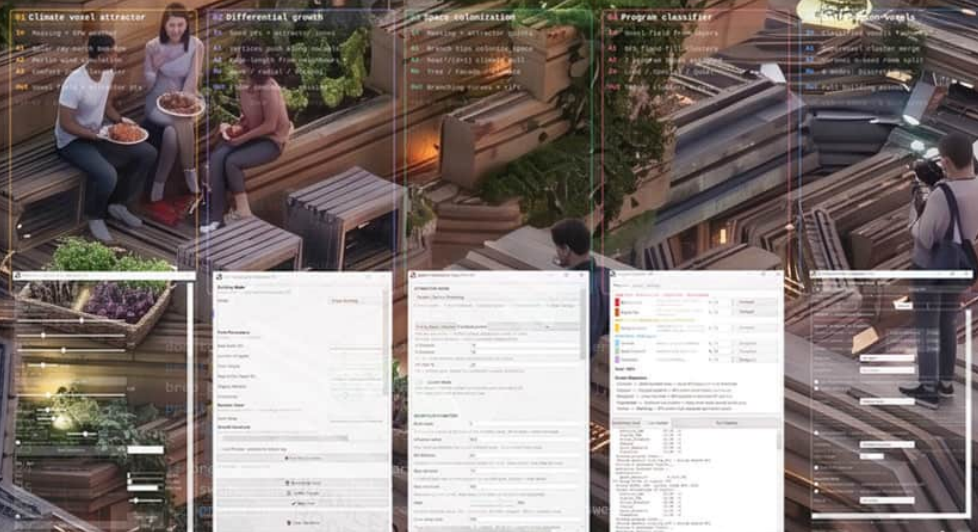
LLM | VIBE CODING WORKFLOW

Multi-agents | MCP | Markdown | Feedback loop | Rhino Python



Text prompts, image references, and MCP tools feed into an LLM agent, generating successive script iterations. Failed outputs return as screenshot feedback, closing the loop.

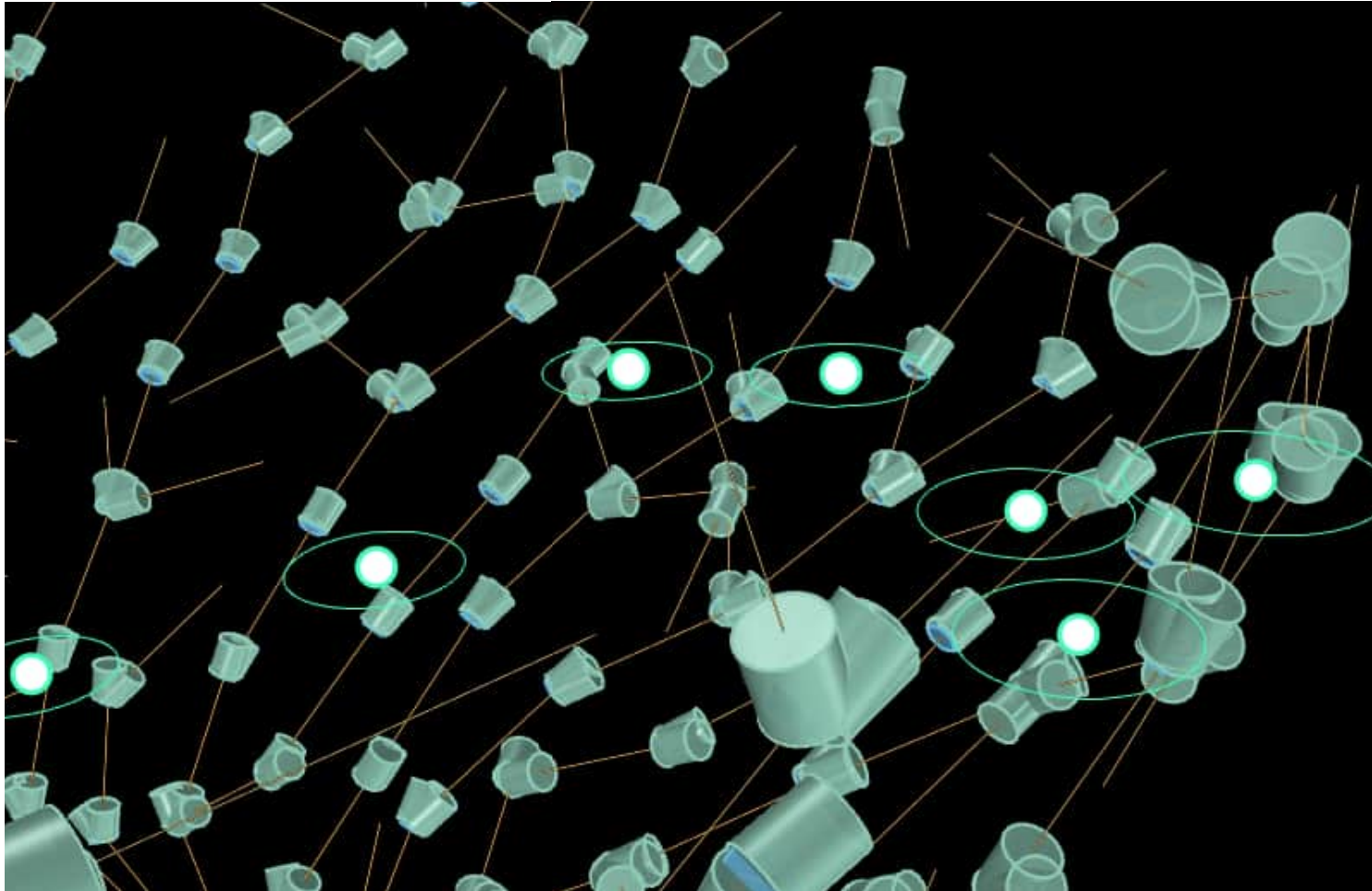
Script pipeline - 5 scripts generating architecture



// ROOFTOP_A

Recreation | Green space

Coding development



This section presents the full technical development of the five Python scripts that constitute the project's computational pipeline — from the AI training loop methodology that produced them, through the individual script architectures, to the final rendered outputs of the assembled building.

Every script in this project was developed through an LLM-assisted agentic workflow. Text prompts, image prompts, reference scripts in Python and Markdown, Grasshopper script references, and MCP tool integrations were fed to an LLM agent — Claude, Gemini, and Runchat — which generated successive script versions. When a version failed in Rhino, a screenshot of the error or output was returned to the LLM as feedback, triggering the next iteration. When a version succeeded, it closed the loop and became the input for the following script. The AI Training Loop diagram in this section documents this process explicitly: the feedback path from Rhino back to the LLM is as architecturally significant as any script output.

The five scripts are presented in pipeline order, each one taking the previous script's output as its input, and none of them knowing what the final form will be.

Script 01 — Climate Voxel Attractor processes Melbourne EPW weather data using solar ray-march simulation (8am–4pm), Perlin wind particle simulation, and comfort zone classification to generate a voxel attractor field. Red voxels indicate peak solar load and structural density; blue voxels indicate passive, low-density zones. The attractor points produced here seed every subsequent script.

Script 02 — Differential Growth Form Finding uses the attractor zones as seeds for an organic massing algorithm — individual vertices push outward along surface normals while resisting edge length compression, producing the wrinkled, extended upper footprint that shades the building in summer. Multiple massing iterations were scored against the climate data; this section documents the evaluation process and the selected mass.

Script 03 — Space Colonisation Algorithm carves the branching rift through the massing. Branch tips grow from root points toward unvisited attractor nodes, consuming each attractor on contact and leaving behind a network of vascular branching curves. Applied at building scale, this produces the crack — the canyon-like negative space that channels indirect light and wind through the full height of the building. Development iterations for this script are documented here in detail, including early urban-canopy SCA tests at 1000×1000m scale and the intermediate connector-agent studies.

Script 04 — Program Classifier reads the voxel field from named layers and uses BFS flood-fill to group connected voxel components into clusters, then tags each cluster with one of seven program types based on size, shape, and SCA proximity: Robotics Lab, Digital Fab, Social Breakout, Seminar, Quiet Research, Classroom, and Rooftop. The classifier's zone logic — Loud at the base, Atrium at the core, Quiet in the upper levels — is documented alongside the script's UI, showing how programmatic zoning emerges without manual floor planning.

Script 05 — Distribution Voxels Component is the most complex script in the pipeline, operating across six modes from the same voxel field input: Discrete mode places combinatorial A/B/C element geometry by Z-level, adjacency, cluster, or random assignment; Room mode generates slabs, walls, and windows using Voronoi cluster grouping; Facade mode stamps exterior faces using Voronoi cluster splits; Structure mode places column breps at vertical edges and beam breps at horizontal edges; Ornament mode applies rotation, scale, and spiral effects driven by attractor distance; and Circulation + Skywalk mode traces the adjacency path graph between floors to generate the spiral stair, with width scaled by attractor weight. The script reached V15 at 348KB — the version shown is the final production build.

The section concludes with the full assembled building output: spatial renders of each program zone from Classroom to Robotic Lab, the Social Breakout spiral, and the rooftop — all generated procedurally from the same voxel field, each space tectonic in a different way, all of them produced without a single piece of manually placed geometry.

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Script pipeline – 5 scripts generating architecture

Each output becomes next input · no script knows the final form

01 Climate voxel attractor

In Massing + EPW weather

A1 Solar ray-march 8am-4pm

A2 Perlin wind simulation

A3 Comfort zone classifier

Out Voxel field + attractor pts

ref V7 · 161KB · lat -37.8°



02 Differential growth

In Seed pts + attractor zones

A1 Vertices push along normals

A2 Edge-length from neighbours

RU Wave / radial / Voronoi

Out Floor contours → massing

ref V5 · 55KB · seed-controlled



03 Space colonization

In Massing + attractor points

A1 Branch tips colonize space

A2 heat²/(d+1) climate pull

Mo Tree / Facade / Climate

Out Branching curves + rift

ref V4 · 348KB · 5 modes



04 Program classifier

In Voxel field from layers

A1 BFS flood-fill clusters

A2 7 program types assigned

Zo Loud / Special / Quiet

Out Tagged clusters + role

ref V7 · 53KB · SCA markers



05 Distribution voxels

In Classified voxels + chunks

A1 SuperVoxel cluster merge

A2 Voronoi n-seed room split

Mo 6 modes: Discrete/Room...

Out Full building assembly

ref V15 · 348KB · 6 dist modes



SITE ANALYSIS

MASS OPERATION

VOXEL FIELD SUBDIVISION

PROGRAM SEPARATION

DISCRETE ELEMENT DISTRIBUTION

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SCRIPT DEVELOPMENT

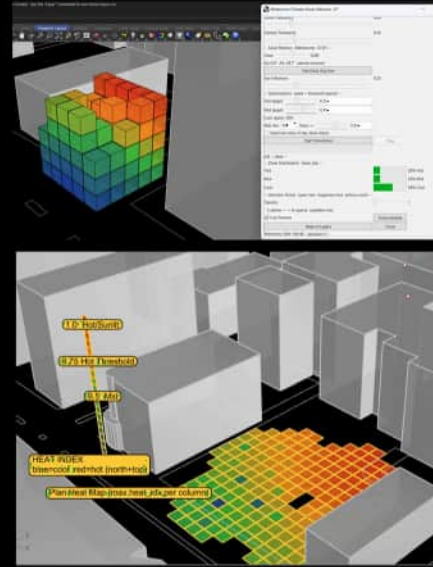
Script 01 - Climate voxel

The script is a Rhino IronPython 2 + Eto.Forms non-modal panel that fuses three independent systems:

(A) **Fern Engine** — sculpts a 3-D voxel field inside a picked site geometry. Each (x, y, z) grid cell gets a noise value from multi-octave fractal noise. Cells above the density threshold survive as voxels.

(B) **Melbourne EPW Climate Parser** — reads 8760-hour annual weather data (GHI, DNI, DHI, temperature), aggregates by month, and produces a climate_factors dict that modulates the noise amplitude, density bias, and sun-influence weighting. Solar position is computed for Melbourne lat ~37.8° using a declination/hour-angle model. Each surviving voxel receives a heat_idx (0=cool/shaded/ground → 1=hot/sun-facing/top) computed from the dot product of the sun vector with the voxel's effective face normal.

(C) **Sits Containment System** — the voxel grid spans the union bounding box of all selected geometry. A compute_mask() builds a set of cell indices that fall inside any of the picked breps, so voxels only appear inside the real building volumes — even across disconnected buildings. The result is a 3-D voxel mass whose colour, density pattern, and attractor marker positions are all driven by real Melbourne climate data.



SITE ANALYSIS

MASS OPERATION

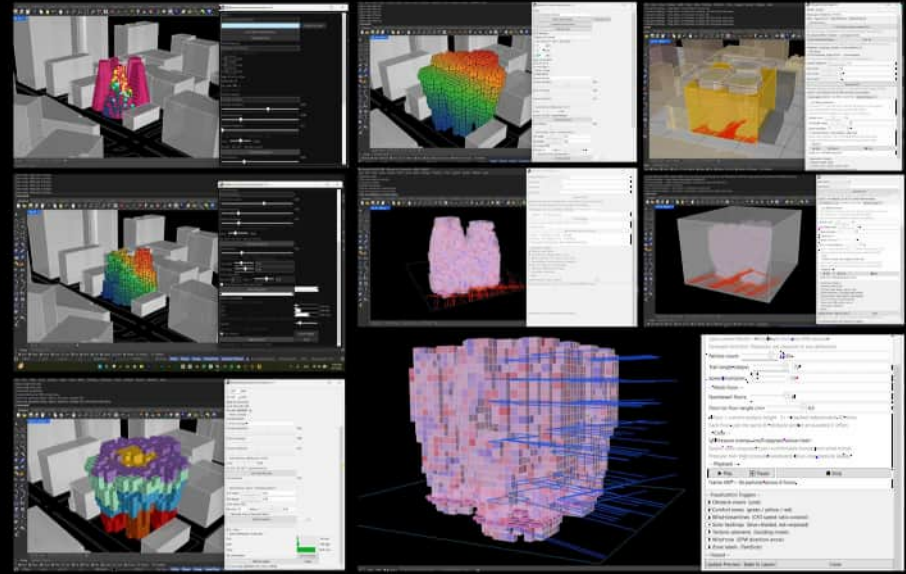
VOXEL FIELD SUBDIVISION

PROGRAM SEPARATION

DISCRETE ELEMENT DISTRIBUTION

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SITE ANALYSIS

MASS OPERATION

VOXEL FIELD SUBDIVISION

PROGRAM SEPARATION

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MASSING GENERATION

Script 02 - Differential Growth Form finding

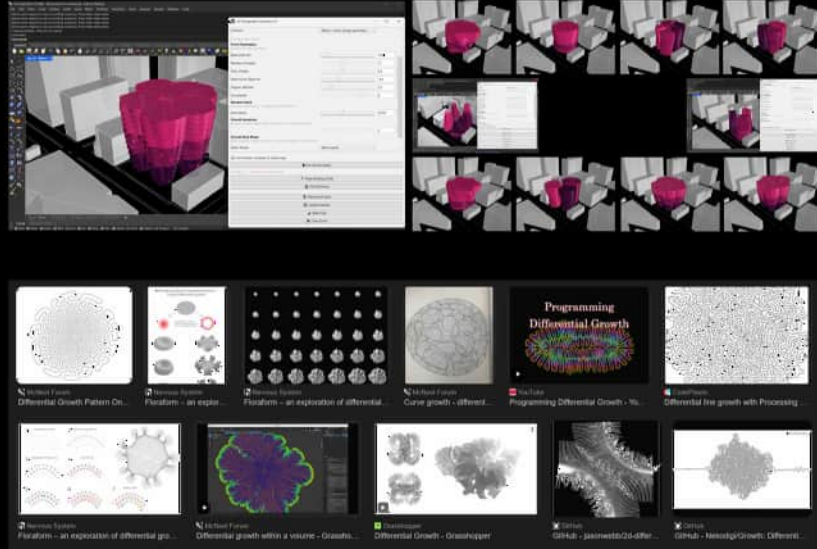
Given Melbourne's climate extremes, multiple massing iterations were tested and evaluated using an EPW-integrated climate comfort agent — scoring each layout against environmental performance.

Script 03 - Description

Description & How It Works: Simulates organic growth by iteratively moving vertices outward along surface normals while maintaining edge length constraints. Creates fluid, biomimetic forms through local agent rules (each vertex "grows" independently, constrained by neighbors).

Multi-agent system:

Agents: Individual vertices
Behavior: Push outward (growth) + resist compression (edge length preservation)
Stigmergy: Vertex positions influence neighbor growth via shared edges
Emergence: Winkled, tensile forms (no central planner)



SITE ANALYSIS

MASS OPERATION

VOXEL FIELD SUBDIVISION

PROGRAM SEPARATION

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MASSING GENERATION

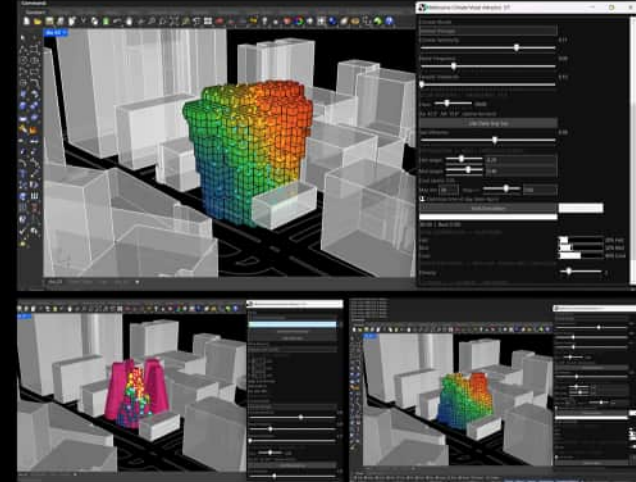
Script 03 - Climate

This allowed me to score each layout against environmental performance.

Climate Attractor points

- Red : High density
- Yellow : Medium density
- Blue : Low density

The climate field is applied to the base massing to generate attractor points, which inform voxel distribution in the next stage. All voxels are populated with attractor data drawn from the urban context.



SITE ANALYSIS

MASS OPERATION

VOXEL FIELD SUBDIVISION

PROGRAM SEPARATION

DISCRETE ELEMENT DISTRIBUTION

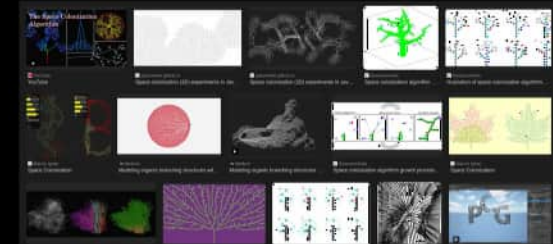
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MASS OPERATION

Script 03 - Colonization Algorithm

The Space Colonisation Algorithm (SCA) was then applied to carve a rift through the mass, mimicking canyon and cliff formations found in nature, channelling indirect light deep into the building.

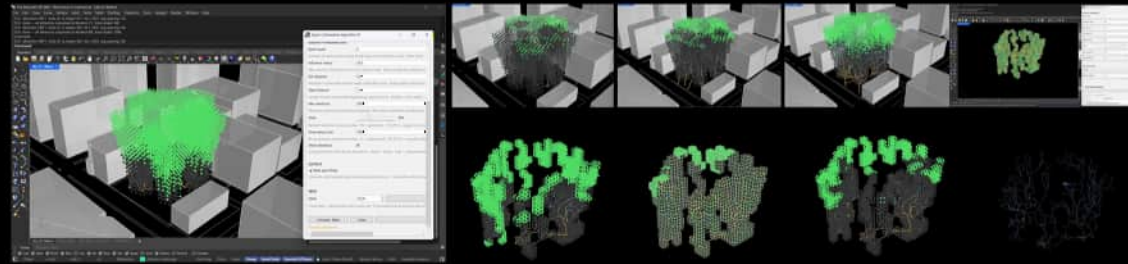


Script 03 - Colonization Algorithm

Description & How It Works: Models how roots or veins branch—agents "colonize" free space by occupying branches toward nearest unvisited space nodes. Outputs branching tree curves (not volumes).

Multi-agent system:

Agents: Individual vertices
Behavior: Push outward (growth) + resist compression (edge length preservation)
Stigmergy: Vertex positions influence neighbor growth via shared edges
Emergence: Winkled, tensile forms (no central planner)



SITE ANALYSIS

MASS OPERATION

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VoxelField

Final Massing

The final massing carries an extended upper footprint that shades the building in summer, while the carved crack allows indirect light and wind to penetrate throughout — creating passive ventilation.

Script 04 - VoxelFieldGenerator

Multi-agent system:

Agents: Individual voxels evaluating noise fields
Behavior: If noise(x,y,z) > threshold → occupy site → empty
Stigmergy: Neighboring voxel density influences threshold decisions (optional clustering post-processing)
Emergence: Organic porosity + connectivity without explicit form-sculpting



MASS MODEL + ATTRACTOR
Immediate Neighbours



GENERATED GEOMETRY
Positive space



SCA
Crack Negative



Keyplan
Section A



Section B



Section C

SITE ANALYSIS

MASS OPERATION

VOXEL FIELD SUBDIVISION

PROGRAM SEPARATION

DISCRETE ELEMENT DISTRIBUTION

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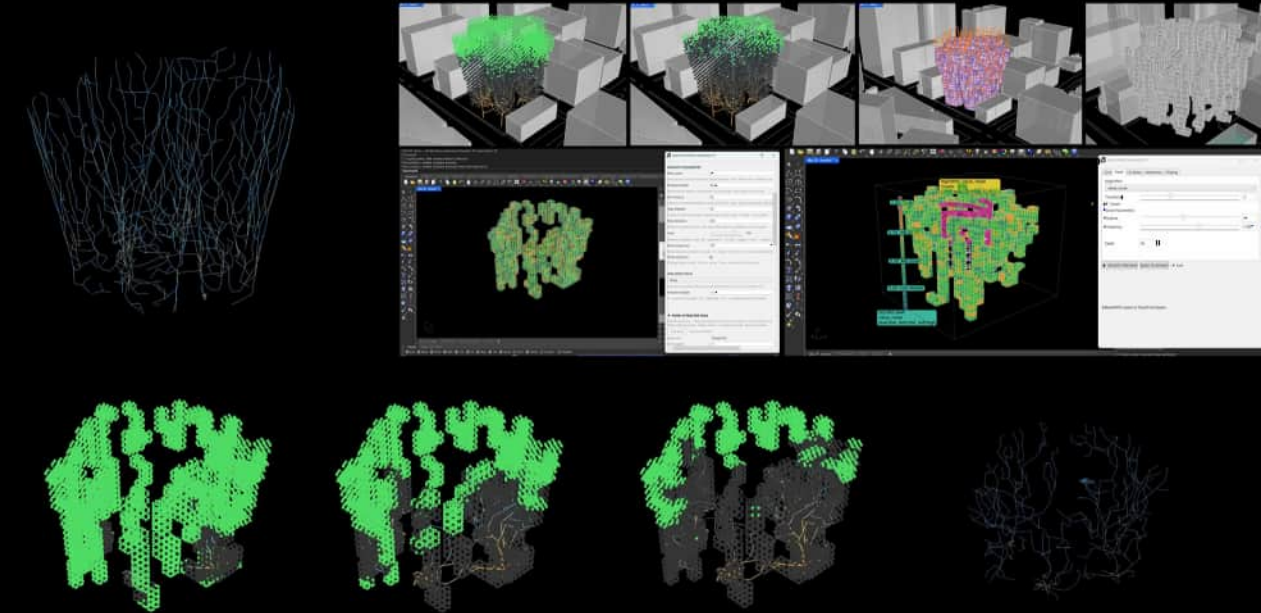
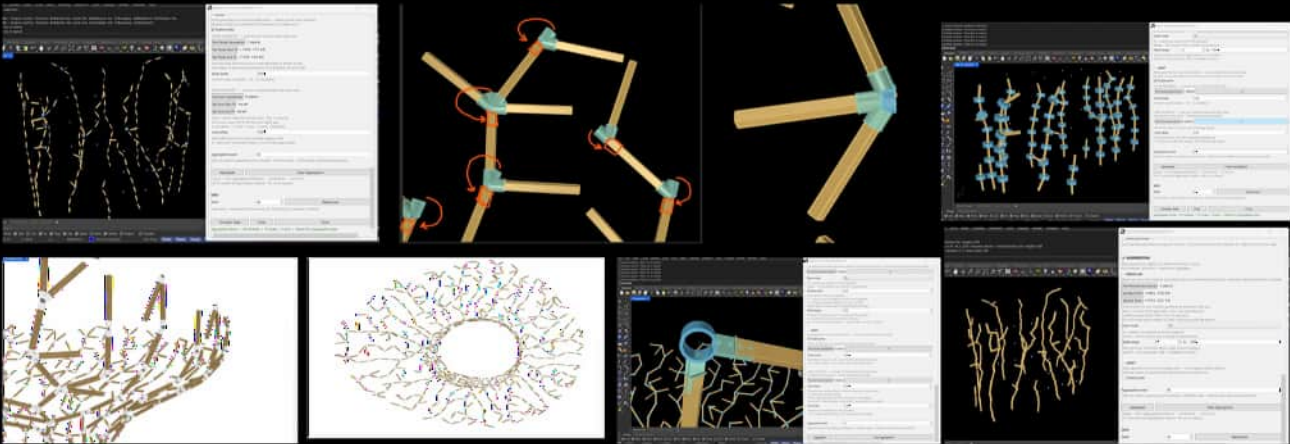
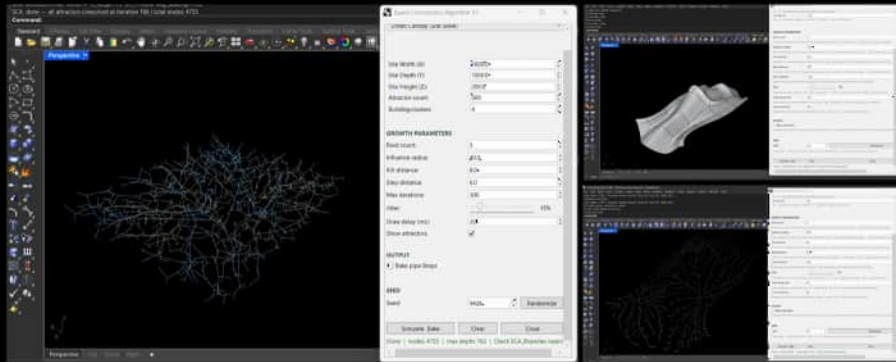
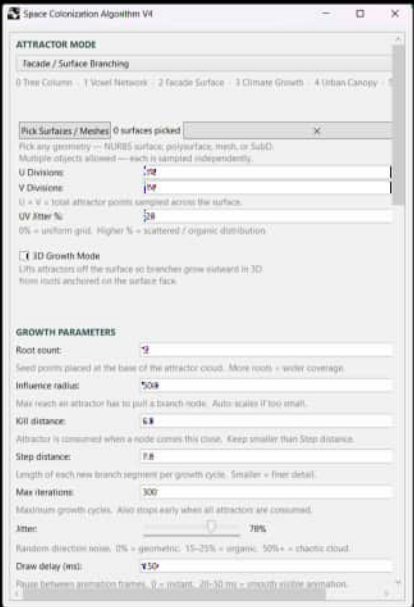
SCRIPT DEVELOPMENT

Script 03 - SPACE COLONIZATION

The Space Colonization Algorithm (SCA) simulates organic branching growth, inspired by biological vascular networks (tree root systems, fungal networks, bone tissue). The algorithm works by:

- **Attractors** — points scattered in 3D space representing resources/targets
- **Growth nodes** — branch tips that step toward nearby attractors each iteration
- **Consumption** — when a node reaches within kill_distance of an attractor, it is consumed (removed), and growth continues organically
- **Output** — a depth-tagged branching tree baked to Rhino layers

The scripts are single-file Rhino Python tools running in Rhino 8 CPython 3, using RhinoCommon (geometry), Eto.Forms (GUI), DisplayConduit (real-time animation), and scriptcontext for Rhino document access.



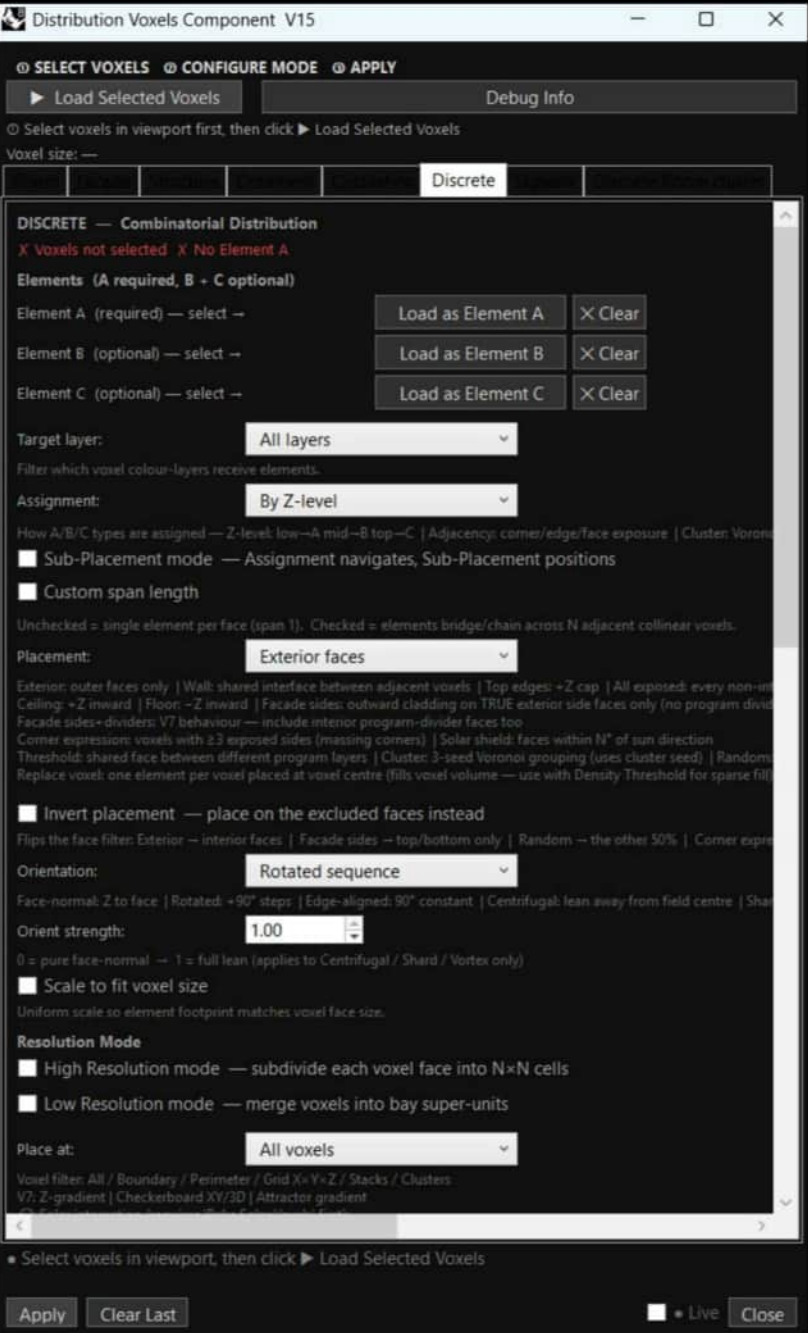
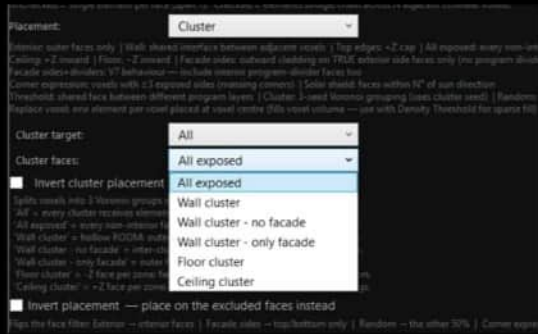
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SCRIPT DEVELOPMENT

Script 05 - Distribution Voxels Component

Distribution Voxels Component (DVC) is a parametric architectural design tool built inside Rhino 8 using CPython 3 and an Eto Forms GUI. It reads a 3D **voxel field** — a regular grid of cubes whose colour layers represent architectural program zones — and distributes user-supplied 3D geometry across that field using eight condition-based placement modes. **Core engine:** Every voxel face computes an outward-facing plane (ZAxis = outward normal). rg.Transform.PlaneToPlane(source, target) maps input geometry from its bounding-box frame to each face. Scale, orientation, span-bridging and hi-res/lo-res tiling layer on top. Data travels between cooperative scripts via sc.sticky (Rhino shared in-memory dict) — no file I/O needed.

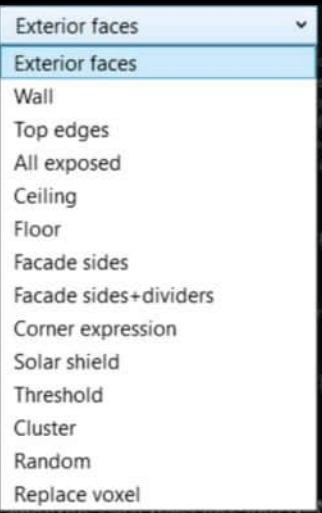
Cluster configuration



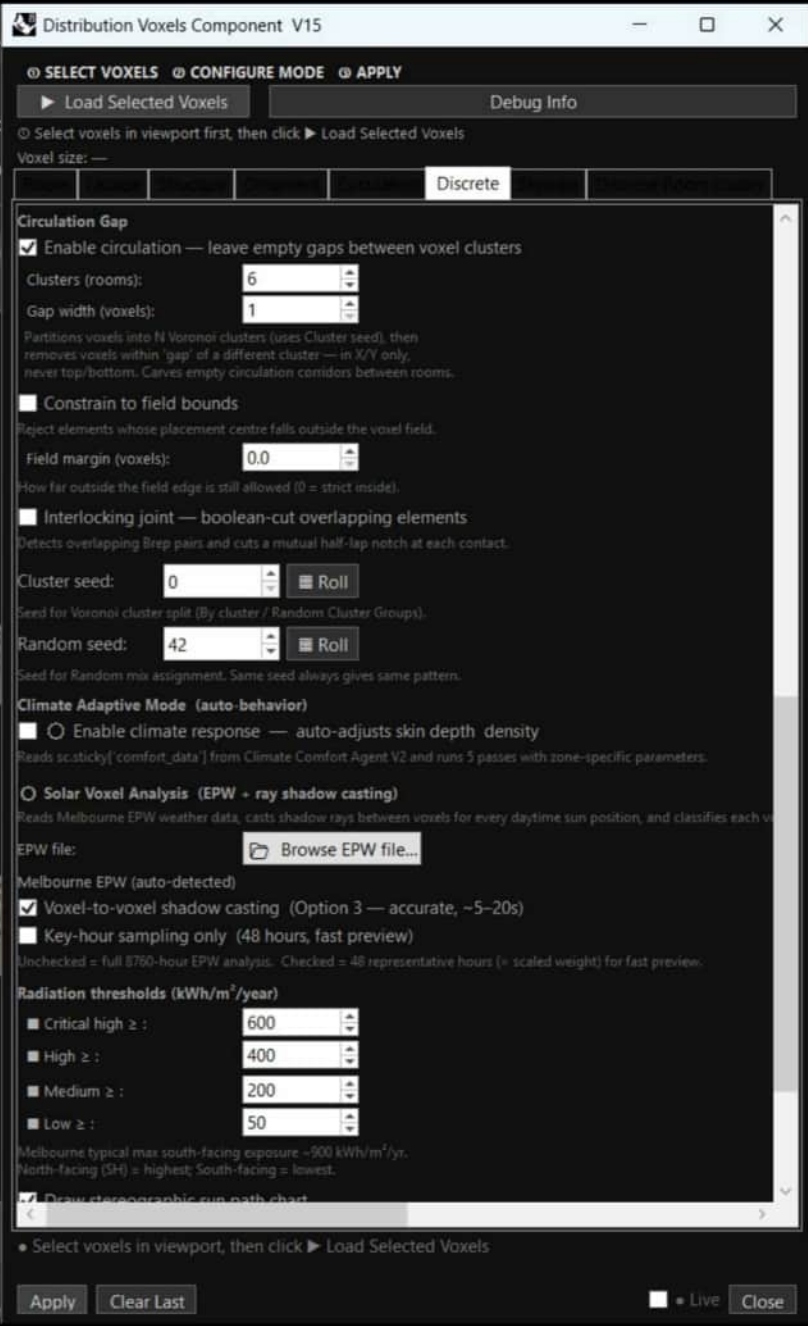
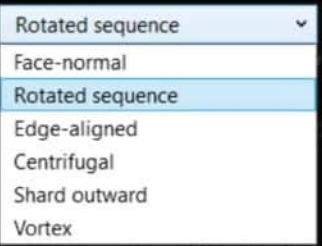
Assignment



Placement : Tectonics



Orientation : Tectonics



DISTRIBUTION VOXELS COMPONENT

Script Development V1 to V16 | RMIT AI-Accelerated Agentic Architecture | 2025-2026

1 — OVERALL SCRIPT APPROACH

Overview

The Distribution Voxels Component (DVC) is a Rhino 8 / CPython 3 parametric architectural tool with an Eto Forms GUI. It reads a 3-D voxel field whose colour layers represent architectural program zones and distributes user-supplied 3D geometry across that field using eight condition-based modes.

The core transform engine maps input geometry from its

Core data flow

Rhino colour layers -> Voxel objects (grid_ijk, center, size, face_types)
face_types -> interior / exterior / edge / corner classification
get_face_plane(vox, dir) -> outward ZAxis plane at face centre
PlaneToPlane(src_plane, tgt_plane) -> geometry oriented to face
Cluster seed -> _split_6_clusters() -> 6-zone Voronoi partition
sc.sticky['comfort_data'] -> Climate Adaptive Mode parameters
EPW file -> parse_epw() -> 8760-hr solar + thermal per voxel

Key design decisions

Non-modal Eto Form — viewport always interactive
Inline pick (Hide->Pick->Show) — no dialog recreate
sc.sticky bridge — voxel data to downstream scripts
PlaneToPlane core — all 14 placement types share one engine
SuperVoxel bays — lo-res Grid X x Y x Z grouping
UITimer 400ms debounce — Live mode real-time response
6-zone Voronoi (_split_6_clusters) — A to F assignment

2 — MULTI-AGENT ECOSYSTEM (cooperative pipeline + internal behaviours)

Output baked -> VOXELGEN_Discrete_A ... F layers in Rhino

A - Climate Comfort Special V1

- Solar analysis on base voxel field
- Outputs ccs_data per-voxel heat_index into sc.sticky['comfort_data']
- DVC Climate Adaptive Mode reads this to auto-adjust skin depth + density per zone (hot/warm/mid/passive/wind)
- Fully decoupled — DVC works without it

B - SCA V2 + Voxel SCA Integrator V1

- Space Colonization Algorithm grows branch networks (5 modes + Influence)
- Influence Field: Climate / Sun / Wind / Gravity / Custom XYZ
- Integrator partitions voxel field into: Remaining / Corridor Void / Atrium Void
- Bridges SCA output to DVC via sc.sticky

C - Diff Growth V3 | Tectonic V5 | Perlin V1

- Diff Growth V3: EPW-scored boundary form-finder; evolves building footprint curve — defines the voxel field shape
- Tectonic Modifier V5: 3-mode mesh modifier, 8 profiles, tiling output; provides facade source geometry for DVC
- Perlin Noise V1: fBm flow fields for Facade and Facade mode variation

D - DVC V16 Internal Agentic Behaviours

- LIVE MODE: UITimer 400ms debounce; any param change auto-clears + re-applies without user clicking Apply button
- CLIMATE ADAPTIVE: reads ccs_data, auto-selects depth/density per zone
- SOLAR VOXEL ANALYSIS: EPW ray casting 8760-hr full or 48-hr key-hour fast

3 — VERSION DEVELOPMENT TIMELINE (V1 to V16)

V1-V2

Foundation
• EPW parser + Perlin 3D engine
• Voxel obj: grid_ijk, center, size
• face_types: int/ext/edge/corner
• PlaneToPlane transform engine
• Eto GUI: tabs + dark theme
• 5 modes: Room/Facade/Struct/

V3

SCA Integration
• SCA Integrator compatibility
• Remaining/Corridor/Atrium Void
• Neighbour connectivity at load
• OUTPUT_LAYERS named layer
• sc.sticky grid handoff to SCA

V4-V6

Assignment+Attraction
• Z-level/adjacency/random assign
• Attractor gradient (dist bias)
• Facade, Structure, Ornament m
• Centrifugal/Shard/Vortex orient
• Orient strength slider 0-1

V7

14 Placement Rules
• Wall (vertical shared faces)
• Corner expression, Solar shield
• Threshold (inter-program faces)
• Invert placement, Custom span
• Z-gradient/Checkerboard assign

V8

Sub-Placement+Cluster
• _split_3_clusters 3-zone Voronoi
• Sub-Placement: bucket -> elem
• forced_label sentinel
• place_inside (interior surface)
• Random + Replace voxel mode

V9-V11

Circ/Skywalk/Res
• Circulation + Skywalk tabs
• Hi-res N x N face sub-grid
• Density threshold random skip
• Solar Voxel Analysis (EPW rays)
• Climate Adaptive Mode scaffold
• Discrete Room Cluster tab (BFS)

V12

Structure+Wall Fix
• Column/Beam/Both custom grid
• Column B alternation per pos
• Wall: vertical faces ONLY fix
• Wall centred on shared interface
• X x Y x Z grid spacing

V13

Circ Gap+Lo-res
• _erode_circulation corridors
• Grid X x Y x Z SuperVoxel bays
• Partial edge bay handling
• Lo-res face tiling (n_u x n_v)
• scale_xyz independent of fit

V14

Wall Cluster+Rooms
• Wall cluster: hollow ROOM logic
• FACADE+DIVIDER+INTERIOR
• no-facade / only-facade variants
• All exposed boundary face fix
• flip_inward: elements into room

V15

Live Mode+Cluster+
• UITimer 400ms debounce Live
• Floor cluster (-Z per zone)
• Ceiling cluster (+Z per zone)
• Invert cluster: gap-zone subject
• All 4 Cluster faces + invert

V16

6 Inputs A to F
• Elements D/E/F added (pick/clear)
• Fallback chain D->C->B->A
• _split_6_clusters 6-zone Voronoi
• Z-level 6 bands / adj 6 bins
• Sub-placement bucket loop x 6
• Output layers Discrete_D/E/F

4 — SYSTEM ARCHITECTURE (V16 Discrete Mode)

DATA INPUTS

Melbourne EPW (8760 hr/yr)
Site Brep / Mesh (colour layers)
Source Geometry A to F (6 slots)
Cluster seed / Random seed
sc.sticky['comfort_data'] from CCS V1
sc.sticky SCA grid from SCA V2
sc.sticky field from Diff Growth V3
Attractor point (optional)

PROCESSING CORE

parse_epw() -> climate_factors
load_voxels() -> face_types + neighbors
get_face_plane() -> outward ZAxis plane
_split_6_clusters() -> 6-zone Voronoi
_check_placement() -> 14 rules + 6 cluster faces
PlaneToPlane(src, tgt) -> oriented geometry
_erode_circulation() -> corridor gap removal
_build_super_voxels() -> lo-res bay merge

CONCURRENT BEHAVIOURS

Live Mode (UITimer)
400ms debounce; param change -> auto clear + re-apply without user click
Climate Adaptive Mode
reads sc.sticky['comfort_data']
auto-sets depth / density per zone
Solar Voxel Analysis
EPW rays 8760-hr full or 48-hr fast

OUTPUTS

Discrete_A ... Discrete_F (6 layers)
Baked geometry groups per label
VOXELGEN_Discrete_{label}_{ts}
sc.sticky -> CCS / SCA / DVC chain
8 output modes: Room / Facade / Structure / Ornament / Circ / Skywalk / Discrete / RC
Hot / Mid / Cool voxels by heat_index
Attractor pts per zone (Cluster target)

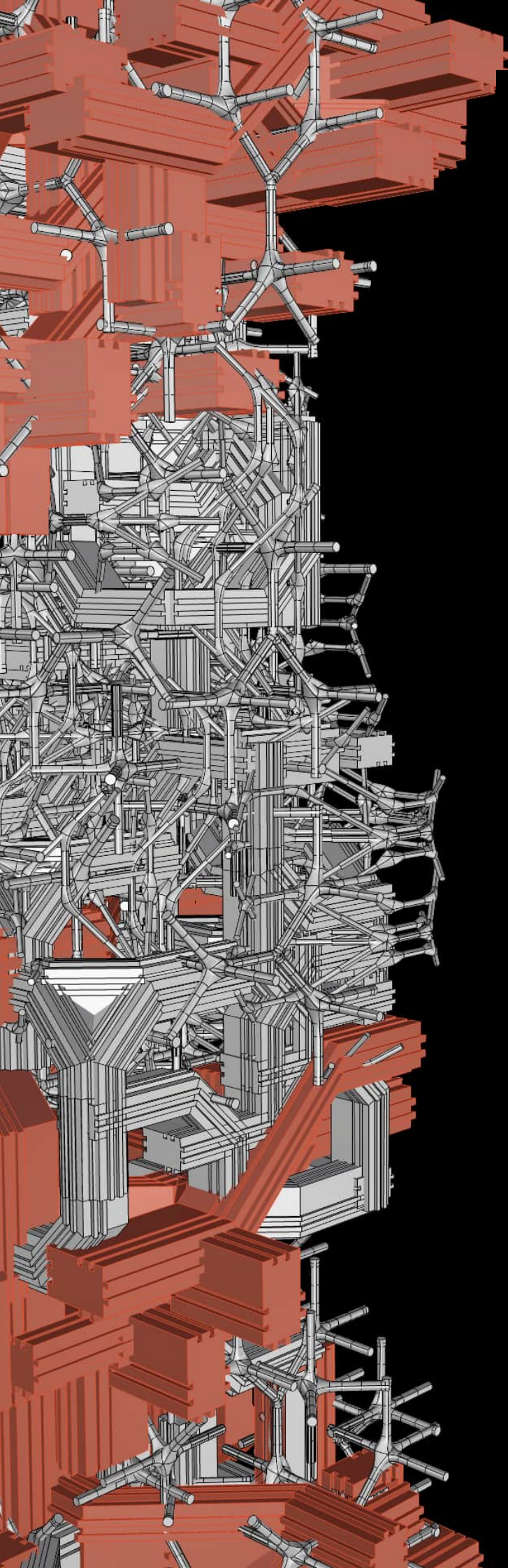
SITE ANALYSIS

MASS OPERATION

VOXEL FIELD SUBDIVISION

PROGRAM SEPARATION

DISCRETE ELEMENT DISTRIBUTION



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Discrete Element Study

Start from voxel field

- Value noise mode
- Difference threshold
- Octaves
- Frequency

Wall Chunks

SITE ANALYSIS MASS OPERATION VOXEL FIELD SUBDIVISION PROGRAM SEPARATION DISCRETE ELEMENT DISTRIBUTION

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Multi-Agent Computational Architecture for Adaptive Learning Environments

SITE ANALYSIS MASS OPERATION VOXEL FIELD SUBDIVISION PROGRAM SEPARATION DISCRETE ELEMENT DISTRIBUTION

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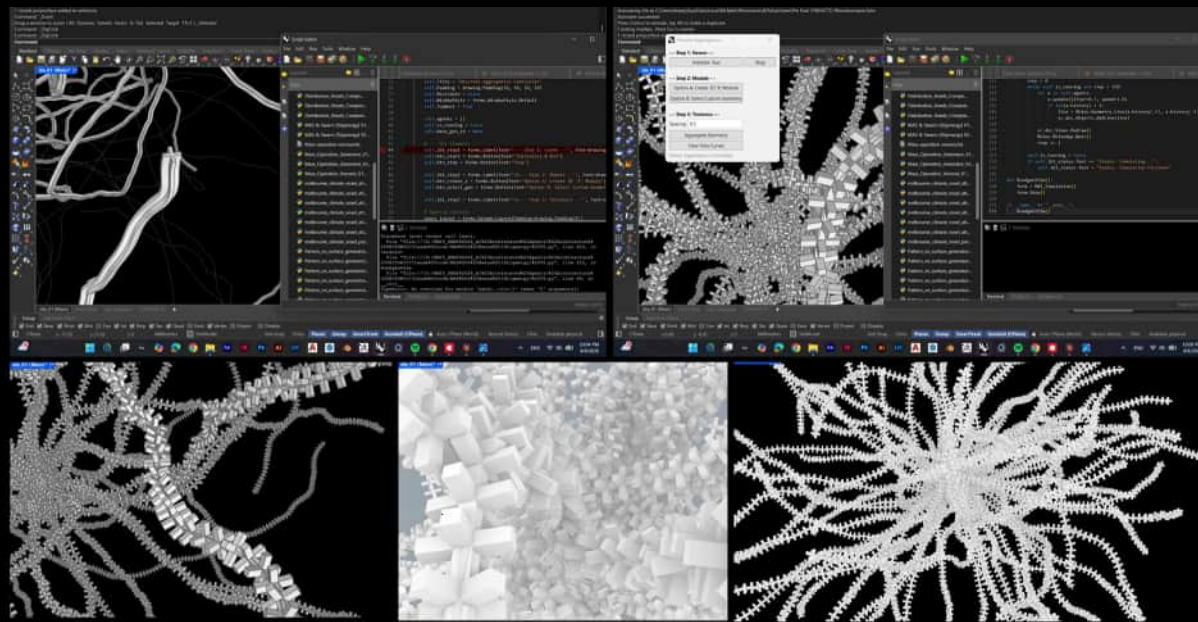
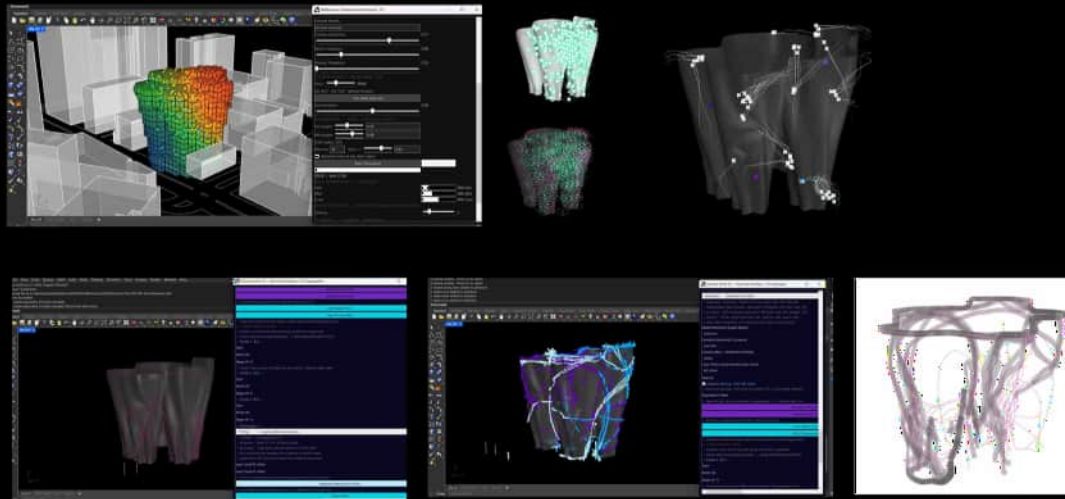
Multi-Agent Computational Architecture for Adaptive Learning Environments

SITE ANALYSIS MASS OPERATION VOXEL FIELD SUBDIVISION PROGRAM SEPARATION DISCRETE ELEMENT DISTRIBUTION

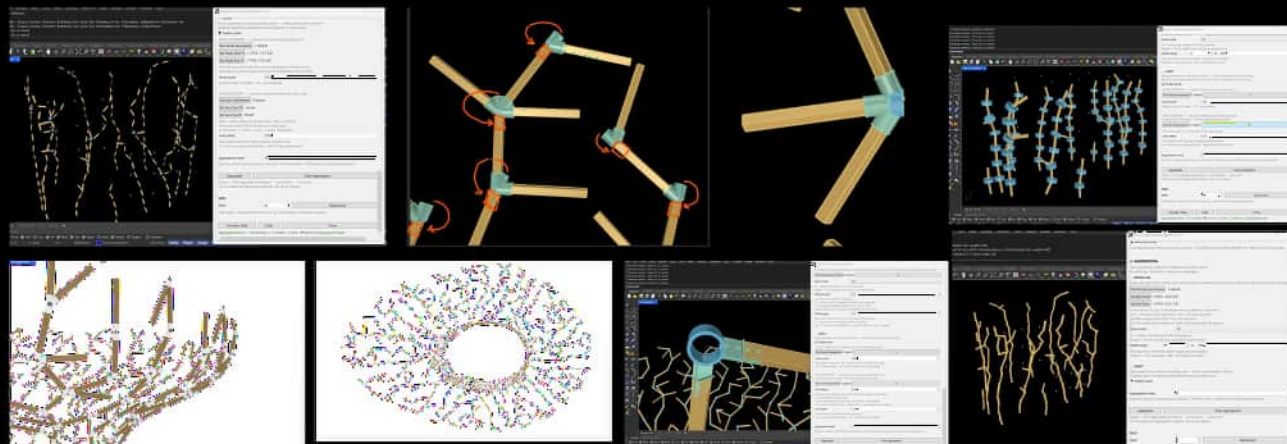
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Under development iterations



AITC

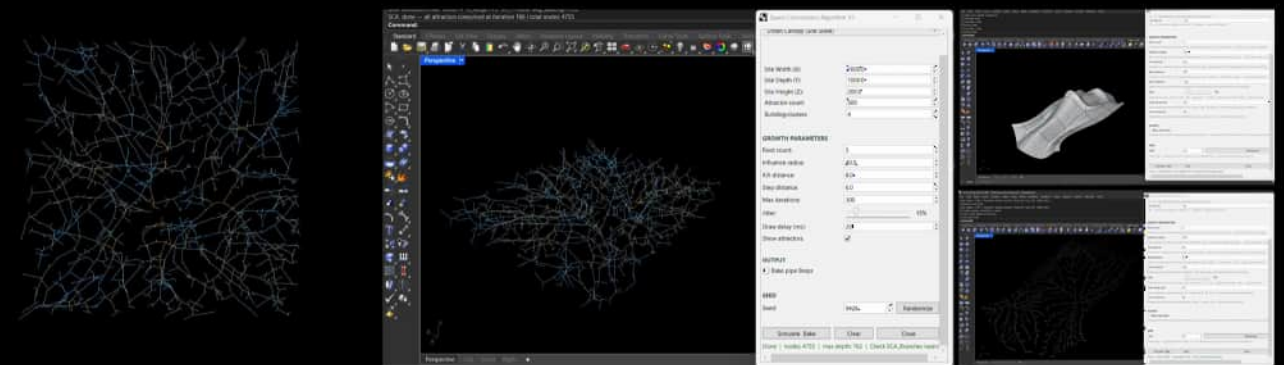


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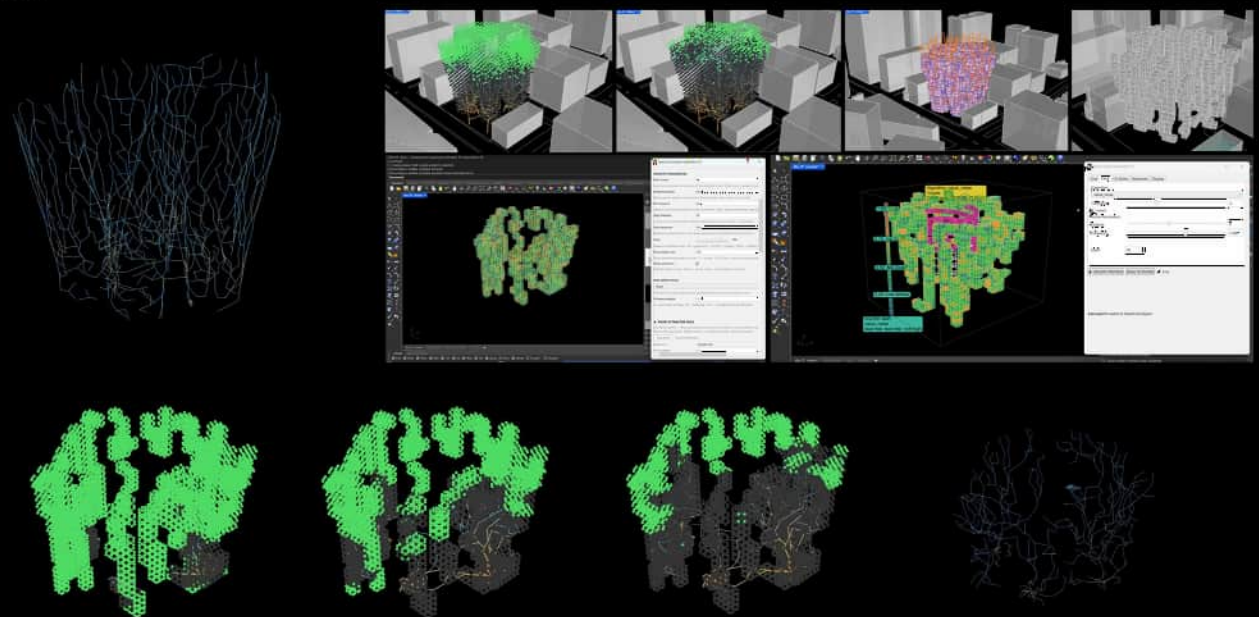
Under development iterations



AITC

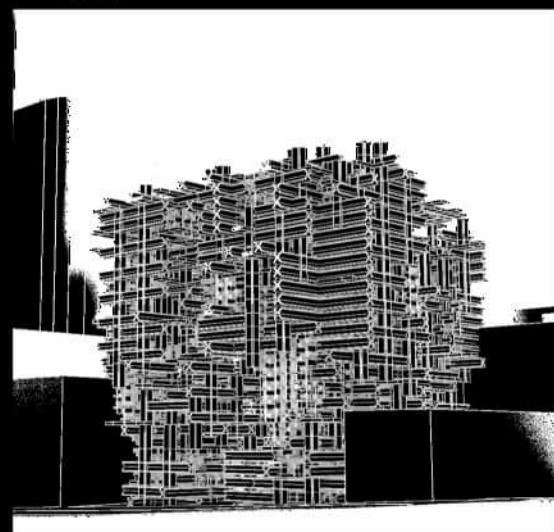


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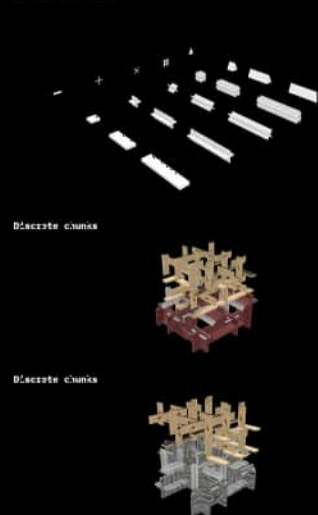


Distribution elements

Street view

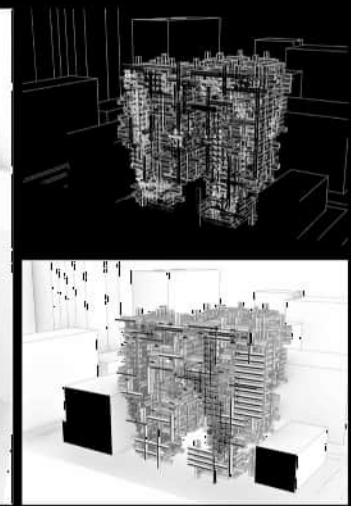
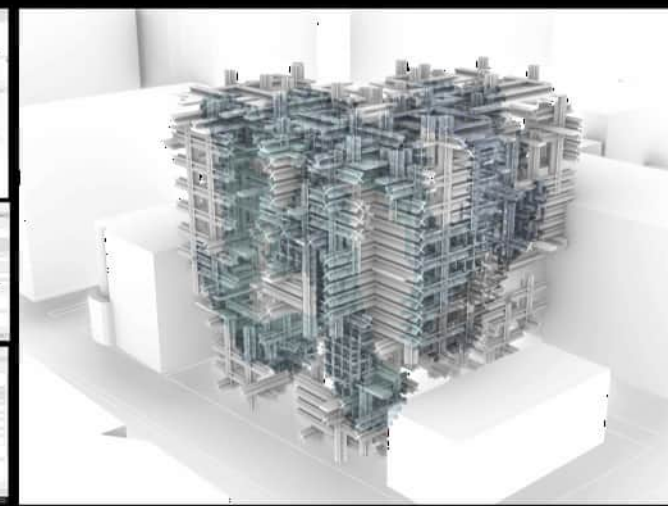
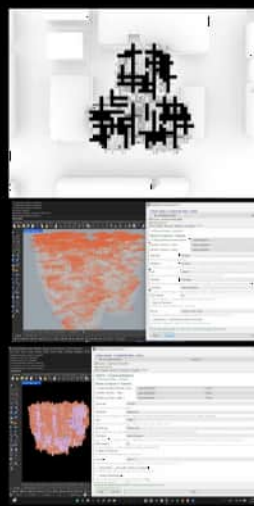


Discrete elements

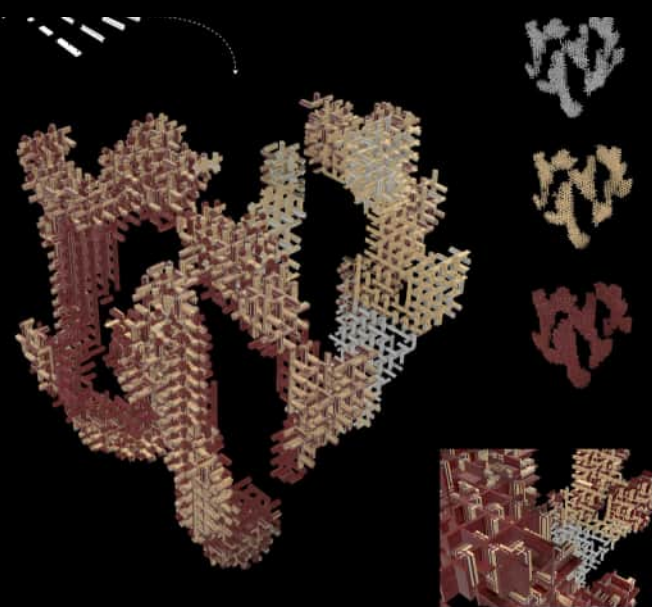
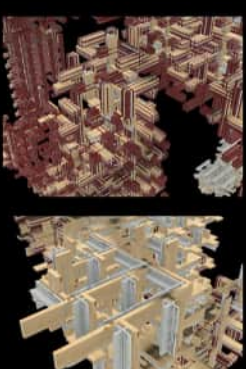


Distribution elements

Script : Voxels field



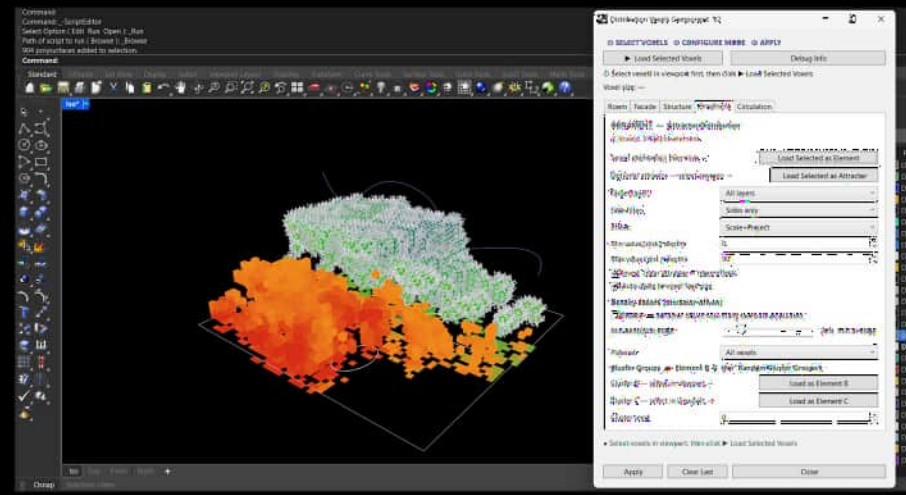
Site Voxels field Data voxels Distribution Discrete modular



Keyframe

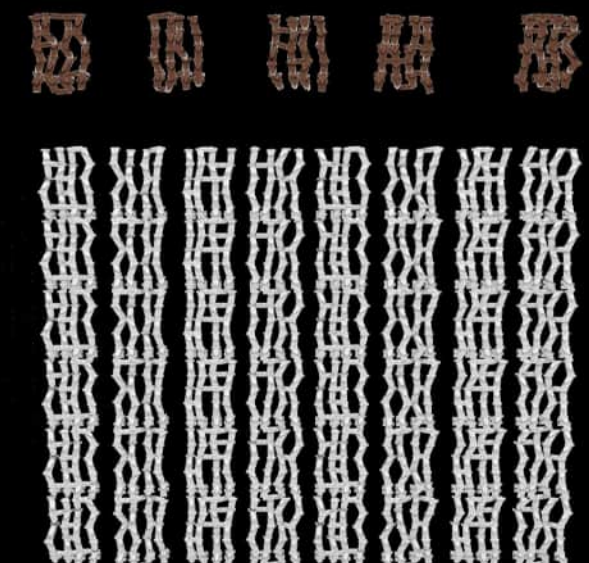
Site Voxels field Data voxels Distribution Discrete modular

DISTRIBUTION METHOD
Contextual site data is gathered and aligned into the spatial database.
data points are prescribed a numerical value, or score.



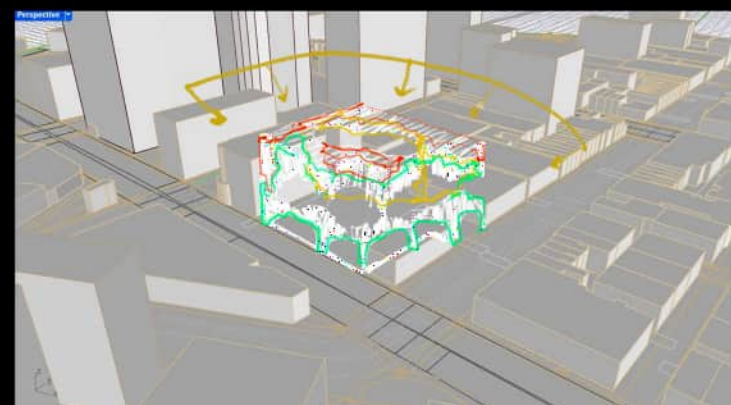
Site Voxels field Data voxels Distribution Discrete modular

DISCRETE MODULAR CATALOG (CHECKS)
Contextual site data is gathered and aligned into the spatial database.
data points are prescribed a numerical value, or score.

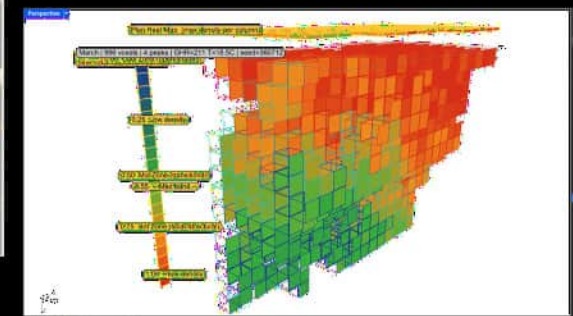
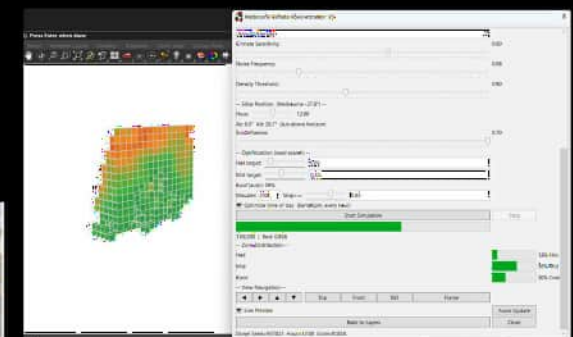


Site Voxels field Data voxels Distribution Discrete modular

DATA VOXELS (FIELD)
data points are prescribed a numerical value, or score.



Site Voxels field Data voxels Distribution Discrete modular



FOLIO SUBMISSION

Kankawee Maksomboon | s4907770

Tasnuma Usrat

AUGMENTED MATERIALRITY

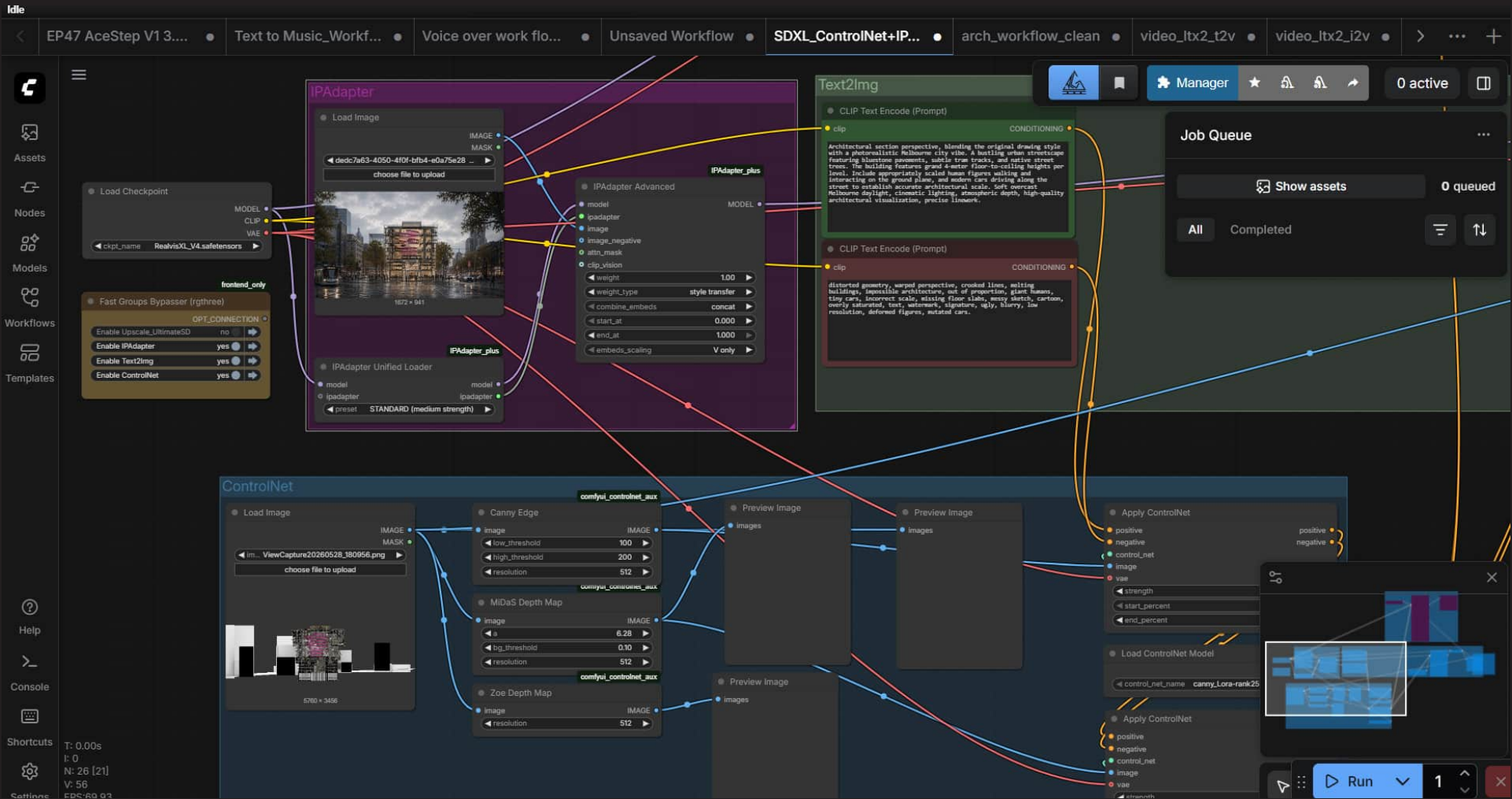
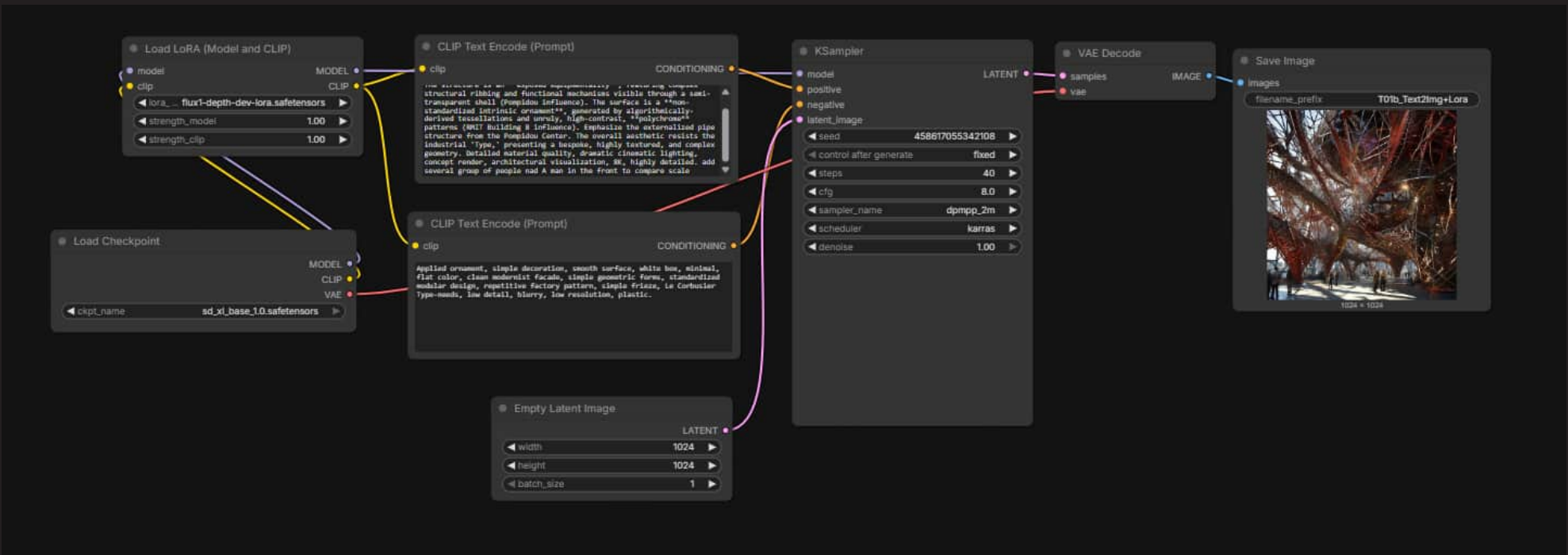
Igor Pantic

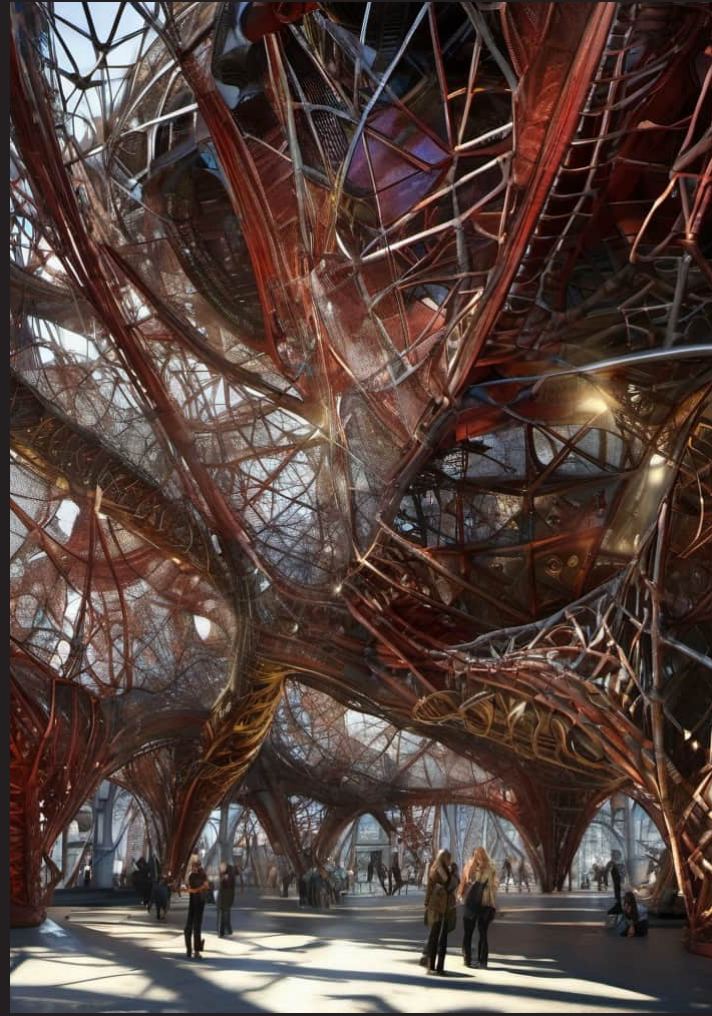


COMFI UI

AI TRAINING

- 01_SDXL_Text2Img
- 02_SDXL_Text2Img+Lora
- 03_SDXL_Img2Img+Lora
- 04_SDXL_ControlNet+IPAdapter
- 05_Flux_Text2Img+LoRA
- 06_Flux_ControlNet_Depth+Multi-LoRA Loader





MATERIAL RESEARCH : USED COFFEE GROUNDS

The primary material for this research is a high-volume waste product that is easily accessible from homes or cafés, particularly within Melbourne's urban context. The aim is to explore the potential of everyday waste as a sustainable construction or design material, reducing landfill impact and promoting circular design practices.



CORE QUALITIES :

Strength Enhancement:

Coffee biochar, made by roasting coffee grounds, increases the 30% strength and durability of concrete. It also reduces the need for sand and cement, lowering costs and environmental impact.

Environmental

Reduces landfill waste by diverting coffee grounds. Decreases reliance on finite resources like mined sand. Lowers the carbon footprint of concrete production.

Real-World

Trials are active in Melbourne, including a **footpath and a road project**.

Researchers have also developed **sustainable bricks** from coffee waste, baked at lower temperatures.

Ongoing

RMIT University is exploring the ideal ratios of coffee grounds, sand, and **cement for maximum performance**.

Research is also looking into how coffee biochar can improve **concrete's thermal and acoustic properties**.

Source : <https://www.rmit.edu.au/news/all-news/2023/aug/coffee-concrete>



MATERIAL TESTING

| Introduction: Bio-base material approaches using waste coffee grounds as the strater, and bio miture glues

MATERIAL TESTING
|Type A process



TYPE A : SOLID COFFEE
Pure bio-material | Solid wasted cofe ground composite

TYPE B : TRANSLUCENT COFFEE
Pure bio-material | Translucent | Thin | Flixible

Material criteria

- Physical strength
- Remaining shape
- Castability
- Curing duration
- Material finish



TYPE A : SOLID COFFEE

TYPE B : TRANSLUCENT COFFEE

Bio-material base
Shrink | Fungus | Slow cure



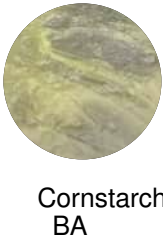
Gelatine
AA



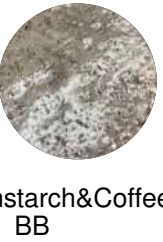
Double Gelatine
AB



Constarch
AC



Cornstarch
BA



Cornstarch&Coffee
BB

Hybrid-material base
Soft | Midium cure



PVA & Cornstarch
AD



Quick-dry PVA & Cornstarch
AE

TYPE C : PLASTER COFFEE

Chemical base
Retain shape | Stiff | Fast cure



PVA
AF



Quick-dry PVA
AG



Polyurethane
AH



Plaster & Coffee
CA

MATERIAL TESTING

| Type A : Ingredient ratio test

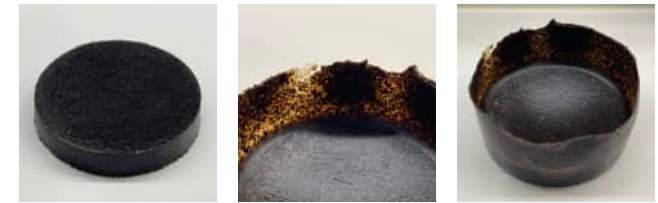
TYPE AA
water : gelatin : syrup : cooking oil : coffee ground
10 : 1.5 : 1.5 : 0.1 : 3

Takes longer to dry, but remains controllable
Note: Rougher surface compared with Type AB



TYPE AB
water : gelatin : syrup : cooking oil : coffee ground
10 : 3 : 1.5 : 0.1 : 3

Ideal ratio, proper shape retention, fast drying, smooth surface
Note: Requires a quick casting process



AUGMENTED MATERIALITY | S2 RMIT UNIVERSITY

MATERIAL TESTING

| Type A : Ingredient ratio test

TYPE AD
PVA : Constarch water ratio (1:4) : coffee ground
3 : 1 : 3

Medium strength, Mid-cure 6 Hr.
Note: Adding constarch mixture for liquid mixing purpose ,while the material break when demold



TYPE AF
PVA : coffee ground
1 : 1

Medium strength, Mid-cure 4 Hr.
Note: Mid-cure can result in cracking.



AUGMENTED MATERIALITY | S2 RMIT UNIVERSITY

MATERIAL TESTING

| Type A : Ingredient ratio test

TYPE AH ●
Polyurethane : coffee ground
1 : 1

Retain shape, stiff & high strength, fast cure 1 hr.
High gloss finish, Perfect for casting
Note: Expandable size when open casting. Suitable for pressed mold





AUGMENTED MATERIALITY | S2 RMIT UNIVERSITY

TYPE AC
water : starch : vinegar : coffee ground
10 : 3 : 2 : 3

Using cornstarch as a substitute for gelatin — a cheaper option, resulting in a rough and matte finish
Note: Breakable when casting in a non-stiff mold



Result

Strength
AA
AB
AC

Casting ability
AA
AB
AC

Transparency
AA
AB
AC1

Smoothness
AA
AB
AC

Surface dry
AA 2 Hr.
AB 1 Hr.
AC 30 min

Material dry
AA 1 week
AB 1 week
AC N/A

TYPE AE
Quick-Dry PVA : Constarch water ratio (1:4) : Coffee ground
3 : 1 : 3

Medium strength, Mid-cure 4 Hr.
Note: Adding constarch mixture for liquid mixing purpose, while constarch effect material strength



TYPE AG
Quick-Dry PVA : Coffee ground
1 : 1

High strength, Mid-cure 2 Hr.
Matt finish, Perfect for casting
Note: Edge condition



WASTED COFFEE GROUNDS

MATERIAL DEVELOPEMENT

Bio Material update

Week 01

Week 03

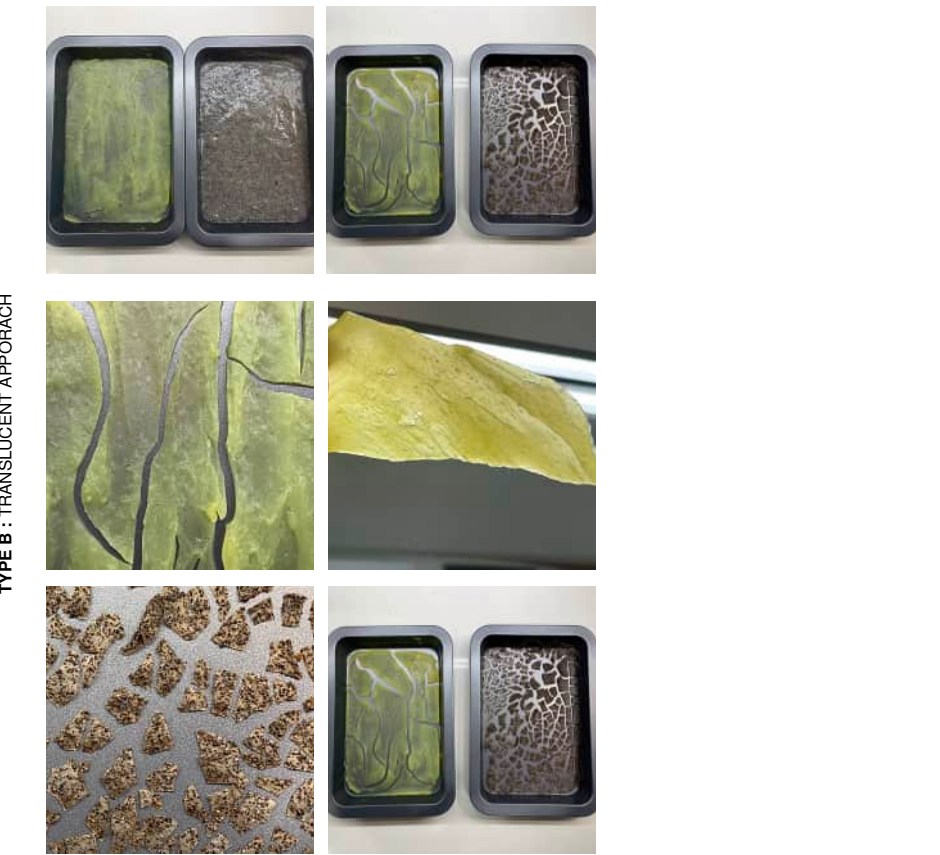
TYPE A : SOLID APPORACH



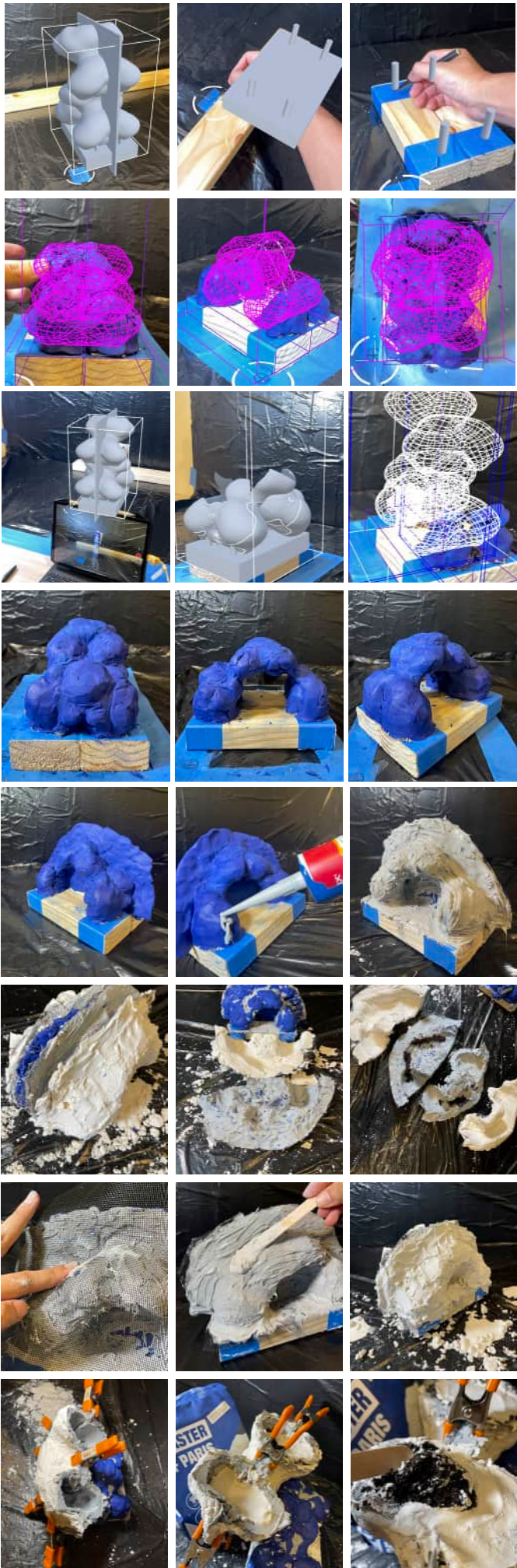
Week 01

Week 03

TYPE B : TRANSLUCENT APPORACH



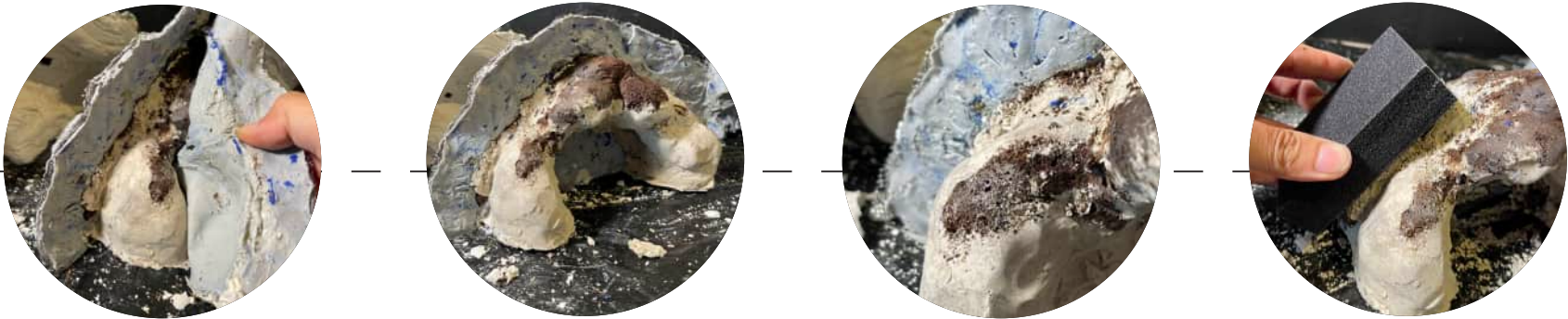
WASTED COFFEE GROUNDS



CASTING TEST

Composite material casting test :
Solid coffee & polyurethane mixture (AH) and Plaster coffee (CA)

Demolding



AUGMENTED MATERIALITY | S2 RMIT UNIVERSITY



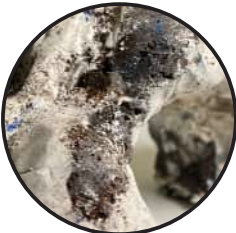
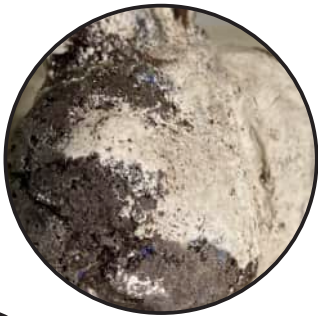
WASTED COFFEE GROUNDS

CASTING TEST

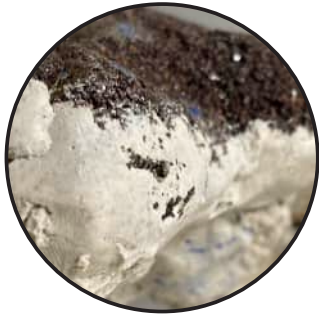
Material's characteristic

Material transition between coffee and plaster

Using plaster as main material, while
coffee only add texture.



Raw coffee material gesture



Cracking part shows
inner layer of coffee material.



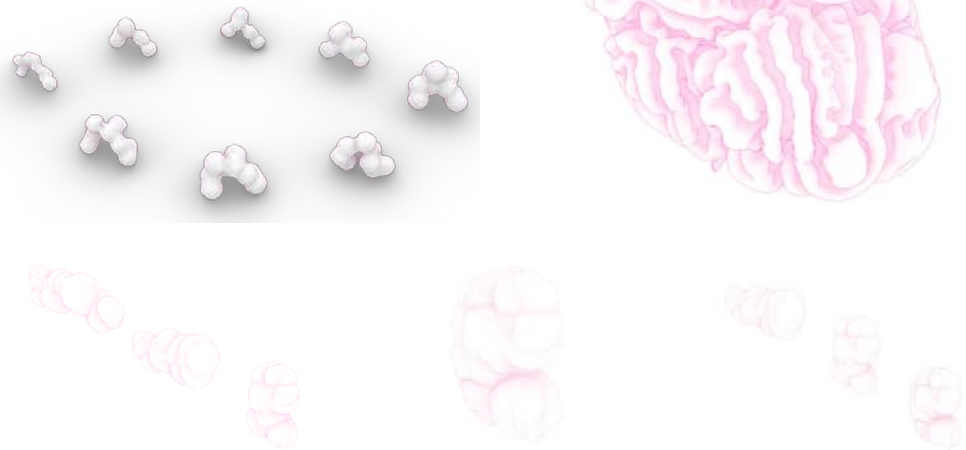
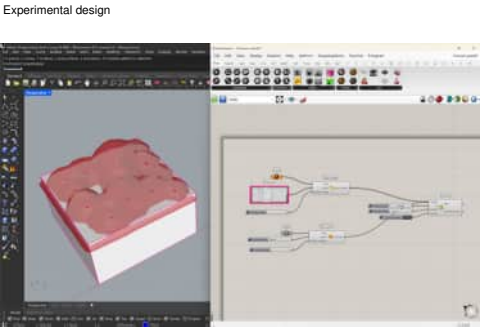
Smooth coffee material gesture
shape edge condition

AUGMENTED MATERIALITY | S2 RMIT UNIVERSITY

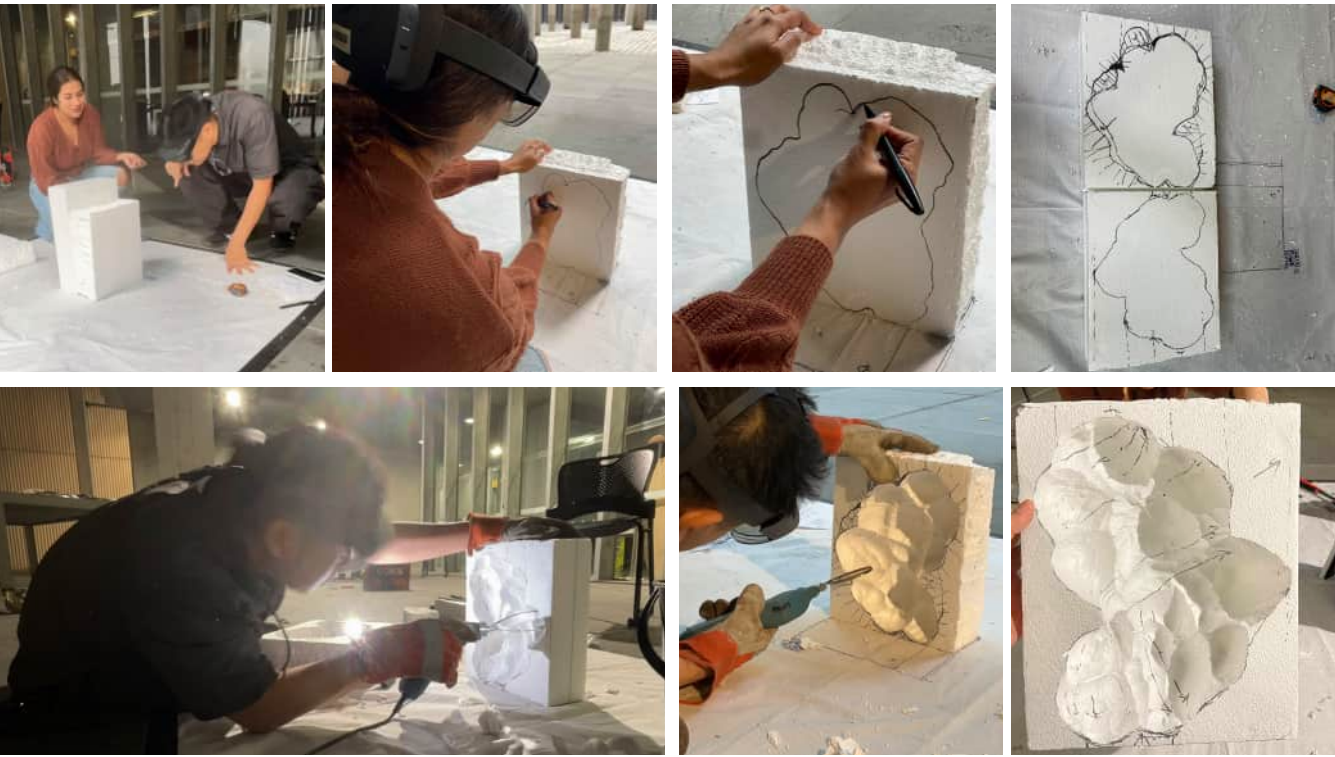
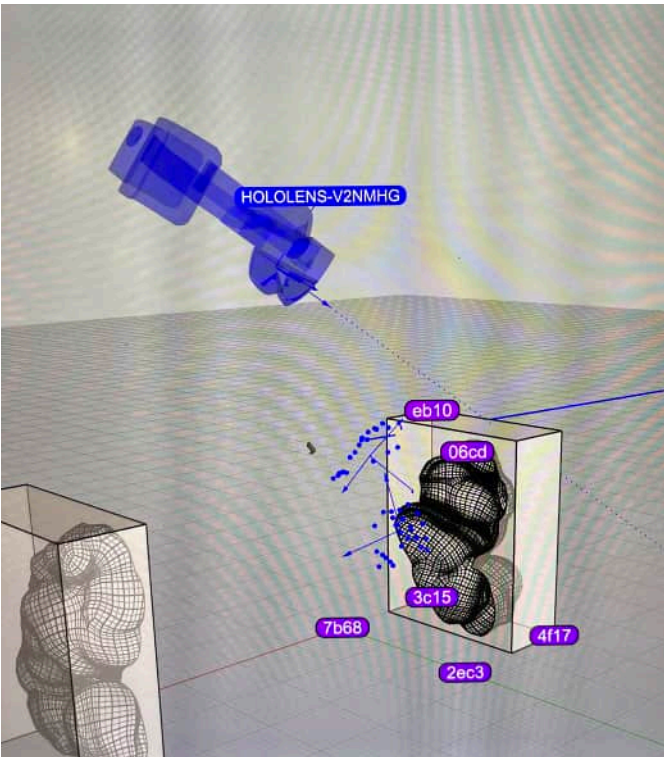
WASTED COFFEE GROUNDS

GEOMETRY DESIGN

Craving mould, using hololen as a guide line



CRAVING AR/MR



CASTING TEST

Positive molding
Base geometry (oil clay) --> Silicone mold (2 days curing) --> Plaster mold (2 hours) -->Material casting (1 day)

MODLDING & CASTING TEST



2-PART 3D PRINTING MOULD PROCESS



Base geometry

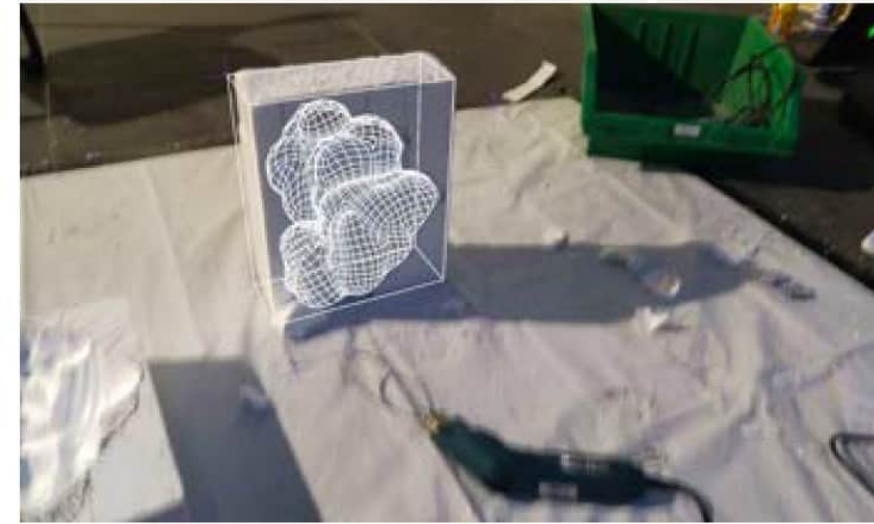
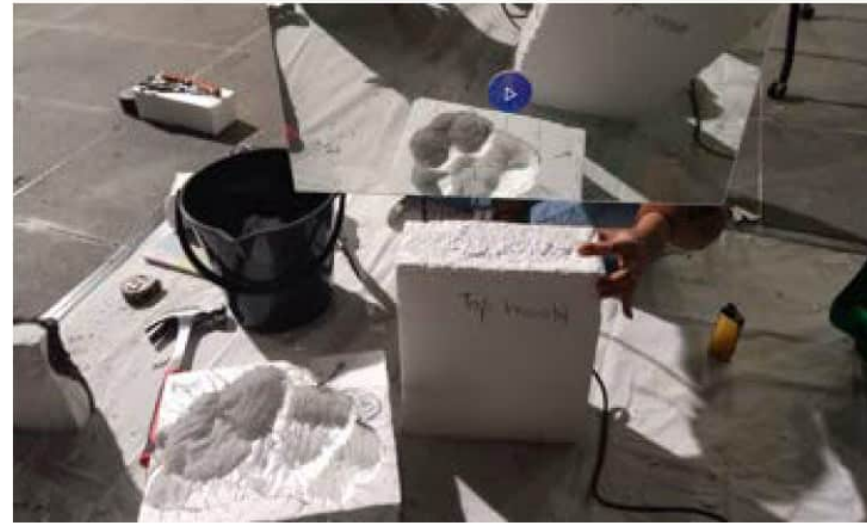
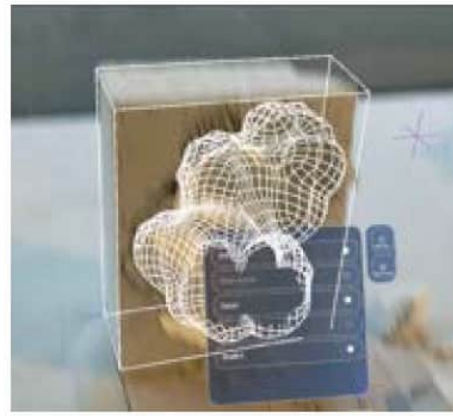
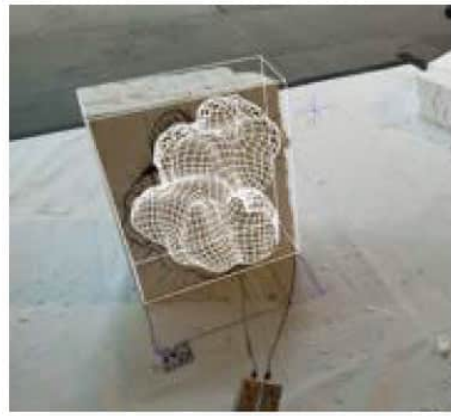


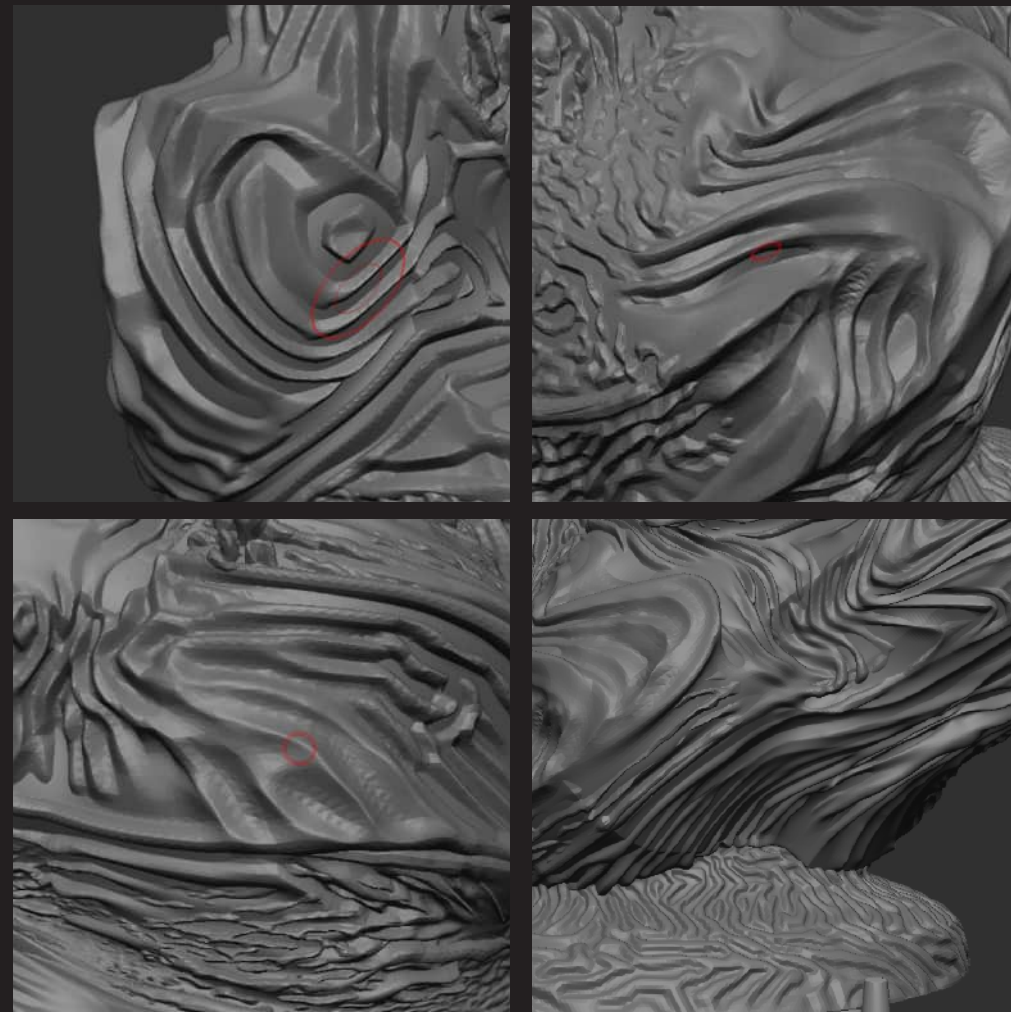
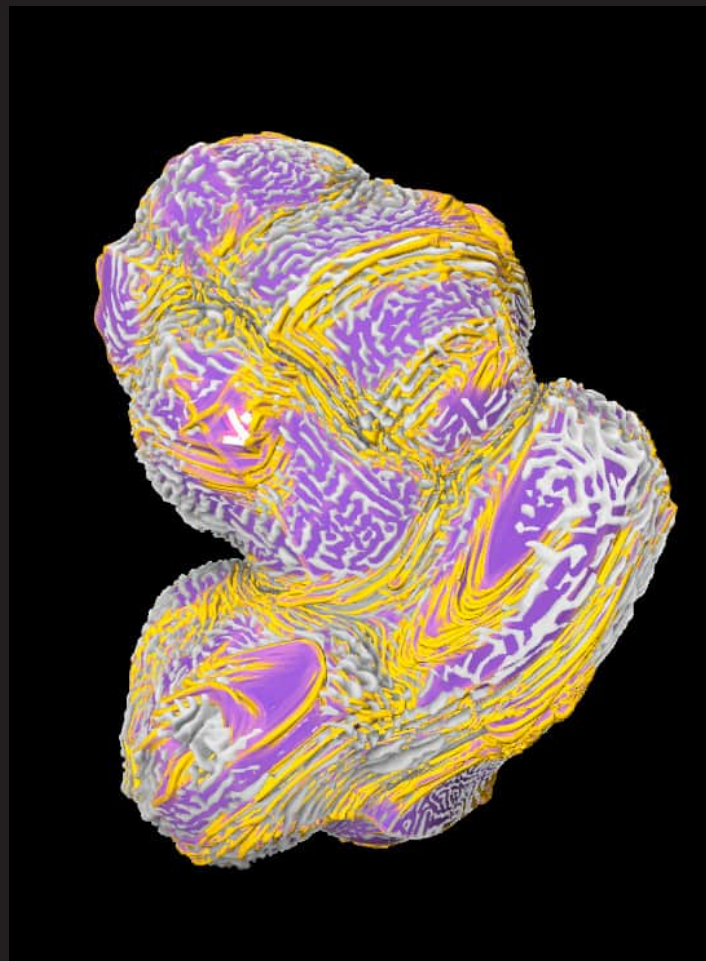
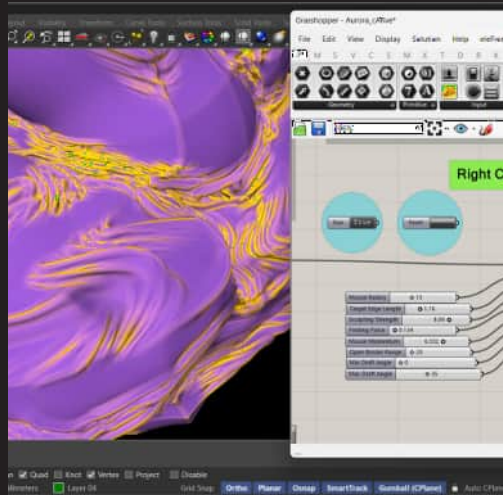
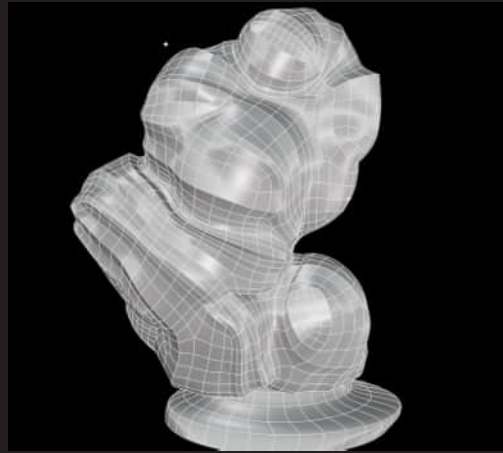
Plaster pouring



Casting&Demolding







PROTOTYPE 3 RE-ATTEMPT

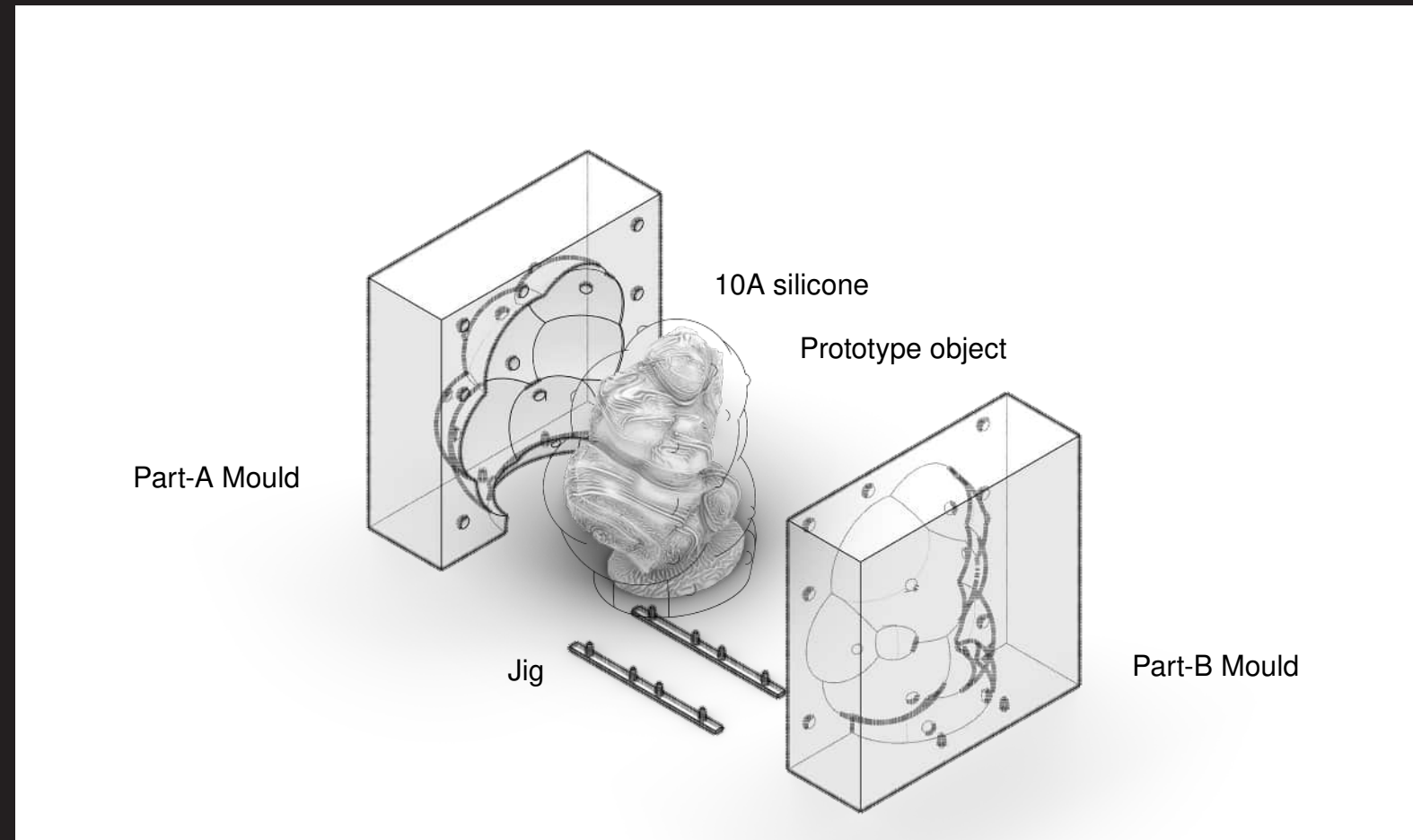
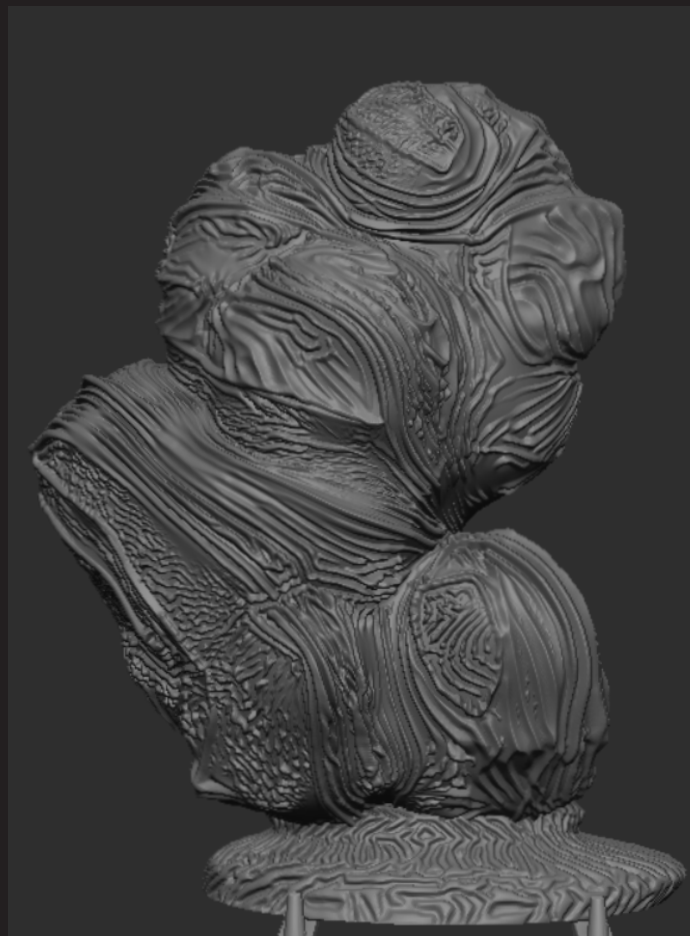
This current project phase, which is actively in the "Testing in Progress" stage, demonstrates a comprehensive digital-to-physical fabrication pipeline. It outlines the methodology for generating organic, high-resolution geometries and physically casting them using a wasted coffee ground mixture.

METHODOLOGY

Stage 1: Base Generation
Create the initial "blob shape" using the Cocoon algorithm.

Stage 2: Modification
Apply Subdivision (Sub-d) modifiers to manipulate and structure the base geometry.

Stage 3: Refinement
Finalize the high-resolution digital iteration by adding complex details using Aurora and ZBrush.



PHYSICAL FABRICATION AND MOULDING STRATEGY

The finalized digital geometry was translated into the physical realm via a 3D-printed mould.

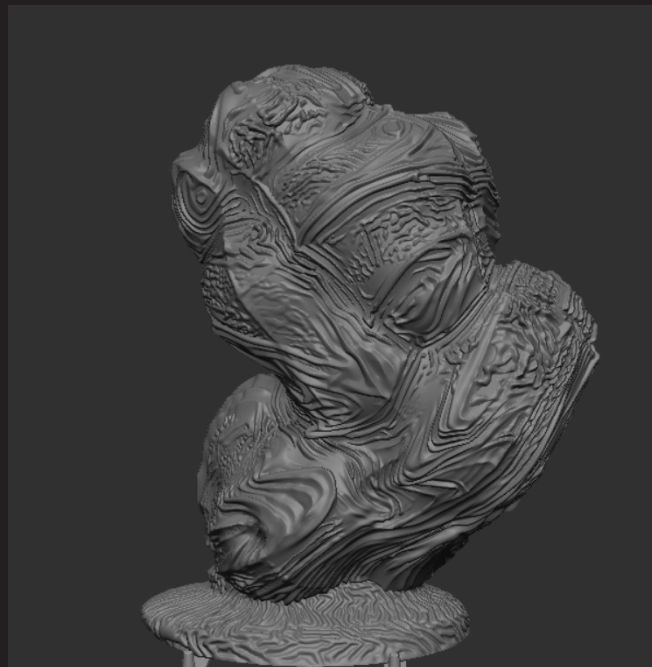
The physical casting assembly is highly controlled, utilizing a Part-A mould, a Part-B mould, and a supporting jig.

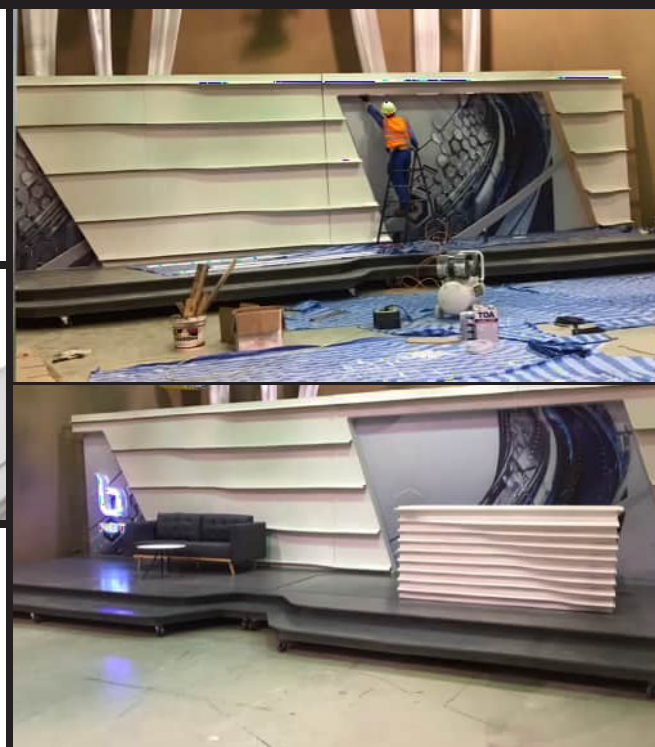
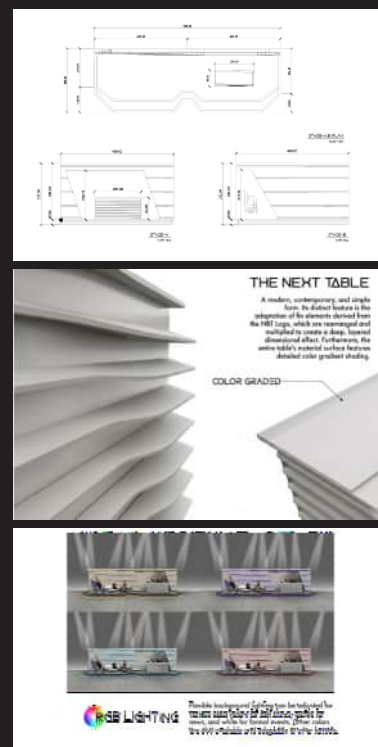
10A silicone is poured to encapsulate the prototype object.

A specific silicone layer is applied to act as a reliable release agent, ensuring the successful demoulding of the cast wasted coffee ground mixture.



ONGOING PROCESS





DIGITAL FABRICATION WORK

THE NEXT STUDIO

KANCHANABURI, THAILAND

Year : 2020

Client: National Broadcasting Television

The NEXT Studio offers an innovative interpretation of broadcast set design, representing a new-modern approach that harmoniously blends highly adaptable functionality with contemporary, organic aesthetics.

Rooted in a commitment to efficiency and sustainability, the design aims to streamline the production process through advanced prefabrication technology and offsite construction. This approach thoughtfully considers the logistics of transportation management from the offsite workshop to the on-site studio, ensuring a seamless installation that aligns with the network's operational vision.

Through careful study of sustainable manufacturing, key design elements such as advanced CNC fabrication and the use of upcycled timber from repurposed wooden pallets are seamlessly integrated. Paired with a refined, stark white color palette, this results in a stage that champions eco-conscious materiality while offering a highly flexible broadcasting environment.

The lighting design emphasizes an integrated RGB illumination system, strategically hidden within the layered architectural fins to create dynamic focal points. Because the primary structure is designed in white, the set acts as a blank canvas, allowing the lighting to completely transform the studio's mood—using yellow for talk shows, purple for news broadcasts, and white for formal ceremonies. This controlled lighting not only adds depth to the tiered, layered presentation tables but also instantly evokes the perfect atmosphere for any specific program.

The studio's key feature is its innovative, moveable stage system, which is thoughtfully divided into two distinct functional zones: a formal news reporting desk and a relaxed talk show interview area. Equipped with wheels for effortless mobility, the stages offer producers a variety of spatial options, allowing the modules to be separated for individual use, stored efficiently, or combined to maximize the studio footprint. This ensures ultimate flexibility to adapt to the client's diverse broadcasting and operational needs.



